Embedded Systems Laboratory

Catedra de Calculatoare
Facultatea de Automatica si Calculatoare

Ingineria Calculatoarelor
Fault Tolerant Computing

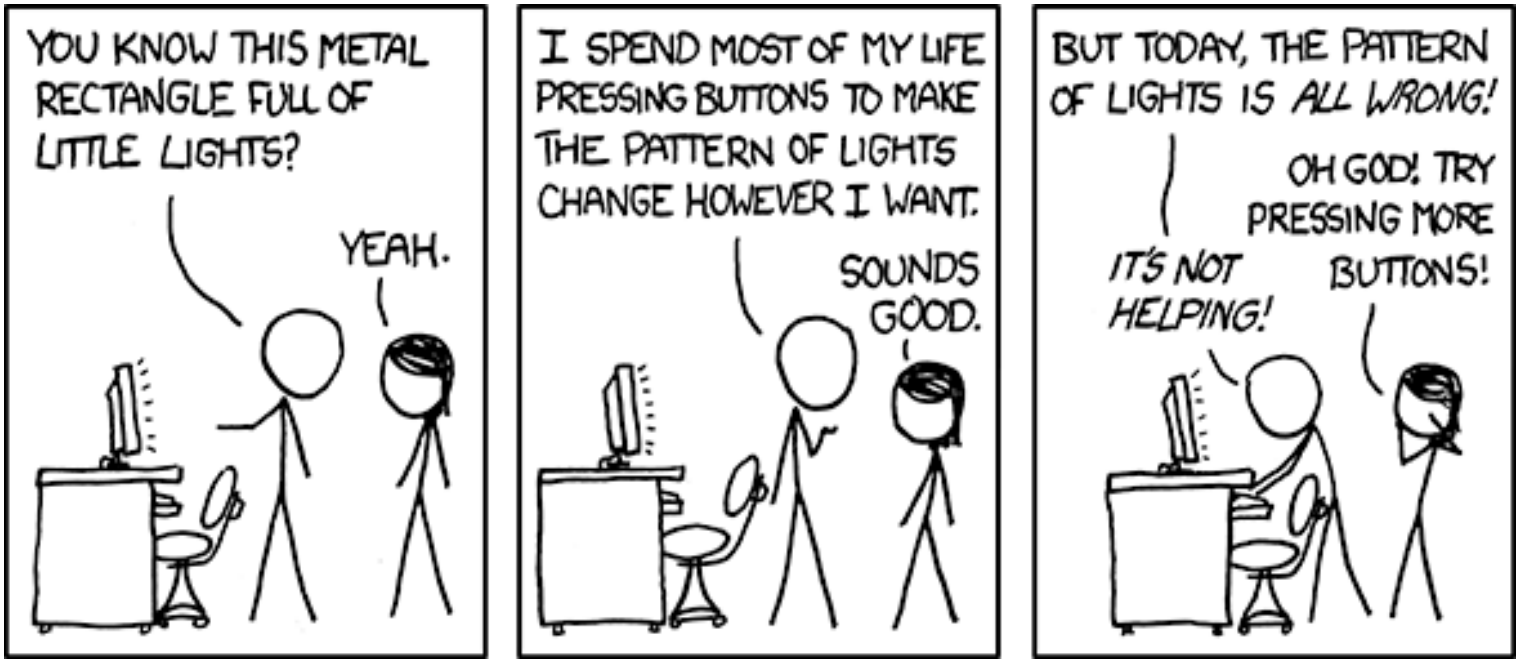
- Cursul 0 -

Facultatea de Automatică și Calculatoare
Universitatea Politehnica București

“If anything can go wrong, it will.”

Murphy’s Law

XKCD of the Day



http://xkcd.com/722/

Important Stuff

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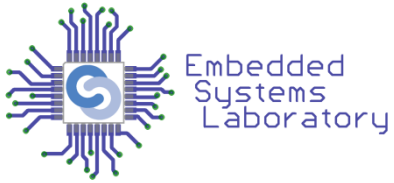
Dan Tudose
- dan.tudose@cs.pub.ro
- dan.s.tudose@gmail.com

Laboratoare: http://elf.cs.pub.ro/ic/wiki

Notare:
- ➢1 punct prezenta laborator
- ➢3 puncte activitatea laborator
- ➢1 punct lucrare laborator
- ➢5 puncte examen final

Cerinte pentru a promova:
- minim 6 prezente la laborator
- minim 2.5 puncte din cele 5 puncte pentru activitatea laborator
- minim 2.5 puncte din cele 5 puncte din examenul final

The Curse of Complexity



Computer engineering is the art and science of translating user requirements we do not fully understand; into hardware and software we cannot precisely analyze; to operate in environments we cannot accurately predict; all in such a way that the society at large is given no reason to suspect the extent of our ignorance.[1]

Microsoft Windows NT (1992): ≈4M lines of code

Microsoft Windows 10 (2015): ≈70M lines of code

Intel Pentium processor (1993): ≈4M transistors

Intel Pentium 4 processor (2001): ≈40M transistors

Intel Xeon Haswell-EP (2015): ≈5500M transistors

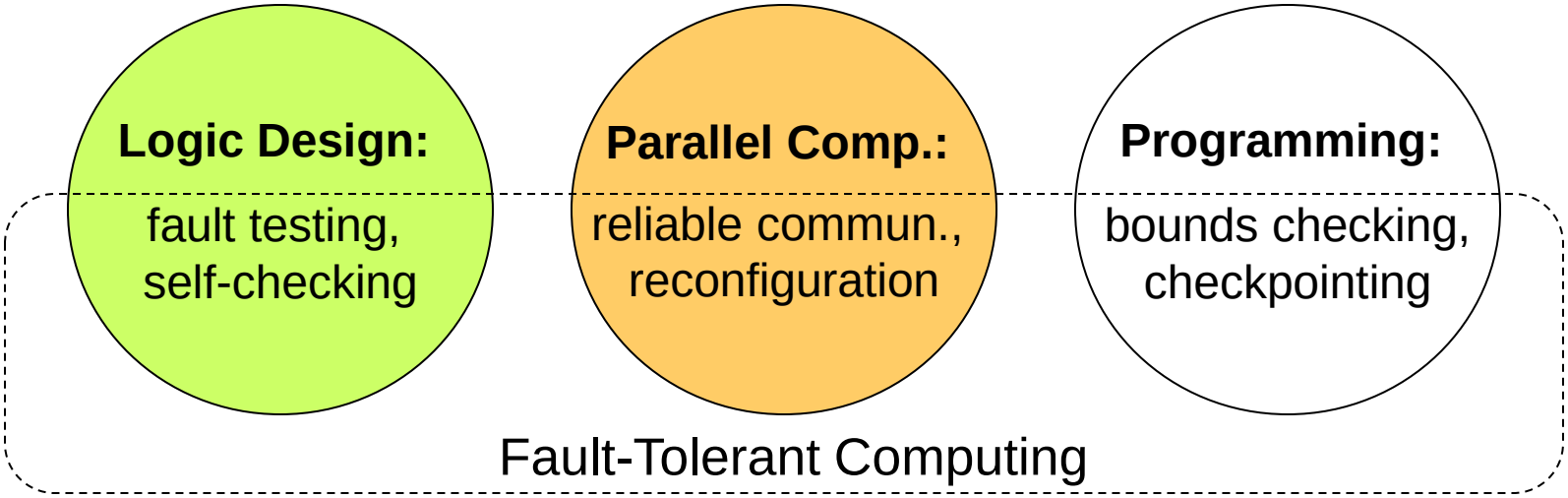
[1]Adapted from definition of structural engineering: Ralph Kaplan, *By Design: Why There Are No Locks on the Bathroom Doors in the Hotel Louis XIV and Other Object Lessons*, Fairchild Books, 2004, p. 229

Why This Course Shouldn't Be Needed

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In an ideal world, methods for dealing with faults, errors, and other impairments in hardware and software would be covered within every computer engineering course that has a design component

Analogy: We do not teach structural engineers about building bridges in one course and about bridge safety and structural integrity during high winds or earthquakes in another (optional) course



Brief History of Dependable Computing

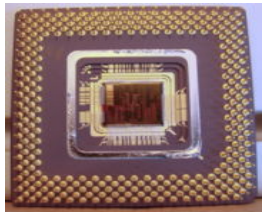
1940s: ENIAC, with 17.5K vacuum tubes and 1000s of other electrical elements, failed once every 2 days (avg. down time = minutes)

1950s: Early ideas by von Neumann (multichannel, with voting) and Moore-Shannon ("crummy" relays)

1960s: NASA and military agencies supported research for long-life space missions and battlefield computing

1970s: The field developed quickly (international conference, many research projects and groups, experimental systems)

1980s: The field matured (textbooks, theoretical developments, use of ECCs in solid-state memories, RAID concept), but also suffered some loss of focus and interest because of the extreme reliability of integrated circuits



1990s: Increased complexity at chip and system levels made verification, testing, and testability prime study topics

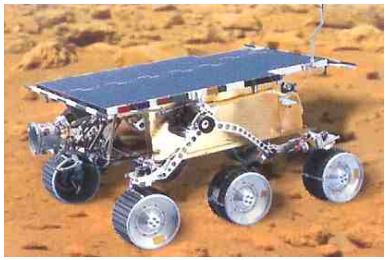
2000s: Resurgence of interest owing to less reliable fabrication at ultrahigh densities and "crummy" nanoelectronic components

Dependable Computing in the 2010s

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There are still ambitious projects; space and elsewhere

- Harsh environments (vibration, pressure, temperatures)
- External influences (radiation, micrometeoroids)
- Need for autonomy (commun. delays, unmanned probes)



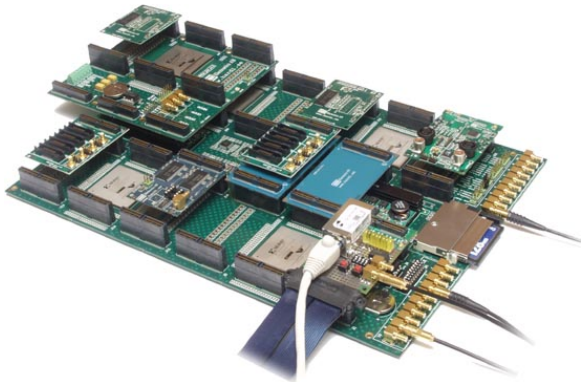
The need is expanding

- More complex systems (e.g., system-on-chip)
- Critical applications (medicine, transportation, finance)
- Expanding pool of unsophisticated users
- Continued rise in maintenance costs
- Digital-only data (needs more rigorous backup)



The emphasis is shifting

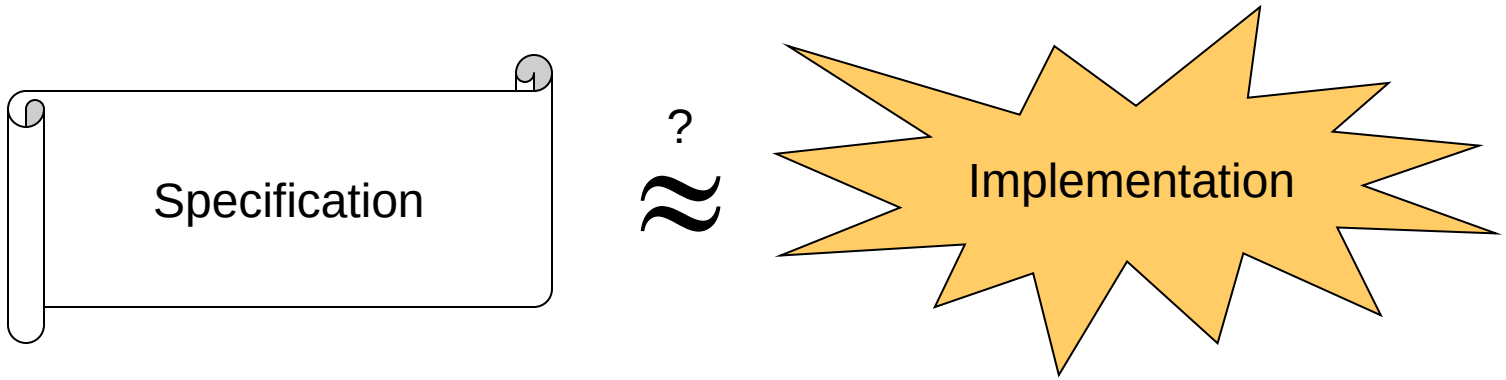
- Mostly COTS-based solutions
- Integrated hardware/software systems
- Entire units replaced (little diagnosis)



Defining Failure

Failure is an unacceptable difference between expected and observed performance.[1]

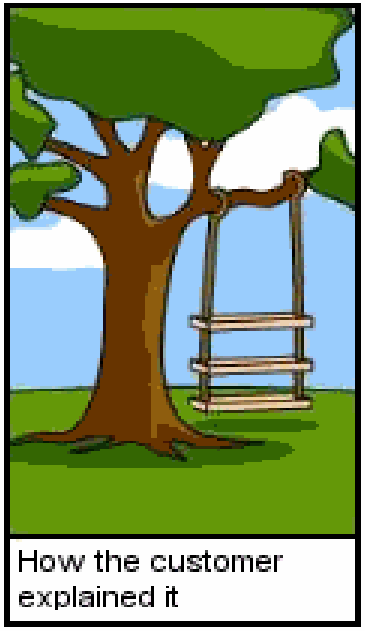
A structure (building or bridge) need not collapse catastrophically to be deemed a failure

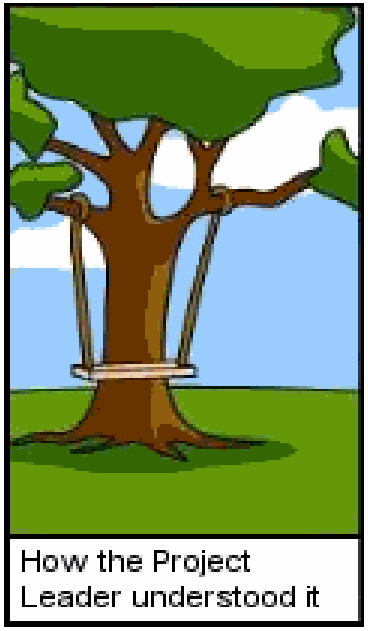


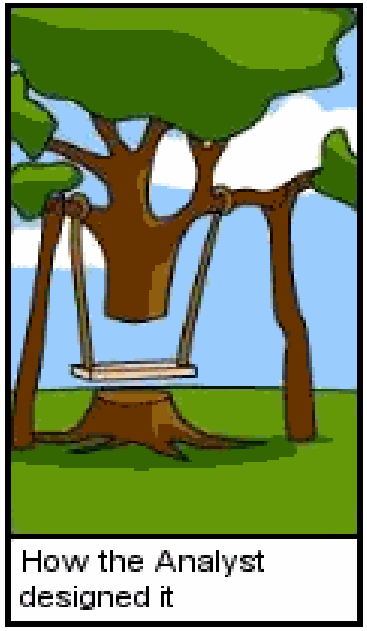
Reasons of typical Web site failures

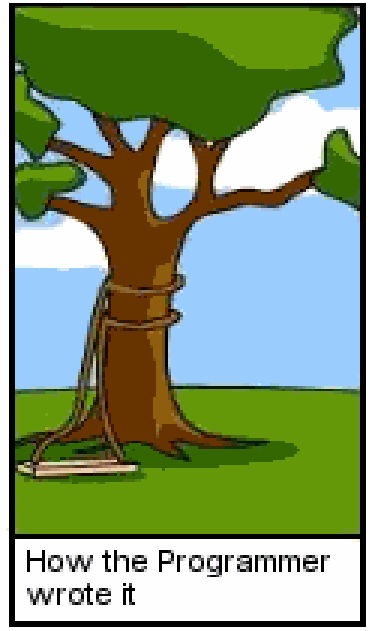
Hardware problems:	15%
Software problems:	34%
Operator error:	51%

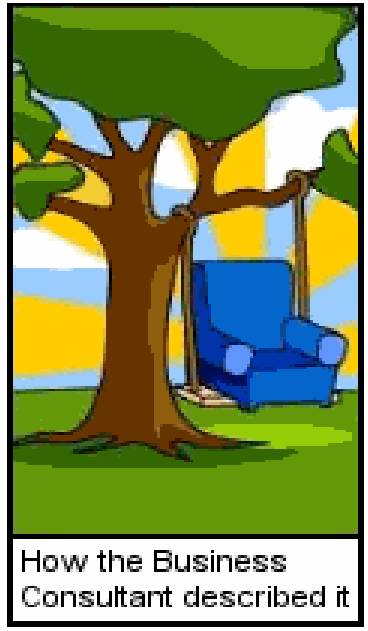
[1] Definition used by the Tech. Council on Forensic Engineering of the Amer. Society of Civil Engineers

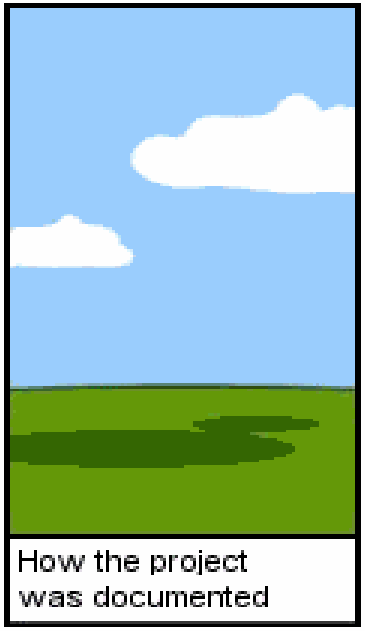

How the customer explained it

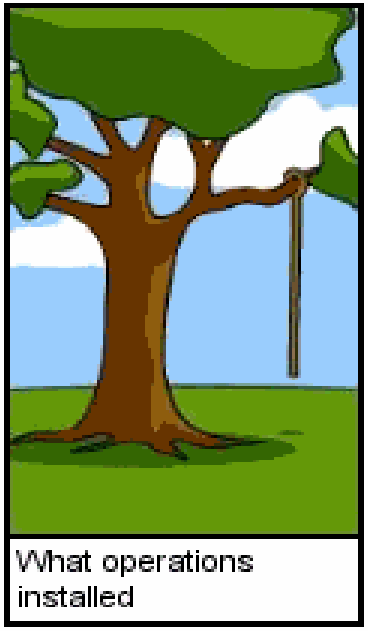

How the Project Leader understood it

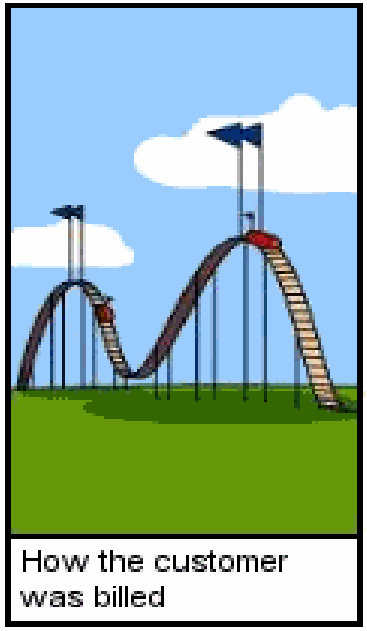

How the Analyst designed it

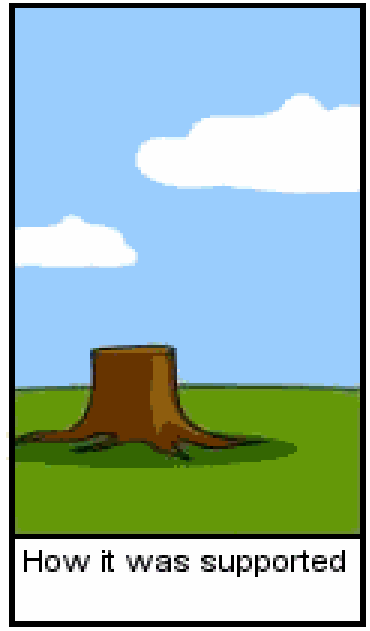

How the Programmer wrote it

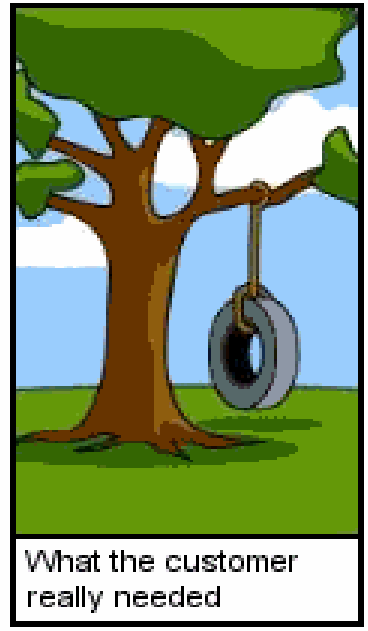

How the Business Consultant described it


How the project was documented


What operations installed


How the customer was billed


How it was supported

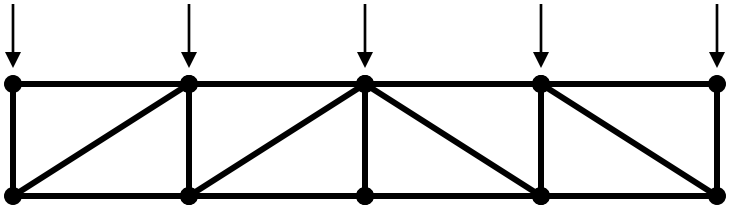

What the customer really needed

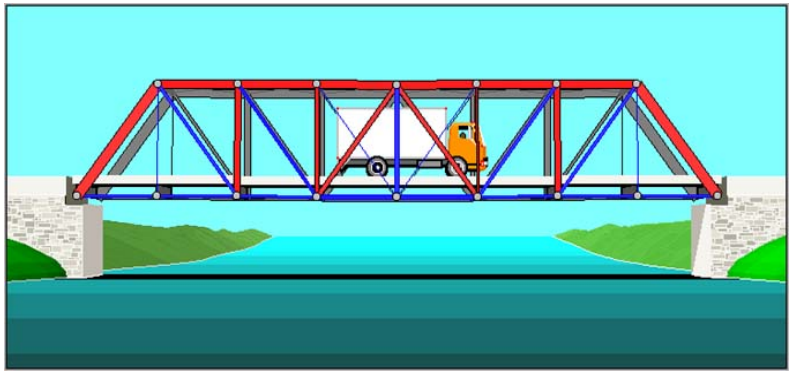
Design Flaws: "To Engineer is Human"[1]

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Complex systems almost certainly contain multiple design flaws

Redundancy in the form of safety factor is routinely used in buildings and bridges





Example of a more subtle flaw:
Disney Concert Hall in Los Angeles reflected light into nearby building, causing discomfort for tenants due to blinding light and high temperature



[1] Title of book by Henry Petroski

Yet Another Example

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The Walkie Talkie melted my Jag! Light reflected from under-construction City skyscraper buckles bodywork and mirror of businessman's car

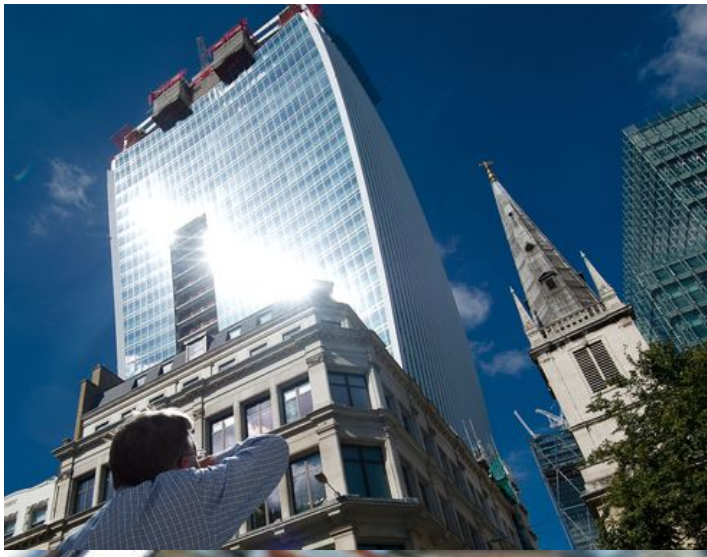
- **Sunlight reflected from skyscraper is causing heat damage to cars beneath**
- **Several panels of a Jaguar XJ had buckled in the glare**
- **Other drivers say their vehicles have wilted in the beam of light**

By SAM WEBB

PUBLISHED: 18:56 GMT, 2 September 2013 | **UPDATED:** 22:39 GMT, 3 September 2013

Share Tweet +1 Share 800 View comments

A £200million skyscraper has had an undeniably dazzling effect on passers-by – but not, unfortunately, the one intended by its architect.



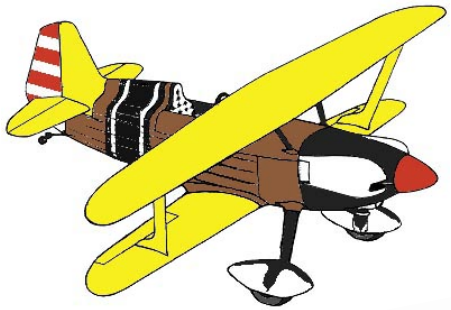


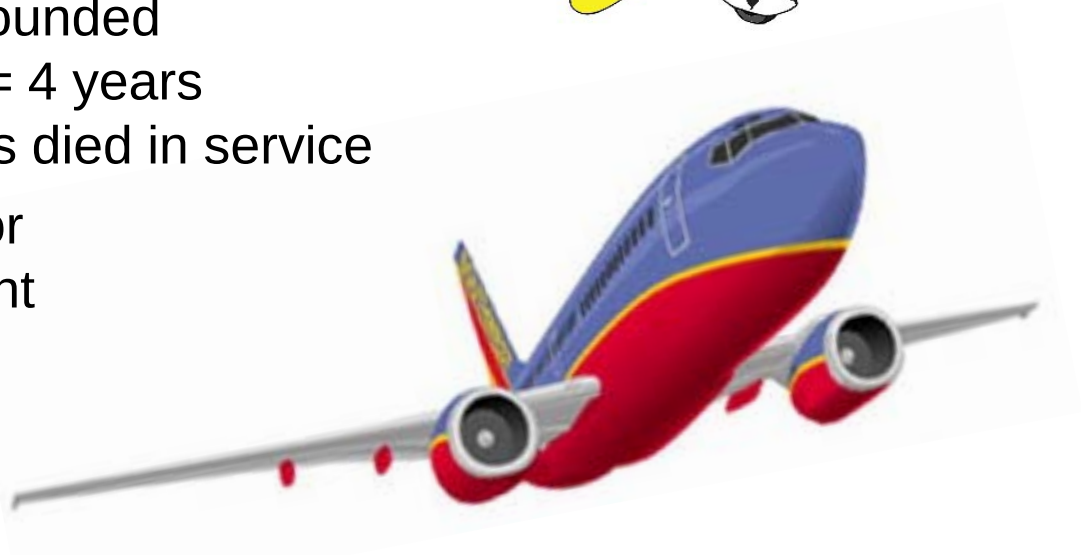


Learning Curve: "Normal Accidents"[1]

Example: Risk of piloting a plane

1903	First powered flight
1908	First fatal accident
1910	Fatalities = 32 (≈2000 pilots worldwide)
1918	US Air Mail Service founded Pilot life expectancy = 4 years 31 of the first 40 pilots died in service
1922	One forced landing for every 20 hours of flight
Today	Commercial airline pilots pay normal life insurance rates





Unfortunately, the learning curve for computers and computer-based systems is not as impressive

[1] Title of book by Charles Perrow (Ex. p. 125)

Mishaps, Accidents, and Catastrophes

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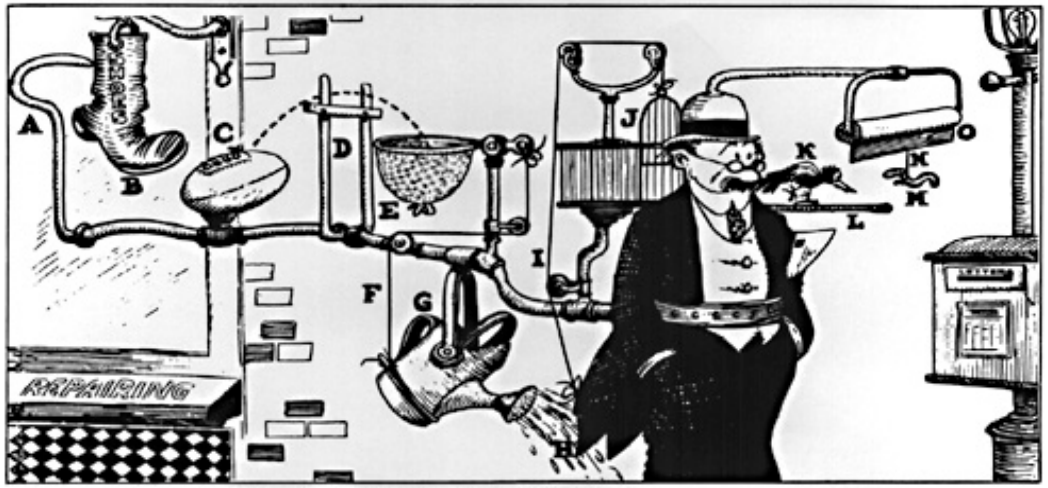
Mishap: misfortune; unfortunate accident

Accident: unexpected (no-fault) happening causing loss or injury

Catastrophe: final, momentous event of drastic action; utter failure

At one time (following the initial years of highly unreliable hardware), computer mishaps were predominantly the results of human error

Now, most mishaps are due to complexity (unanticipated interactions)



Keep You From Forgetting To Mail Your Wife's Letter RUBE GOLDBERG (tm) RGI 049

Rube Goldberg contraptions



The butterfly effect

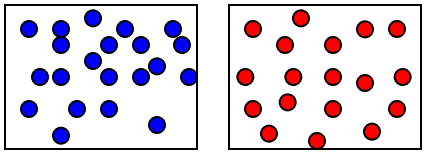
Pretest: Failures and Probabilities

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This test will not be graded or even collected, so answer the test questions truthfully and to the best of your ability / knowledge

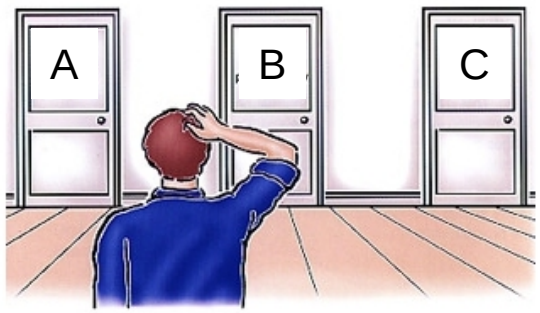
Question 1: Name a disaster that was caused by computer hardware or software failure. How do you define “disaster” and “failure”?

Question 2: Which of these patterns is more random?



Question 3: Which do you think is more likely: the event that everyone in this class was born in the first half of the year or the event that at least two people were born on the same day of the year?

Question 4: In a game show, there is a prize behind one of 3 doors with equal probabilities. You pick Door A. The host opens Door B to reveal that there is no prize behind it. The host then gives you a chance to switch to Door C. Is it better to switch or to stick to your choice?



Pretest (Continued): Causes of Mishaps



Question 5: Does this photo depict a mishap due to design flaw, implementation bug, procedural inadequacies, or human error?

Pretest (Continued): Reliability and Risk

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Question 6: Name an emergency backup system (something not normally used unless another system fails) that is quite commonplace

Question 7: Which is more reliable: plane X or plane Y that carries four times as many passengers as plane X and is twice as likely to crash?

Question 8: Which is more reliable: a 4-wheel vehicle with one spare tire or an 18-wheeler with 2 spare tires?

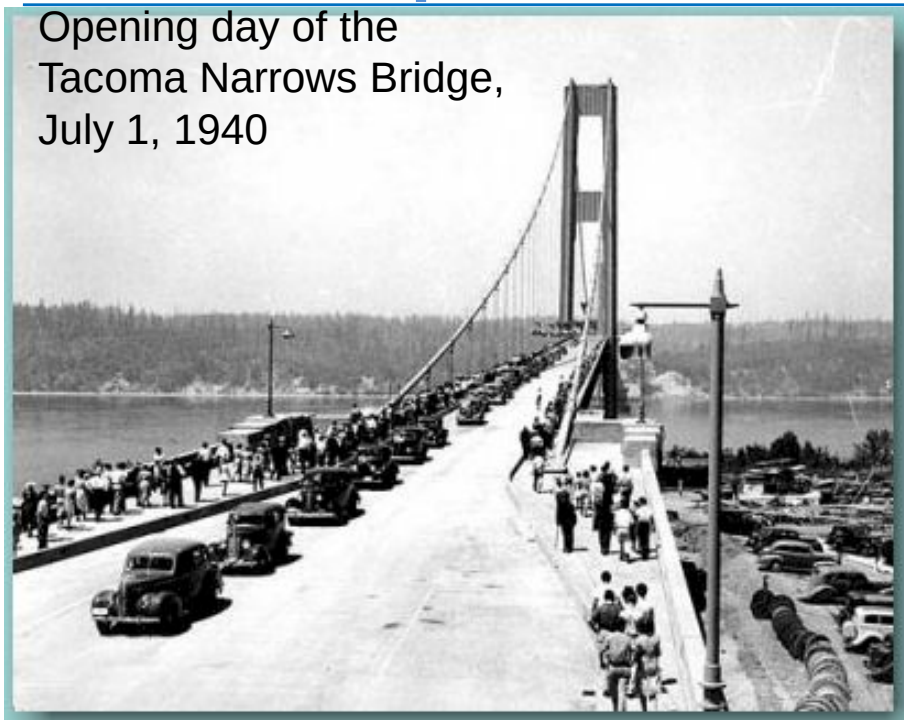
Question 9: Which surgeon would you prefer for an operation that you must undergo: Surgeon A, who has performed some 500 operations of the same type, with 5 of his patients perishing during or immediately after surgery, or surgeon B who has a perfect record in 25 operations?

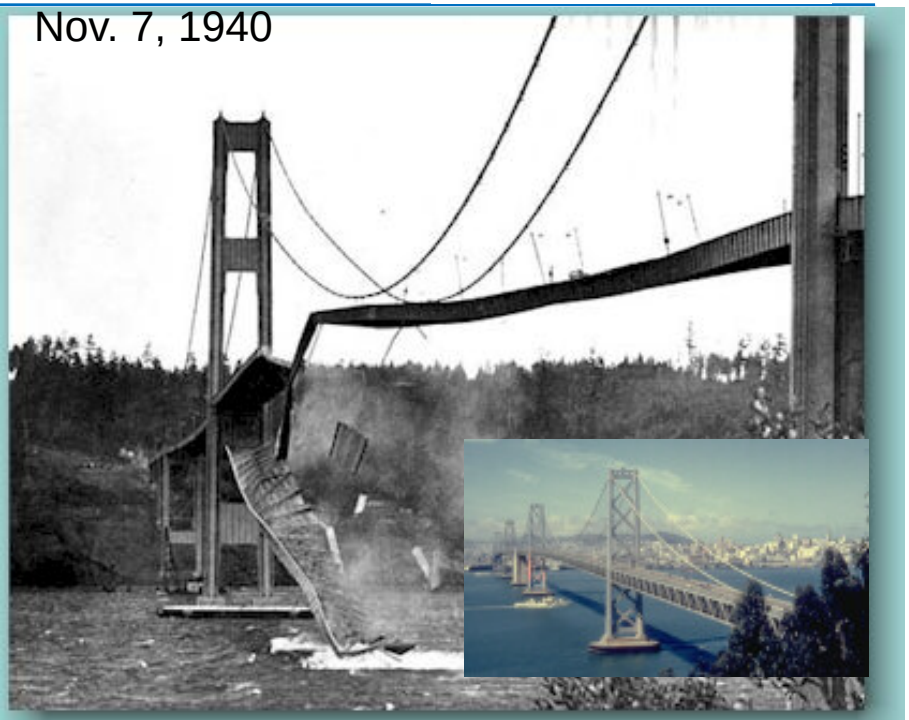
Question 10: Which is more probable at your home or office: a power failure or an Internet outage? Which is likely to last longer?

If you had trouble with 3 or more questions, you really need this course!

What Do We Learn from Bridges that Collapse?

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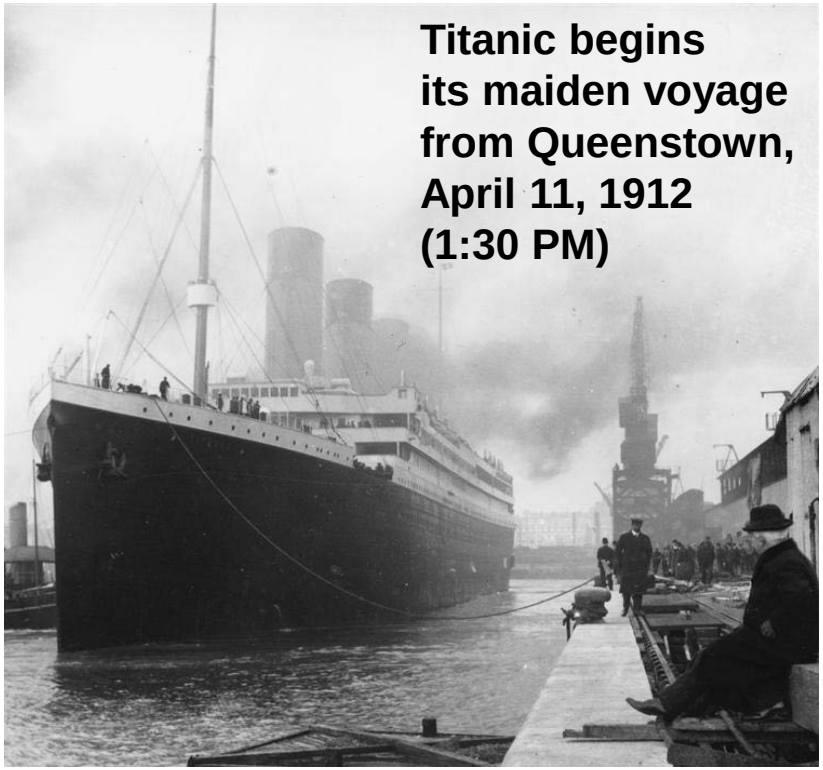
One catastrophic bridge collapse every 30 years or so

See the following amazing video clip (Tacoma Narrows Bridge):
http://www.enm.bris.ac.uk/research/nonlinear/tacoma/tacnarr.mpg

" . . . failures appear to be inevitable in the wake of prolonged success, which encourages lower margins of safety. Failures in turn lead to greater safety margins and, hence, new periods of success."

Henry Petroski, *To Engineer is Human*

. . . or from “Unsinkable” Ships that Sink?



Titanic begins its maiden voyage from Queenstown, April 11, 1912 (1:30 PM)



April 15, 1912 (2:20 AM)

“The major difference between a thing that might go wrong and a thing that cannot possibly go wrong is that when a thing that cannot possibly go wrong goes wrong, it usually turns out to be impossible to get at or repair.”

Douglas Adams, author of *The Hitchhiker’s Guide to the Galaxy*

… or from Poorly Designed High-Tech Trains?





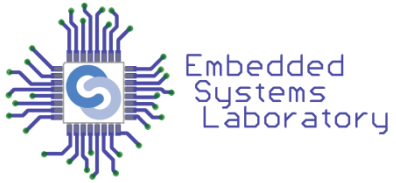
Train built for demonstrating magnetic levitation technology in northwest Germany rams into maintenance vehicle left on track at 200 km/h, killing 23 of 29 aboard

Official investigation blames the accident on human error (train was allowed to depart before a clearance phone call from maintenance crew)

Not a good explanation; even low-tech trains have obstacle detection systems

Even if manual protocol is fully adequate under normal conditions, any engineering design must take unusual circumstances into account (abuse, sabotage, terrorism)

Design Flaws in Computer Systems



Hardware example: Intel Pentium processor, 1994

For certain operands, the FDIV instruction yielded a wrong quotient

Amply documented and reasons well-known (overzealous optimization)

Software example: Patriot missile guidance, 1991

Missed intercepting a scud missile in 1st Gulf War, causing 28 deaths

Clock reading multiplied by 24-bit representation of 1/10 s (unit of time) caused an error of about 0.0001%; normally, this would cancel out in relative time calculations, but owing to ad hoc updates to some (not all) calls to a routine, calculated time was off by 0.34 s (over ≈100 hours), during which time a scud missile travels more than ½ km

User interface example: Therac 25 machine, mid 1980s[1]

Serious burns and some deaths due to overdose in radiation therapy

Operator entered “x” (for x-ray), realized error, corrected by entering “e” (for low-power electron beam) before activating the machine; activation was so quick that software had not yet processed the override

[1] Accounts of the reasons vary

Causes of Human Errors in Computer Systems

1. Personal factors (35%): Lack of skill, lack of interest or motivation, fatigue, poor memory, age or disability

2. System design (20%): Insufficient time for reaction, tedium, lack of incentive for accuracy, inconsistent requirements or formats

3. Written instructions (10%): Hard to understand, incomplete or inaccurate, not up to date, poorly organized

4. Training (10%): Insufficient, not customized to needs, not up to date

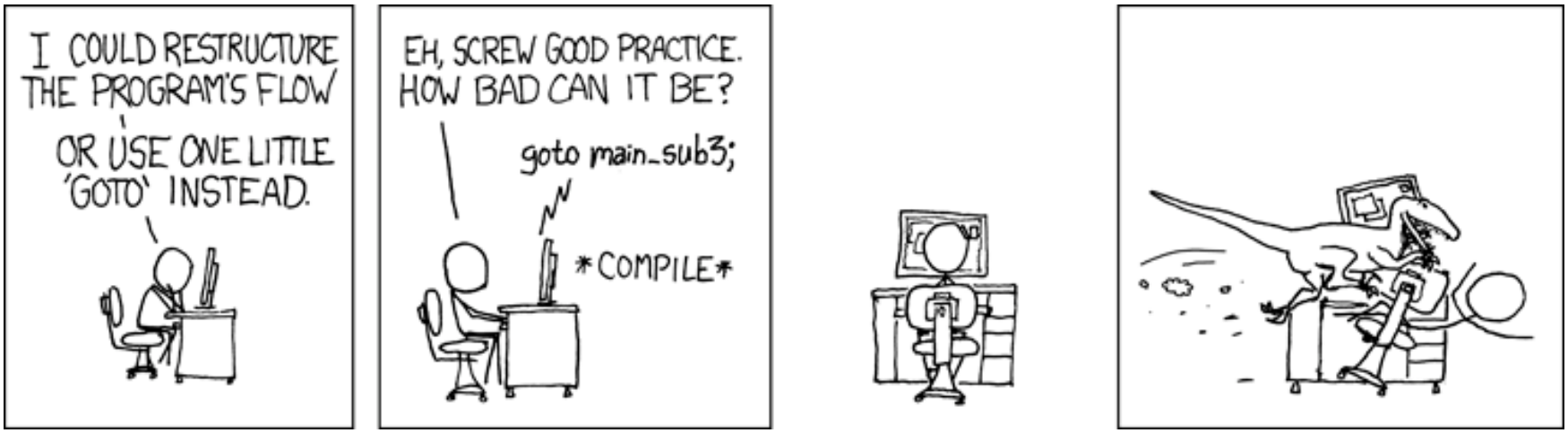
5. Human-computer interface (10%): Poor display quality, fonts used, need to remember long codes, ergonomic factors

6. Accuracy requirements (10%): Too much expected of operator

7. Environment (5%): Lighting, temperature, humidity, noise

Because “the interface is the system” (according to a popular saying), items 2, 5, and 6 (40%) could be categorized under user interface


NO! - Bad User!!!
You've been warned 3 times that this file does not exist.
Now you've made us catch this worthless exception and we're upset.
Do not do this again.
OK
Big error
You really screwed up this time.
I know
What else is new
It's my first day



http://xkcd.com/292/

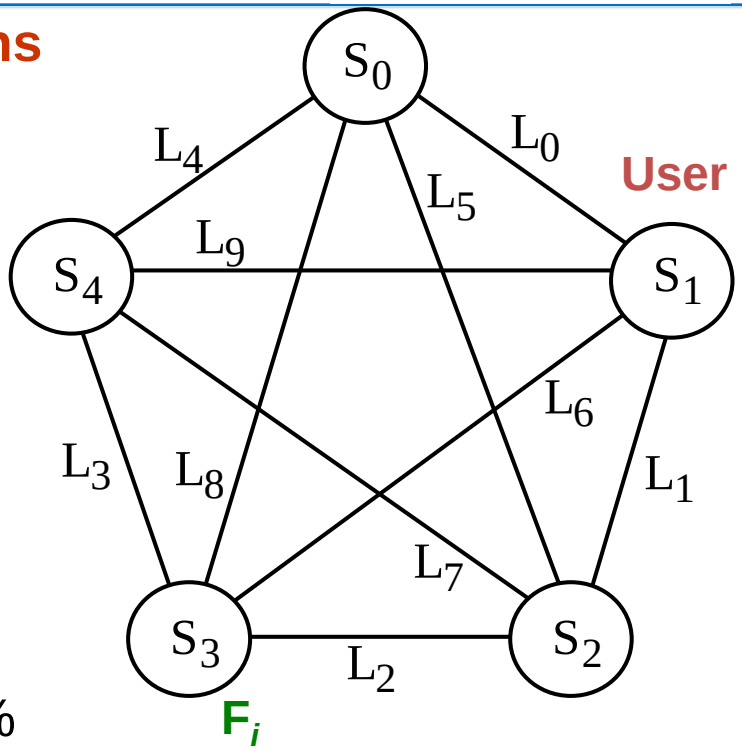
A Case Study

Data availability and integrity concerns

Distributed DB system with 5 sites
Full connectivity, dedicated links
Only direct communication allowed
Sites and links may malfunction
Redundancy improves availability

S: Probability of a site being available
L: Probability of a link being available

Single-copy availability = *SL*
Unavailability = 1 – *SL*
= 1 – 0.99 × 0.95 = 5.95%



Data replication methods, and a challenge

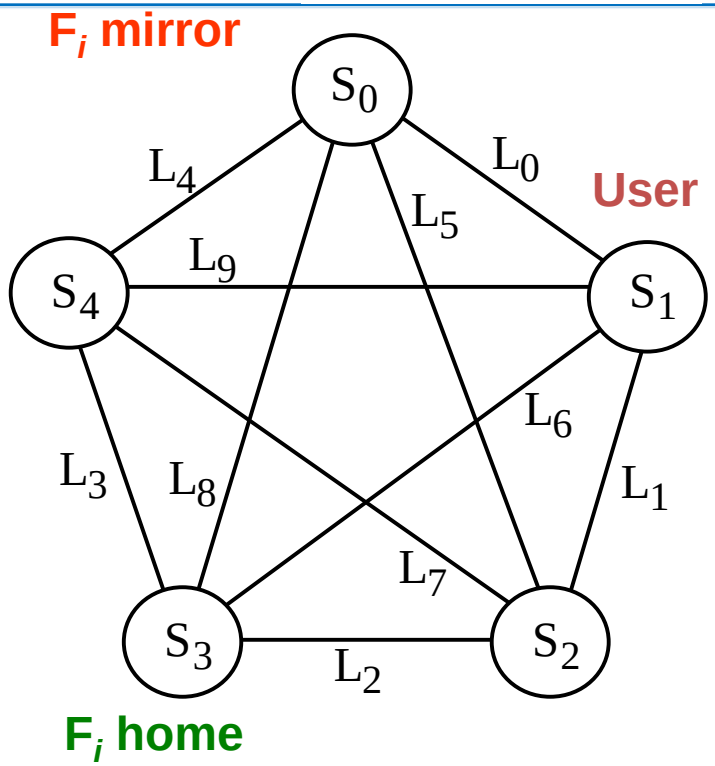
File duplication: home / mirror sites
File triplication: home / backup 1 / backup 2
Are there availability improvement methods with less redundancy?

Data Duplication: Home and Mirror Sites

S: Site availability e.g., 99%
L: Link availability e.g., 95%

$$A = SL + (1 - SL)SL$$

- SL: Primary site can be reached
- SL: Mirror site can be reached
- $(1 - SL)$: Primary site inaccessible



Duplicated availability = $2SL - (SL)^2$
Unavailability = $1 - 2SL + (SL)^2$
= $(1 - SL)^2$ = 0.35%

Data unavailability reduced from 5.95% to 0.35%

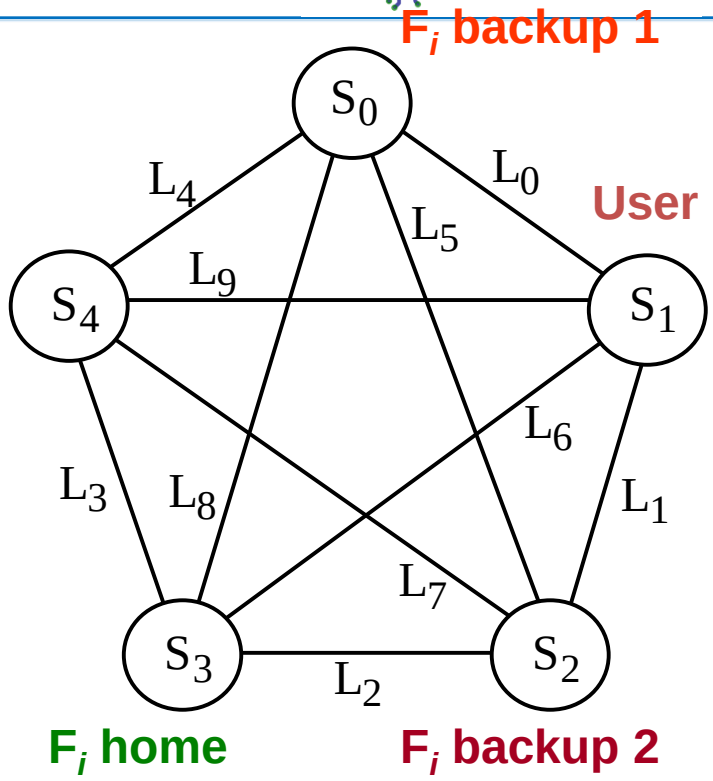
Availability improved from ≈ 94% to 99.65%

Data Triplication: Home and Two Backups

S: Site availability e.g., 99%
L: Link availability e.g., 95%

$$A = SL + (1 - SL)SL + (1 - SL)^2SL$$

- SL: Primary site can be reached
- $(1 - SL)SL$: Primary site inaccessible, Backup 1 can be reached
- $(1 - SL)^2SL$: Primary and backup 1 inaccessible, Backup 2 can be reached

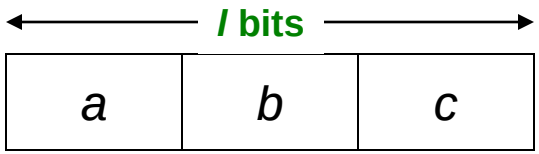


Triplicated avail. $= 3SL - 3(SL)^2 - (SL)^3$
Unavailability $= 1 - 3SL - 3(SL)^2 + (SL)^3$
$= (1 - SL)^3 = 0.02\%$

Data unavailability reduced from 5.95% to 0.02%

Availability improved from ≈ 94% to 99.98%

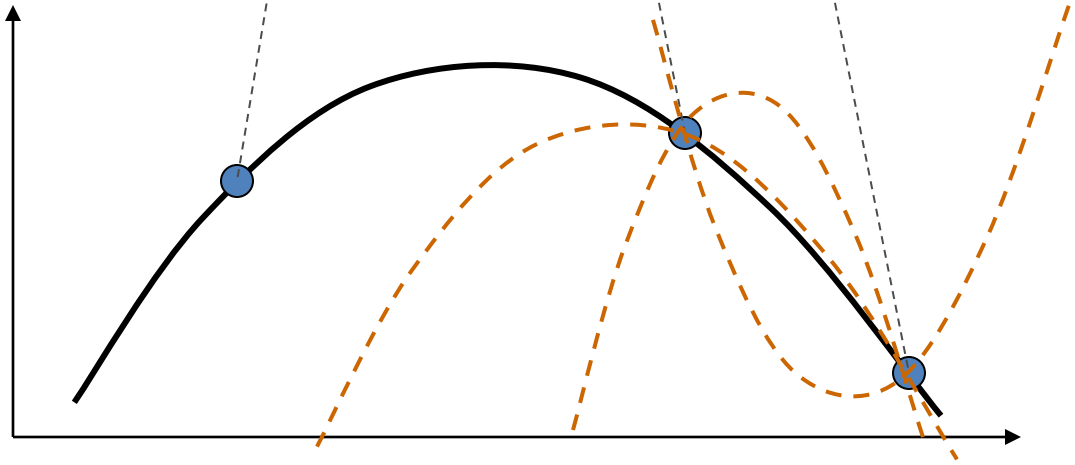
Dispersion for Data Security and Integrity

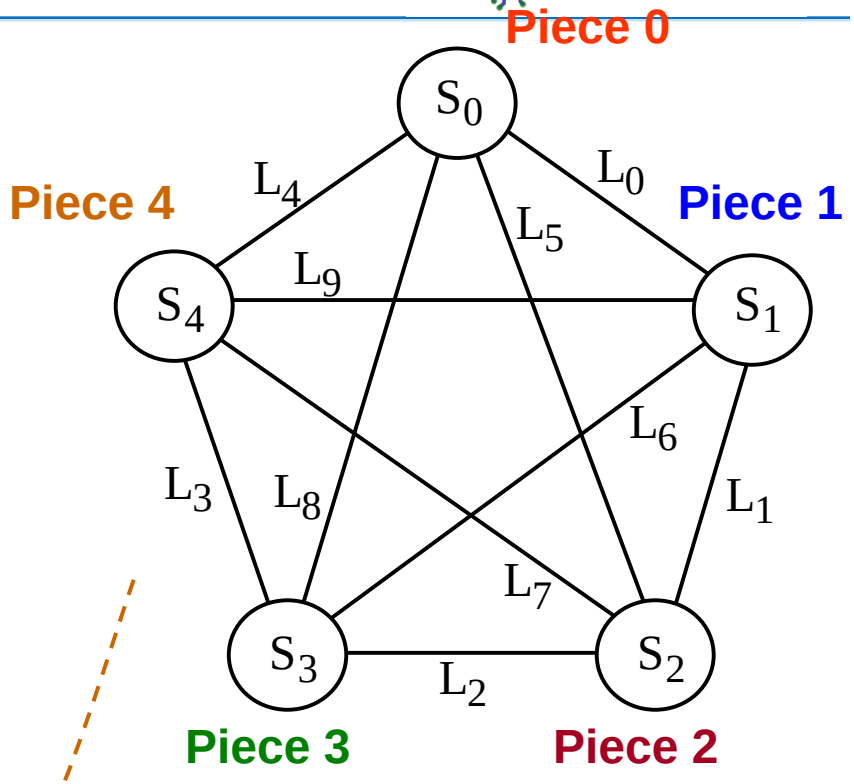


Encoding with
67% redundancy

$f(x) = ax^2 + bx + c$







Note that two pieces would be inadequate for reconstruction

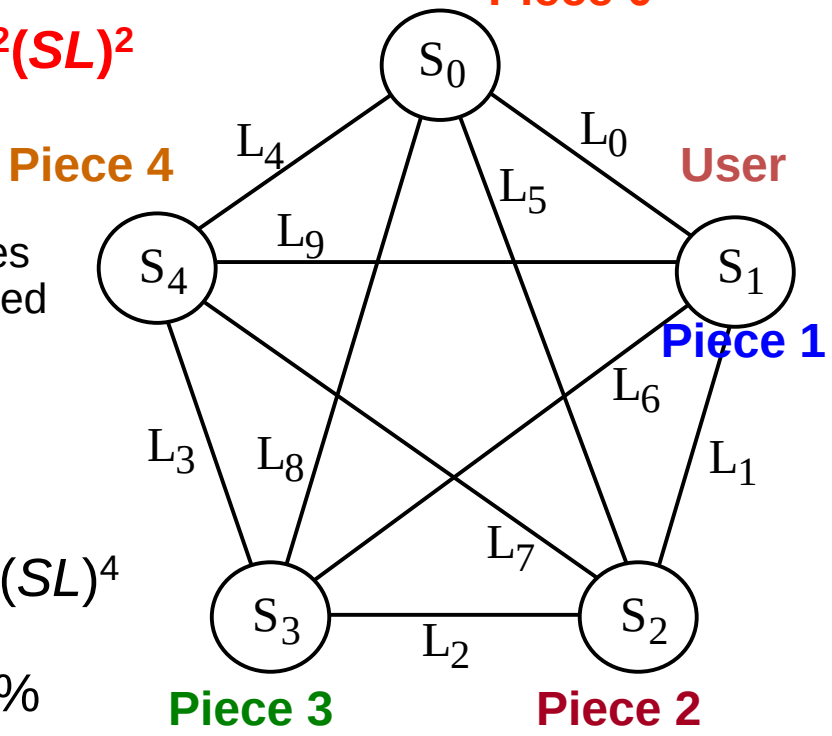
Data Dispersion: Three of Five Pieces

$$A = (SL)^4 + 4(1 - SL)(SL)^3 + 6(1 - SL)^2(SL)^2$$

All 4 pieces can be reached

Exactly 3 pieces can be reached

Only 2 pieces can be reached

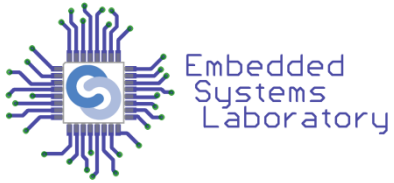


S: Site availability e.g., 99%
L: Link availability e.g., 95%

Dispersed avail. = $6(SL)^2 - 8(SL)^3 + 3(SL)^4$
Availability = 99.92%
Unavailability = 1 – Availability = 0.08%

Scheme →	Nonredund.	Duplication	Triplication	Dispersion
Unavailability	5.95%	0.35%	0.02%	0.08%
Redundancy	0%	100%	200%	67%

Questions Ignored in Our Simple Example



1. How redundant copies of data are kept consistent

When a user modifies the data, how to update the redundant copies (pieces) quickly and prevent the use of stale data in the meantime?

2. How malfunctioning sites and links are identified

Malfunction diagnosis must be quick to avoid data contamination

3. How recovery is accomplished when a malfunctioning site / link returns to service after repair

The returning site must be brought up to date with regard to changes

4. How data corrupted by the actions of an adversary is detected

This is more difficult than detecting random malfunctions

The example does demonstrate, however, that:

- Many alternatives are available for improving dependability
- Proposed methods must be assessed through modeling
- The most cost-effective solution may be far from obvious

Toleranța la defecte

Defecte:

- Software
- Hardware
- Design
- Operational

Eroare Latentă

Activă

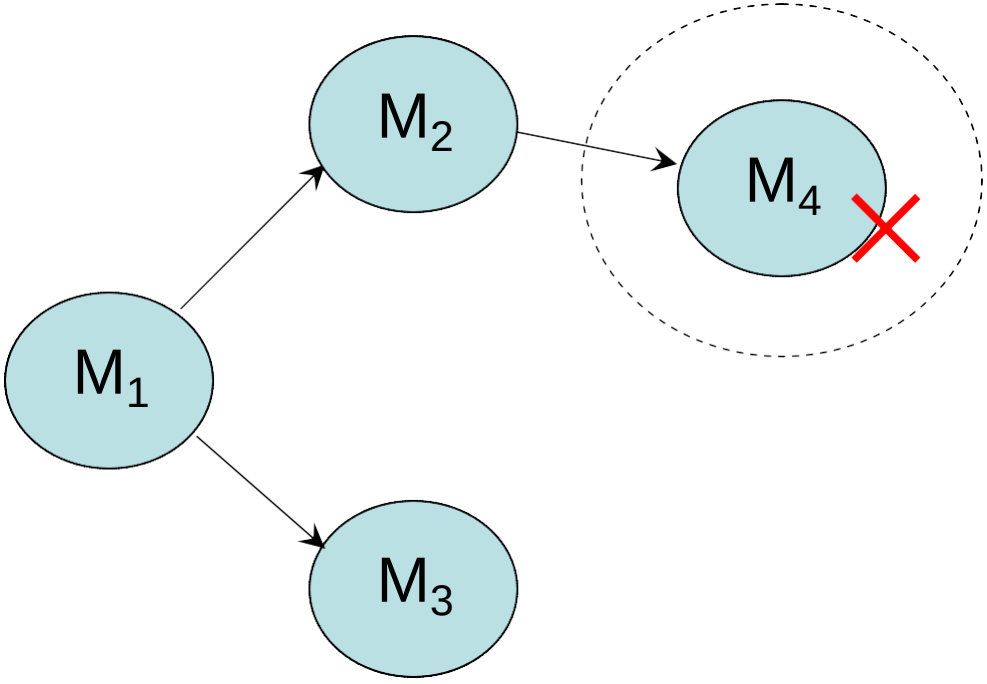


Eroare



Defecțiune

- Componente nefiabile (module).



Exemple

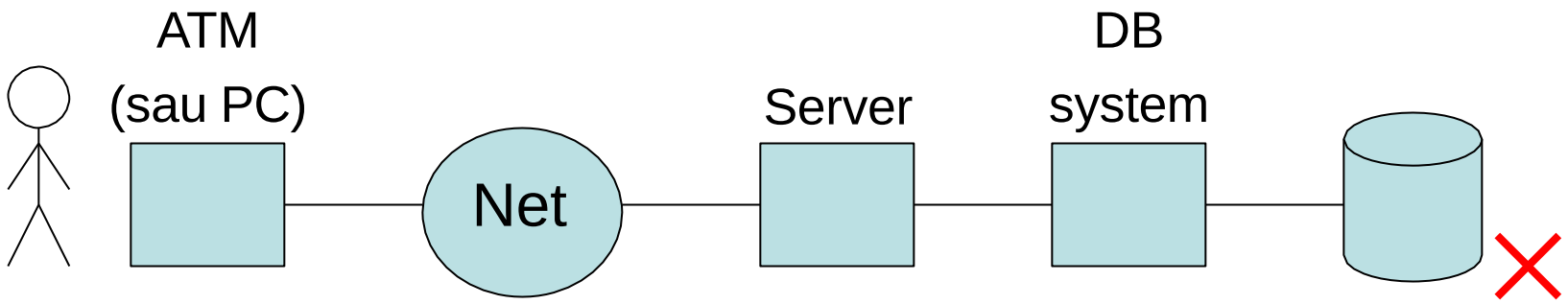
- Bad locking
- Routing
- Packet loss
- Congestion collapse
- DNS

Abordare Sistematică

1) Modularizare
2) Detectarea erorilor
3) Mascarea erorilor

→ Conform cu specificațiile

Redundanță



XFER (from, to, $)

- Fail-stop
- Fail-fast
- Fail-soft
- Fail-safe

Modelare

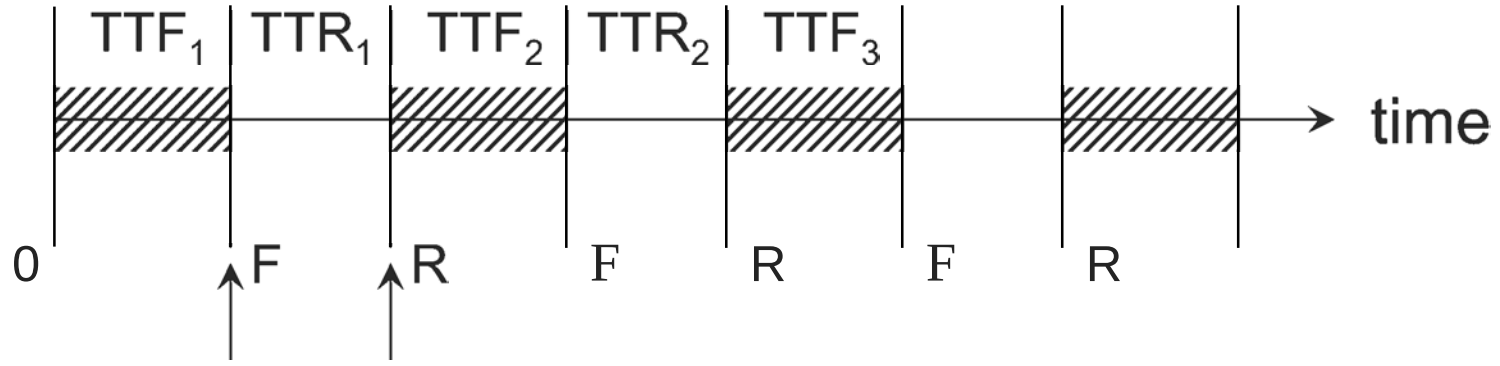
1) Nr. Erori Tolerate

2) Mean Time To Failure (MTTF)

Disponibilitate

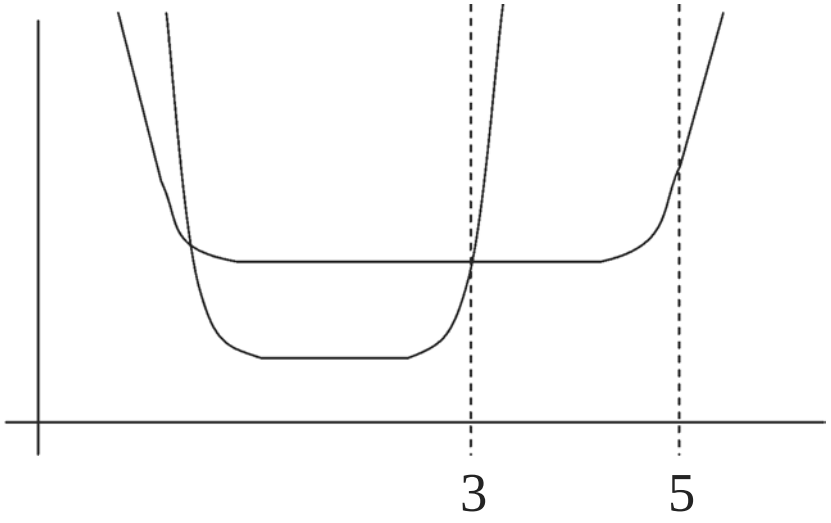
$$\frac{\sum TTF_i}{\sum TTF_i + \sum TTR_i}$$

$$\frac{MTTF}{MTTF + MTTR}$$



Rata defecțiunilor:

$$h(t) = \mathrm{P}(\text{defecțiune în } (t, t + dt] \mid \text{OK @ } t)$$



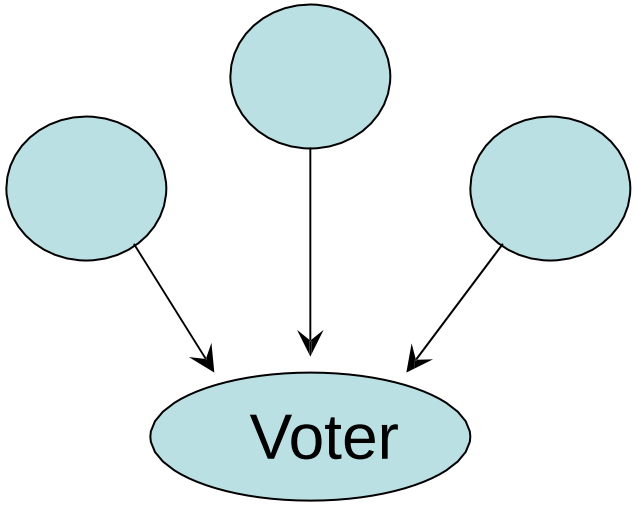
Fiabilitate $R(t) = P(\text{OK @ timpul } t)$

$e^{-(t/MTTF)}$

Exemplu:

1) Redundanță spațială: codare, log-uri, copiere + votare
2) Redundanță temporală: retry / undo

Votare



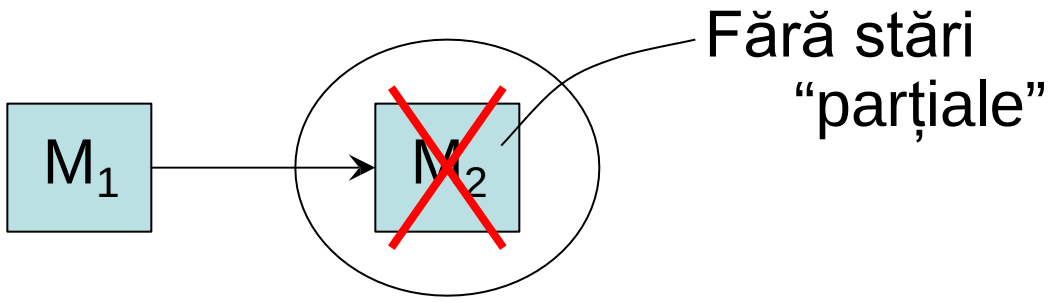
$$R_{3V} = R^3 + 3R^2(1- R) > R$$

când R în (½, 1)

Atomicitate

Defecțiuni:

1) Replicare + votare
2) Recuperabilitate



xfer (from, to, $ amount)

“Totul sau nimic”

RECUPERABILITATE

Concurență

S = 1000

xfer (S, C, 100) : A_1

xfer (S, C, 200) : A_2

A_1 înaintea A_2

SAU

A_2 înaintea A_1

IZOLABILITATE

Atomicitate

- Ascunde faptul că acțiunile sunt compuse
- REC + IZO

- Consistență ->Invariant
- Durabilitate

Recuperabilitate

- Fail-fast
- Repară (recovery)
- Restart

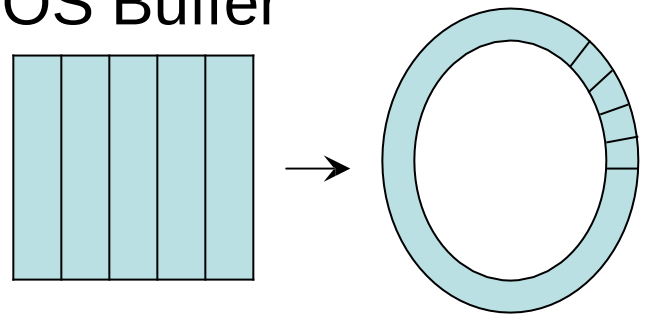
1) Sector recuperabil
2) Version history
3) Log-uri

Model

- Fără concurență
- Fără erori hardware
- Avem erori software

App

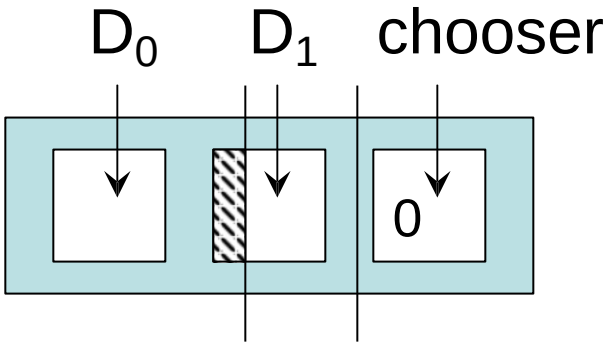
OS Buffer



Model

careful_put (sec, data)
″ _get

Soluție: Copiere



Regula de aur:
Nu modificăm niciodată singura copie.

Commit point

- Înainte -> nici o schimbare nu e vizibilă
- După -> toate schimbările sunt vizibile

Generalizare

begin_ra()



pre-commit

COMMIT()

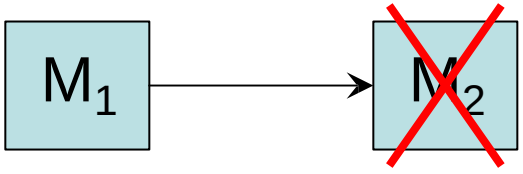


post-commit

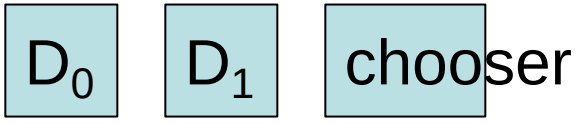
end_ra()

ABORT()

Recuperabilitate



Recoverable sector



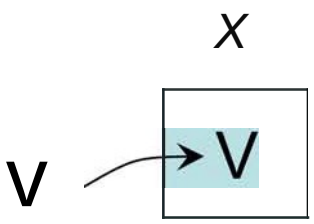
Acțiuni mai mari?

Commit point

```
id <- begin_ra()
                          ABORT()
      ⌇
      COMMIT()
      ⌇
      end_ra()
```

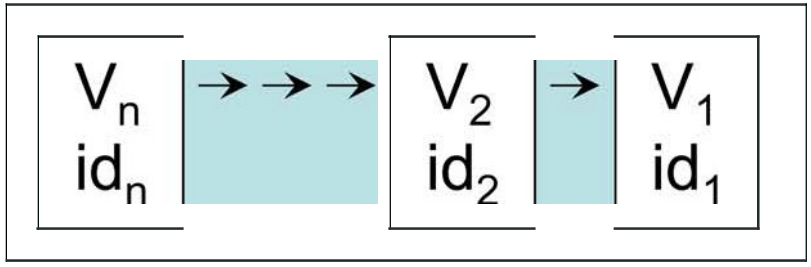
Version history

Cell storage



Install

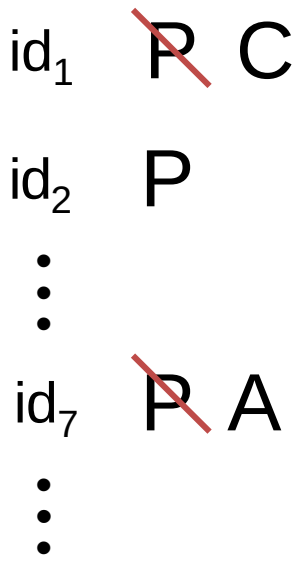
Stocare jurnalizată




```
WriteJournal(item x, v, id)
```

`ReadJournal(item x, id)`

Commit record table



1) Bootstrap
2) Slow

Logging:

Cell st. : read / write

Log : non-volatil + secvențial

Plan:

1) Fail -> Recover from log

-> uncommitted -> back out (undo)

-> committed -> install (redo)

2)ABORT() -> undo cur. action (redo)

Append-only:

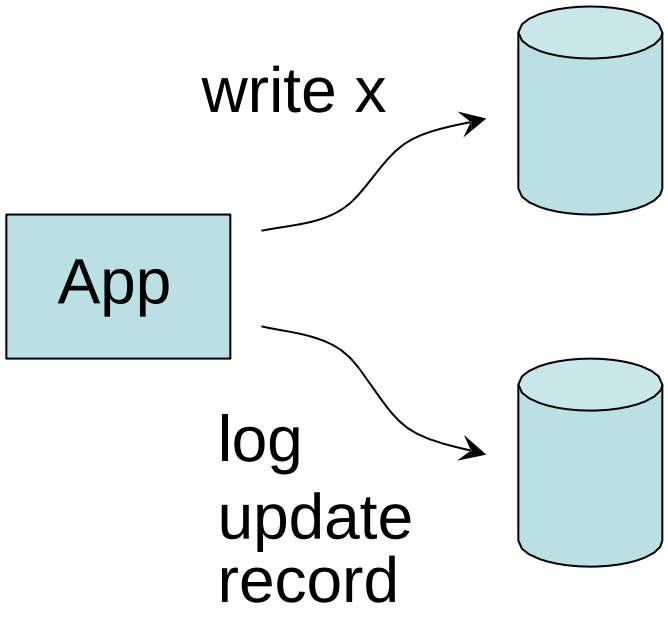
```
type: UPDATE
id: 172
undo: x <- old
redo: x <- new
```

```
type: OUTCOME
id: 174
status:
    COMMITTED
```

1) Când scriem în log?

2) Cum facem recovery?

BD pe disc:



1) WAL protocol:
Write ahead logging

2) Log COMMIT record before returning from **`commit()`**

Recovery:

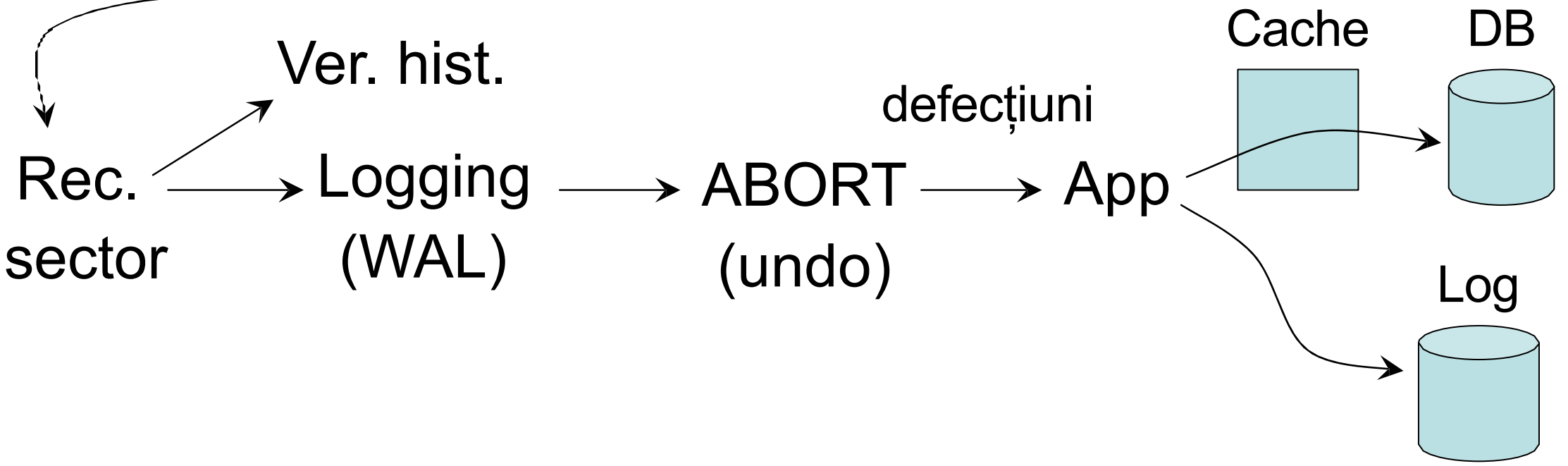
1) Scanează log-ul în sens invers
2) Winners: COMMITTED sau ABORTED
   Losers: Toate celelalte
3) Redo COMM. Winners &

Undo Losers

Atomicitate

Izolabilitate

Recuperabilitate



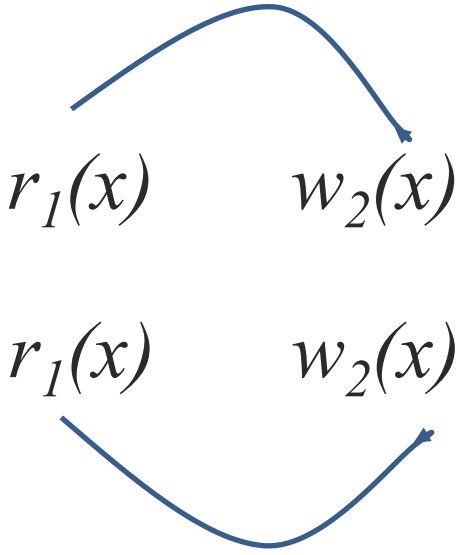
Procedeu recuperare

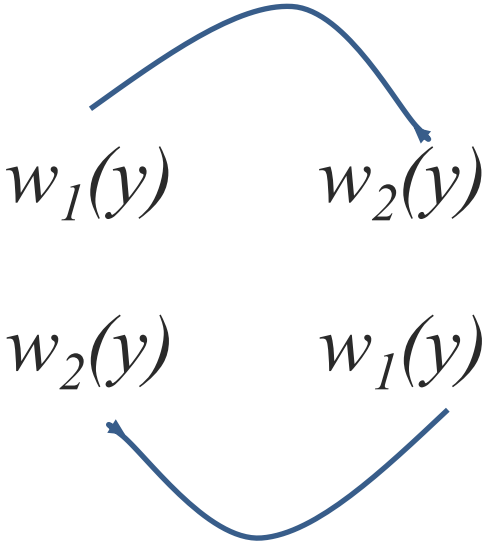
1) Undo
2) Redo

1) Scanează log-ul în sens invers
2) Winners = C + A
3) Losers:
4) Redo C winners + undo losers.

Izolabilitate

T1	T2
1) read x	2) write x
3) write y	4) write y





$r(z)$ $w(z)$
$w(z)$ $r(z)$
$w(z)$ $w(z)$

Serializabilitate:

Săgețile de conflict ale trace-ului sunt în aceeași ordine ca o oarecare execuție secvențială a acțiunilor.

Action Graph

T1	**T2**	**T3**	**T4**
1) $r_1(x)$	2) $w_2(x)$	3) $r_3(y)$	4) $r_4(x)$
5) $w_1(y)$	6) $w_2(y)$	7) $w_3(z)$	

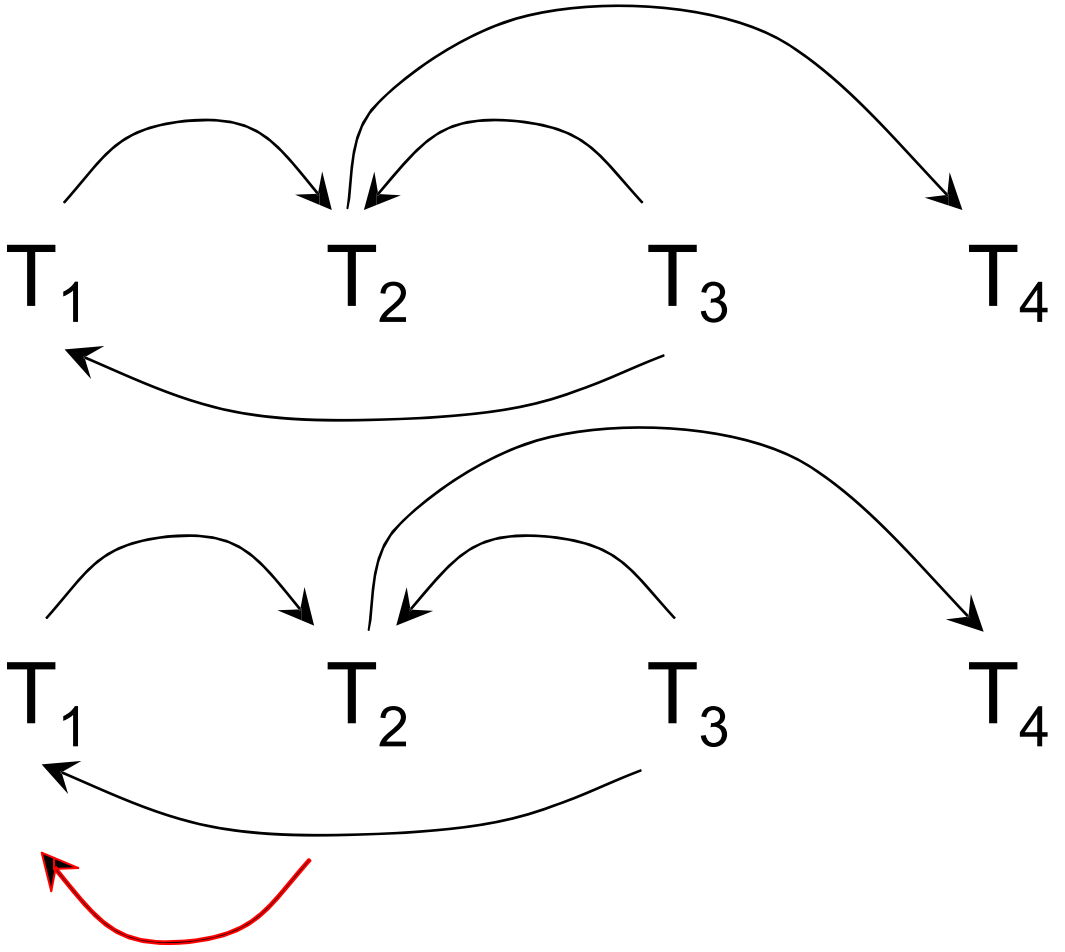
T1	T2	T3	T4
1) $r_1(x)$	2) $w_2(x)$	3) $r_3(y)$	4) $r_4(x)$
5) $w_1(y)$	6) $w_2(y)$	7) $w_3(z)$	

T_1 T_2

T_1 T_3

T_2 T_3

T_2 T_4



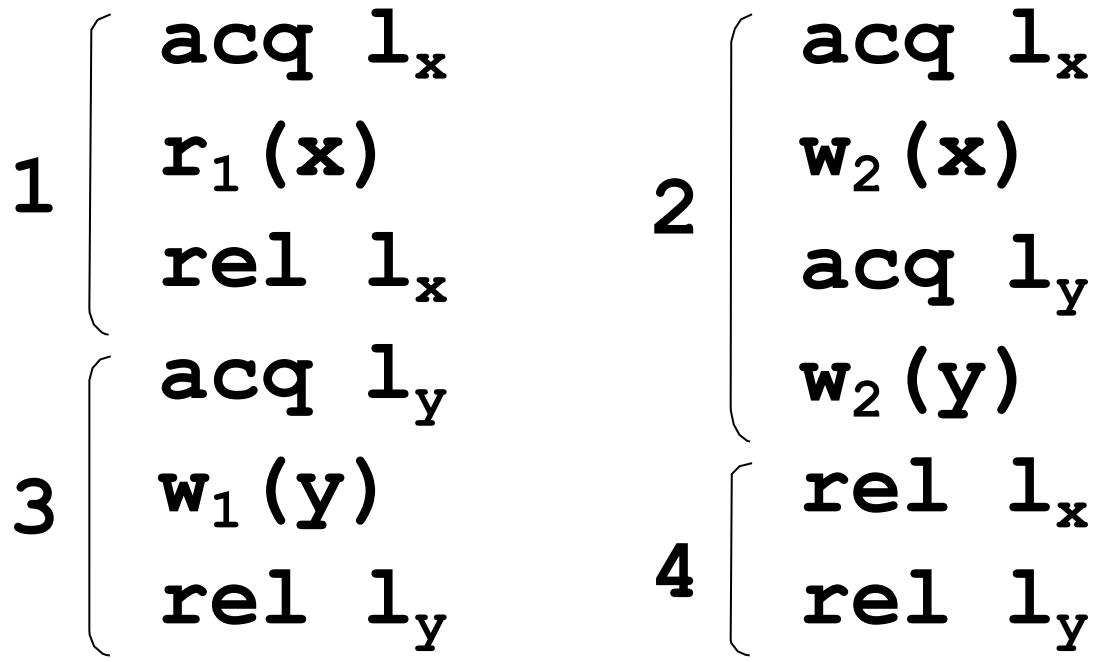
Action Graph este aciclic

<=> trace-ul este serializabil

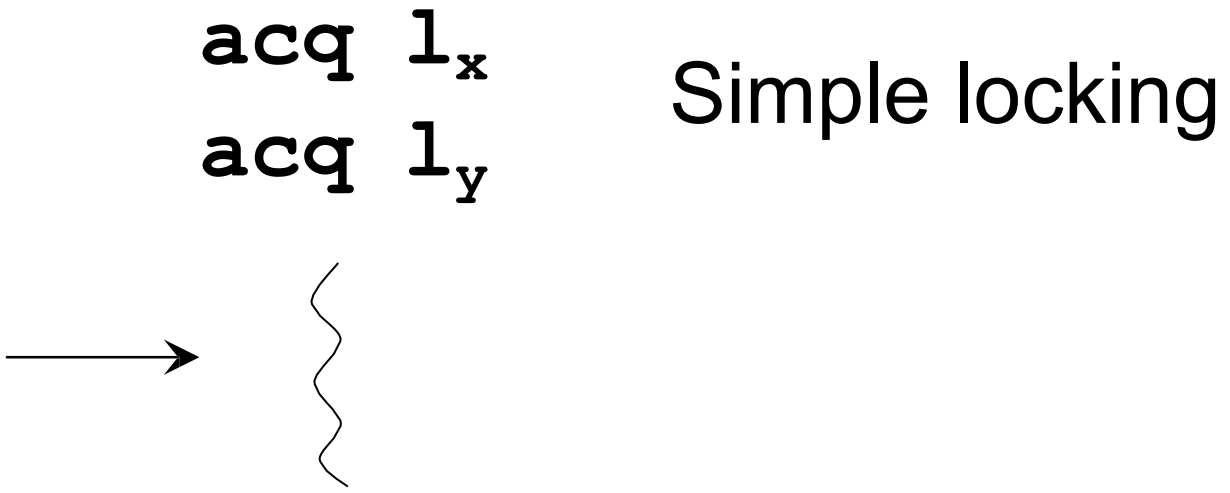
=> : Sortare Topologică

Locking:

```
acq (lock of x)
rel (    ″     )
```



Izolabilitate:



Simple locking

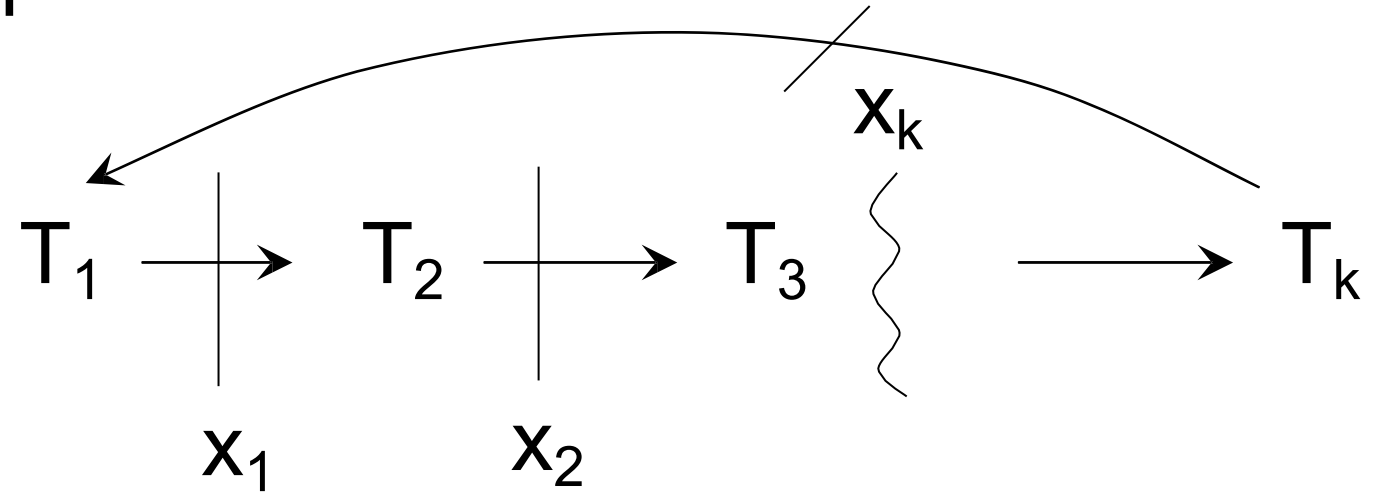
```
acq l_x
r(x)
acq l_y
r(y)
```

Two-phase locking (2 PL)

Nici un release <u>înainte</u> de TOATE acquire-urile

<u>Corect</u>

Demonstrație prin reducere la absurd



rel l_1

acq l_1

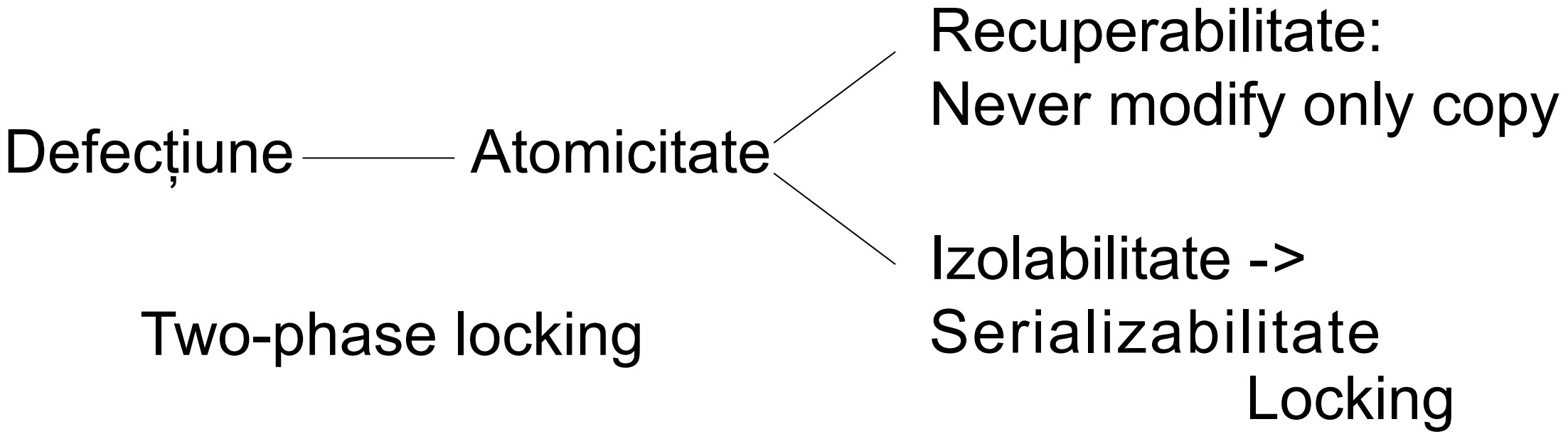
rel l_2

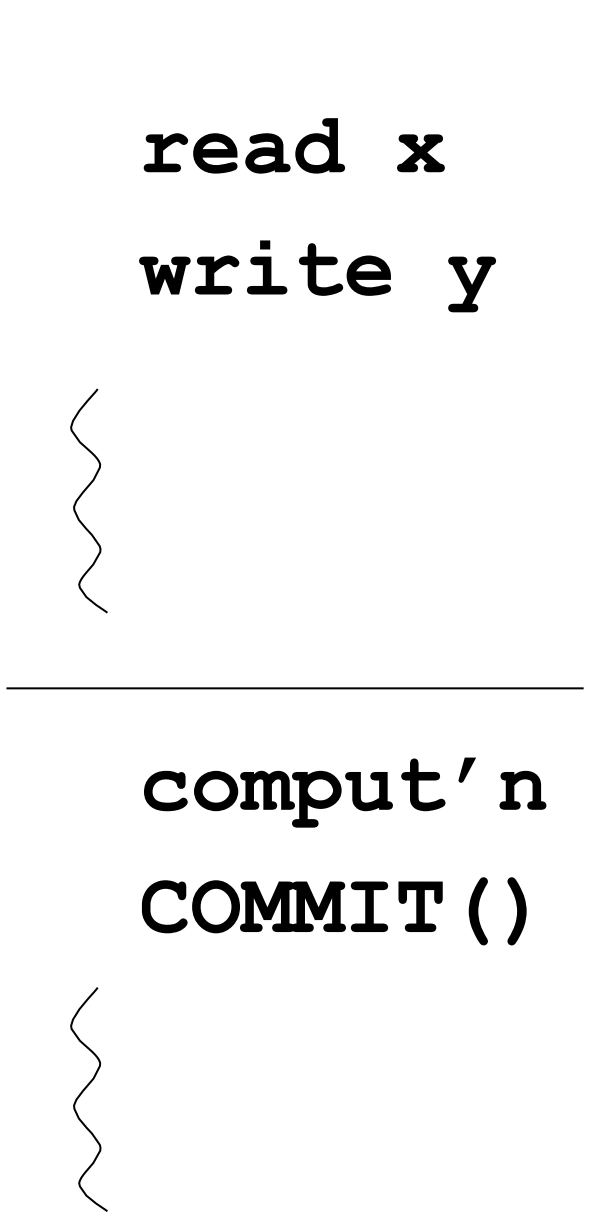
acq l_2

acq l_{k-1}

rel l_k

acq l_k


Defecțiune
Atomicitate
Recuperabilitate:
Never modify only copy
Izolabilitate ->
Serializabilitate
Two-phase locking
Locking

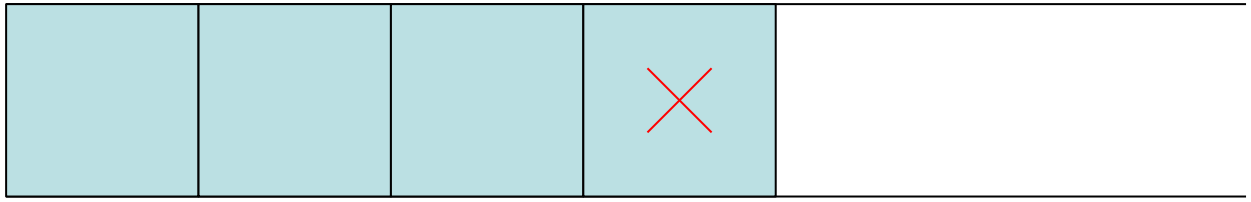


Deadlock-uri

```
   acq l_x                 acq l_y
      read x                  read y
→  acq l_y              →  acq l_x
      write y                 write x
```

1) Timere
2) Waits-for graph

Logs & Locks

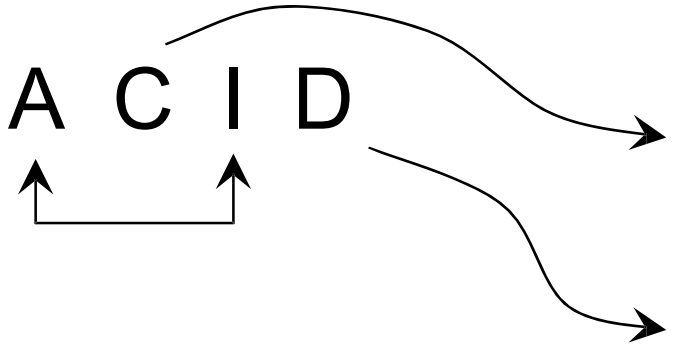


Aplicații

1) Tranzacții
   - Coerență
   - Durabilitate

2) Atomicitate distribuită

Tranzacții



Trebuie să respecte un invariant

Permanență

Centralizat

Reguli de integritate

SID	Nume	DeptID
35	----	43

Dept ID	Nume Dept
43	
25	

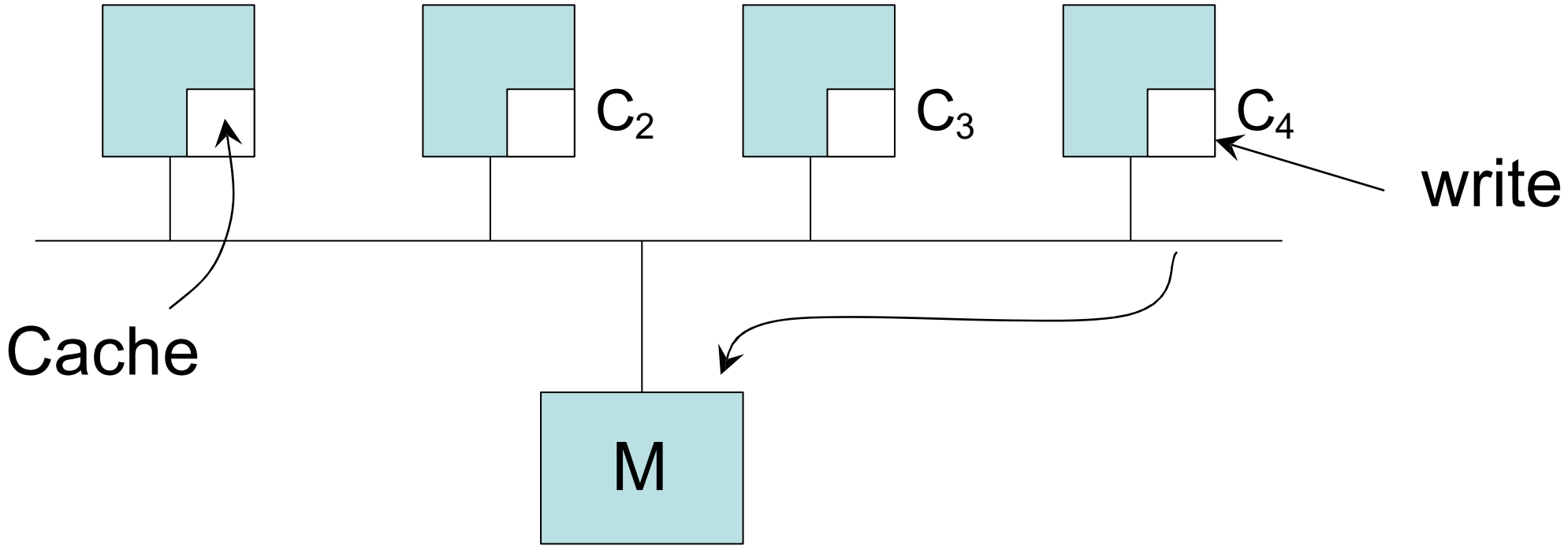
Date Distribuite

- DNS -> Timp de expirare
- Cache web
  - ↳“if-modified-since”

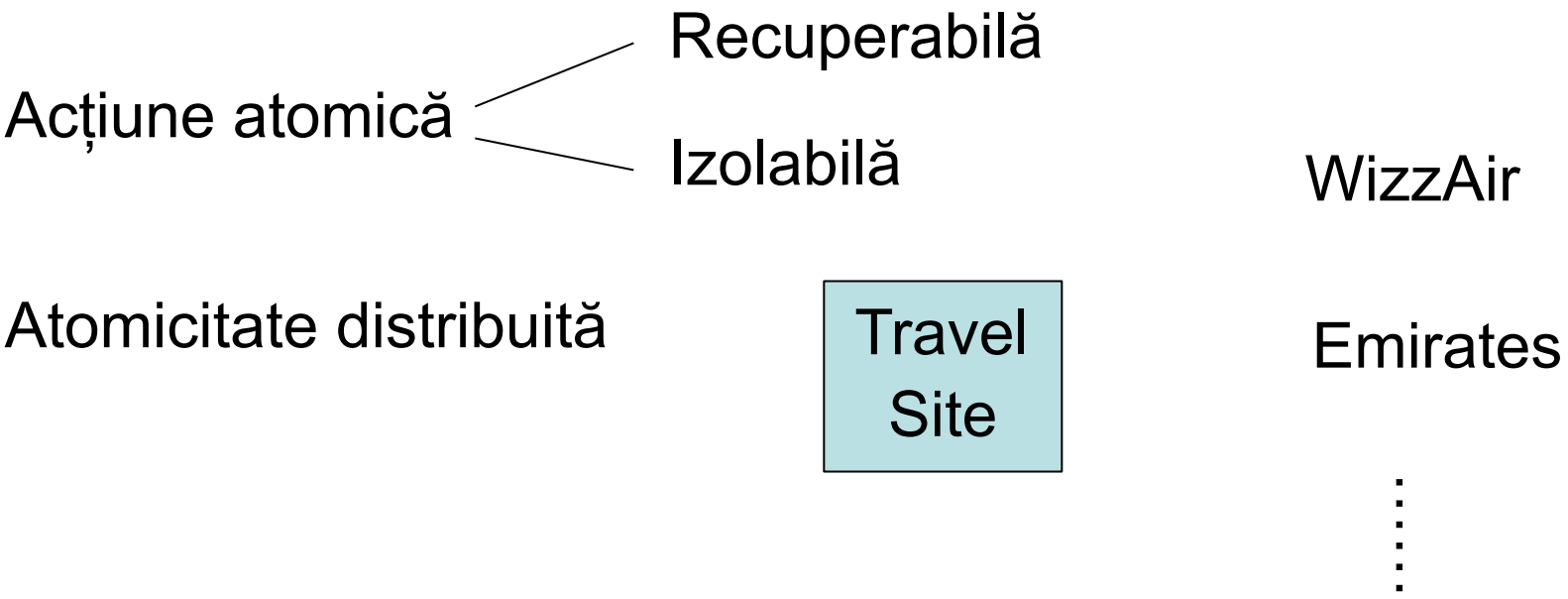
Coerență puternică:

Read întoarce rezultatul ultimului write

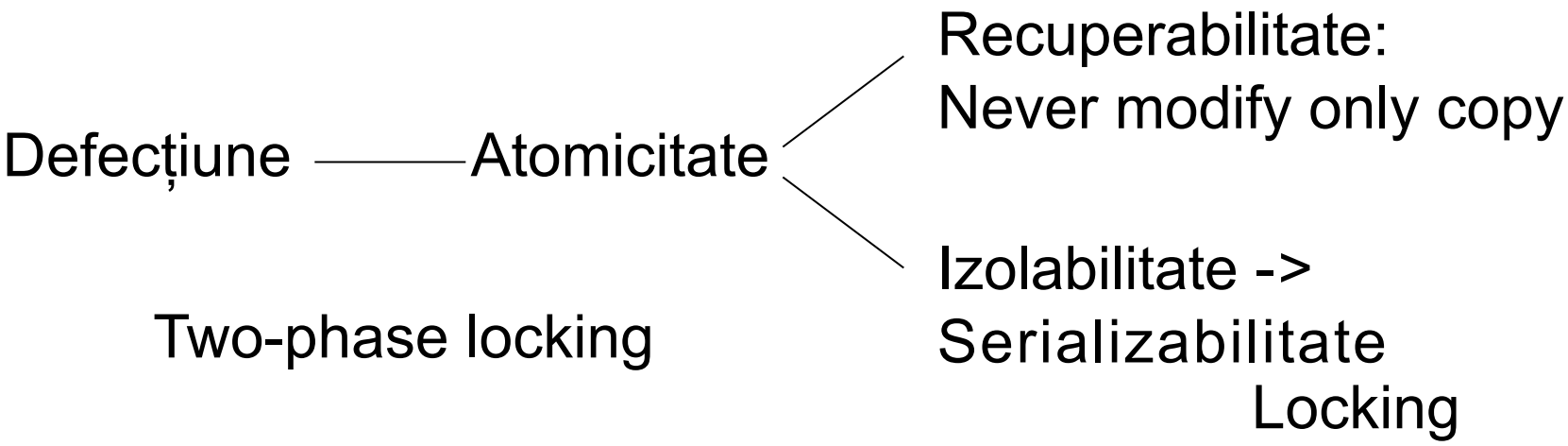
Coerență eventuală



1) Write-thru cache
2) Snoopy cache


Acțiune atomică
Recuperabilă
Izolabilă
Atomicitate distribuită
Travel Site
WizzAir
Emirates

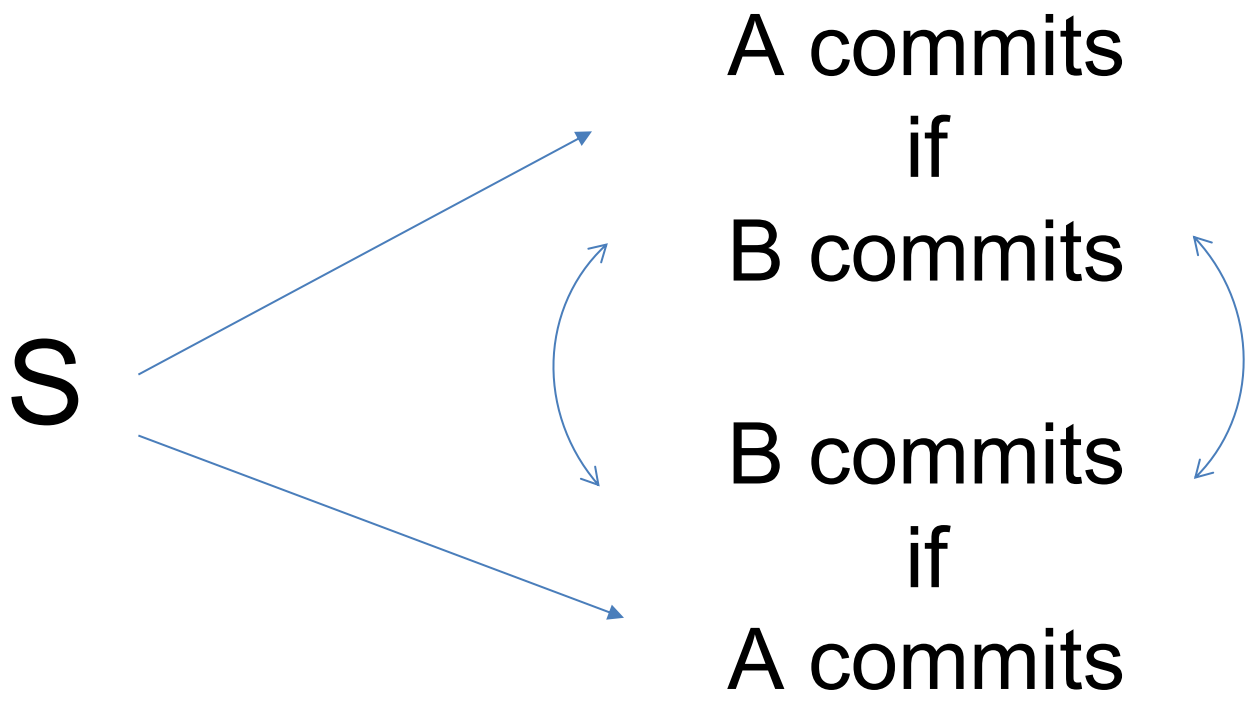
Acțiuni atomice imbricate



Begin

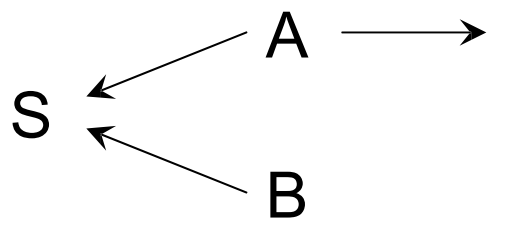
Buy WizzAir	A
Buy Emirates	B

End


S
A commits
if
B commits
B commits
if
A commits

Begin S	Begin A	Begin B	……	T.C. A	T.C. B	Commit S	Release locks

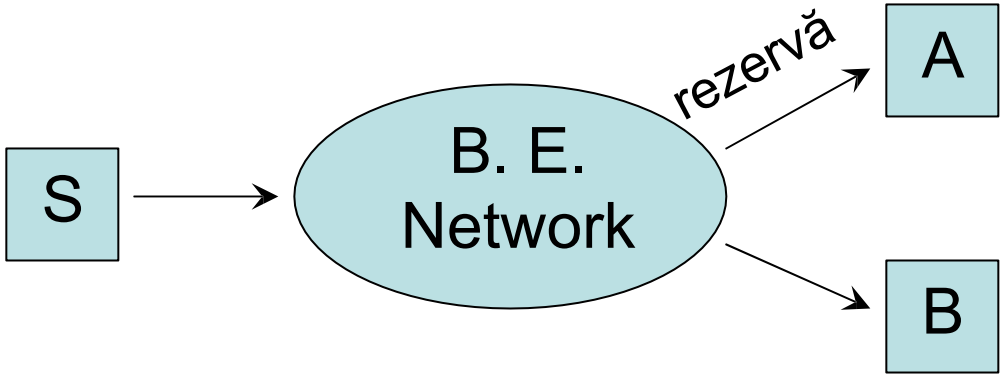
Dacă A și B sunt în conflict?



Când A face tentative commit,

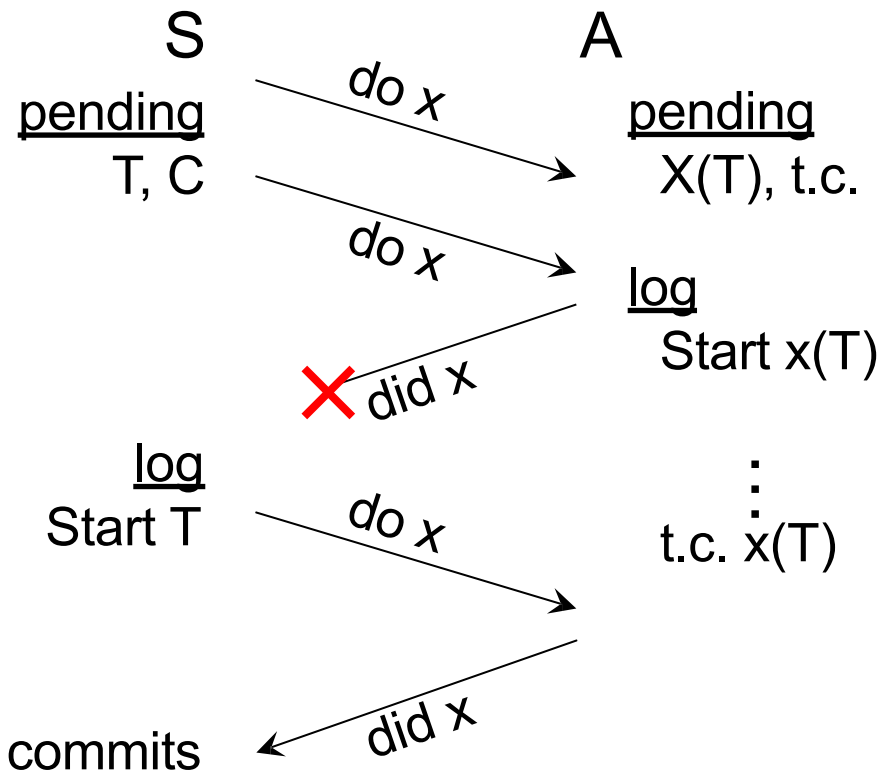
S + tranzacțiile copii văd update-urile lui A

Cazul distribuit

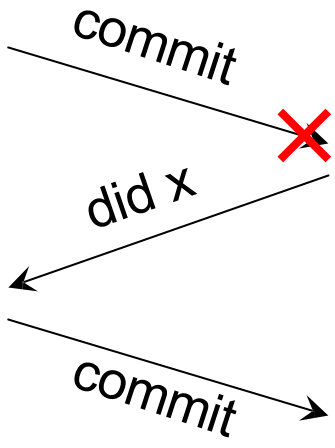


RPC-uri

Two-phase commit


S
A
pending
T, C
do x
pending
X(T), t.c.
do x
log
Start x(T)
did x
log
Start T
do x
t.c. x(T)
did x
commits

Phase 2



S crashes
- before C
  - UNDO T, remembers T aborted
- after C
  - REDO T, remembers T committed

A crashes
- before T.C.
- after T.C. -> up to S

A, B fac vizibile rezultatele simultan?

NU

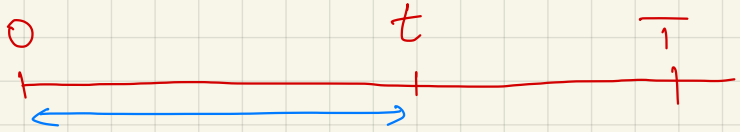
Problema generalilor bizantini

IC - Cursul 6

Notițe de curs

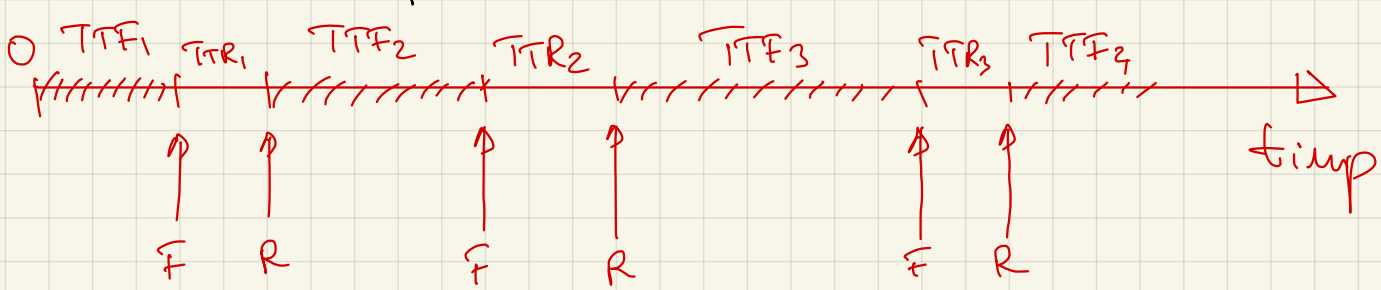
Toleranța la defecte pt. sisteme hardware

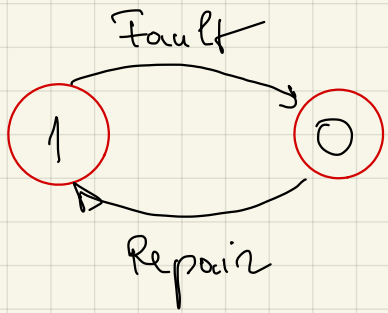
1) **Fiabilitatea** $R(t)$ - reliability



$$R(t) = P(T > t \mid OK@t=0)$$

2) **Medie timpului de bună funcționare** (MTBF)

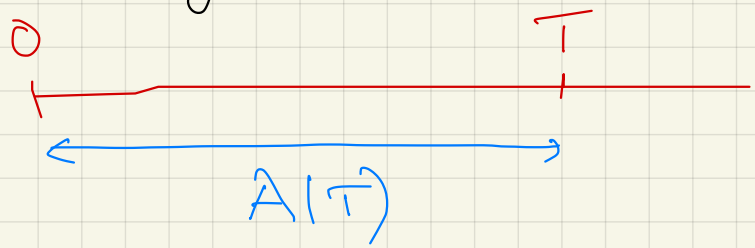




$$MTTF = \sum_i \frac{TTF_i}{n} \qquad MTTR = \sum_i \frac{TTR_i}{n}$$

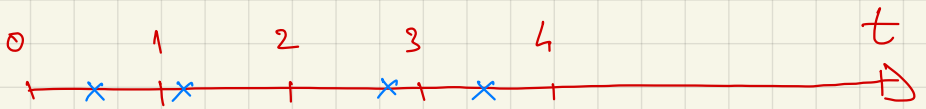
3) **Disponibilitatea** $A(t)$ - availability

$$A(T) = \frac{1}{T}\int_0^T A(t)\,dt$$



$$A(\infty) = \lim_{t\to\infty} A(T) = 0$$

$$A = \frac{MTTF}{MTBF} = \frac{MTTF}{MTTF + MTTR}$$



$MTBF = 1h = 3600s$

$MTTR = 1s$

$$\Rightarrow A = \frac{3599}{3600} = 0,99972 = 99,972\,\%$$

$A = 99,99\,\%$ → downtime ≈ 53 min.

$A = 99,9995\,\%$ → 31,5 s downtime

Notiuni teoria probabilitatilor

$A \rightarrow P(A) \qquad 0 \leq P(A) \leq 1$

$P(\bar{A}) = 1 - P(A)$

$A, B \qquad P(A|B)$

$P(A \cdot B) = P(A|B) \cdot P(B) \Rightarrow P(A|B) = \frac{P(A \cdot B)}{P(B)}$

Dacă A și B sunt independente: $P(A|B) = P(A)$

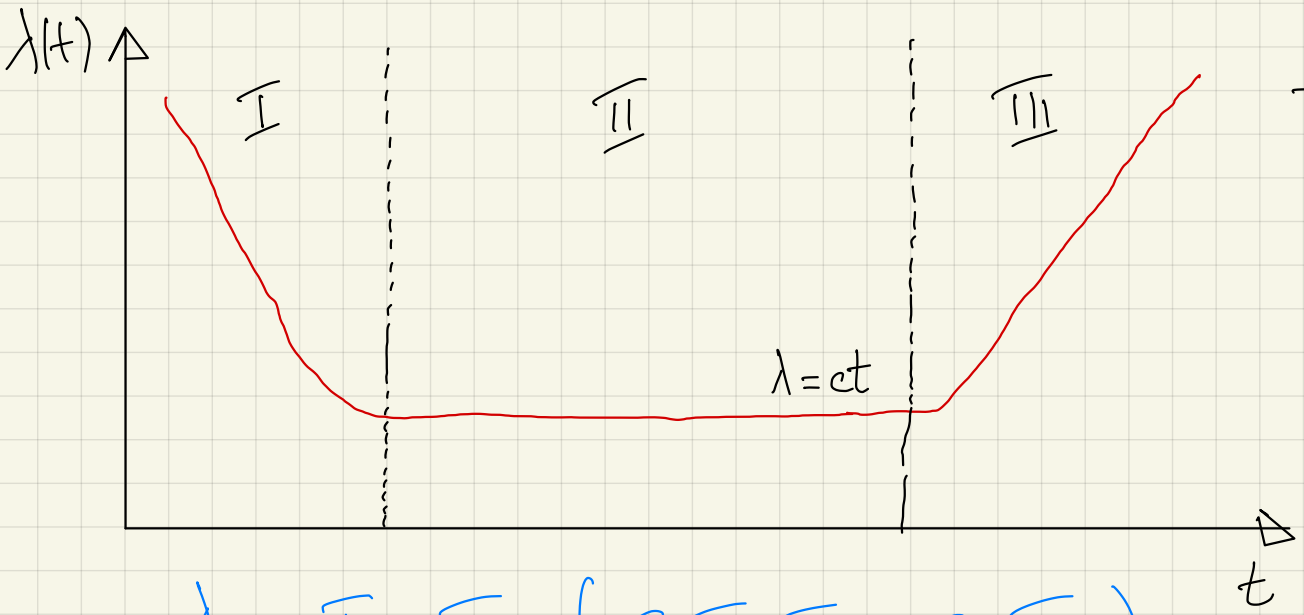
$P(A \cdot B) = P(A) \cdot P(B) = P(B) \cdot P(A) = P(B \cdot A)$

$P(A+B) = P(A) + P(B) - P(A \cdot B)$

Dacă A și B sunt mutual-exclusive

$P(A+B) = P(A) + P(B)$

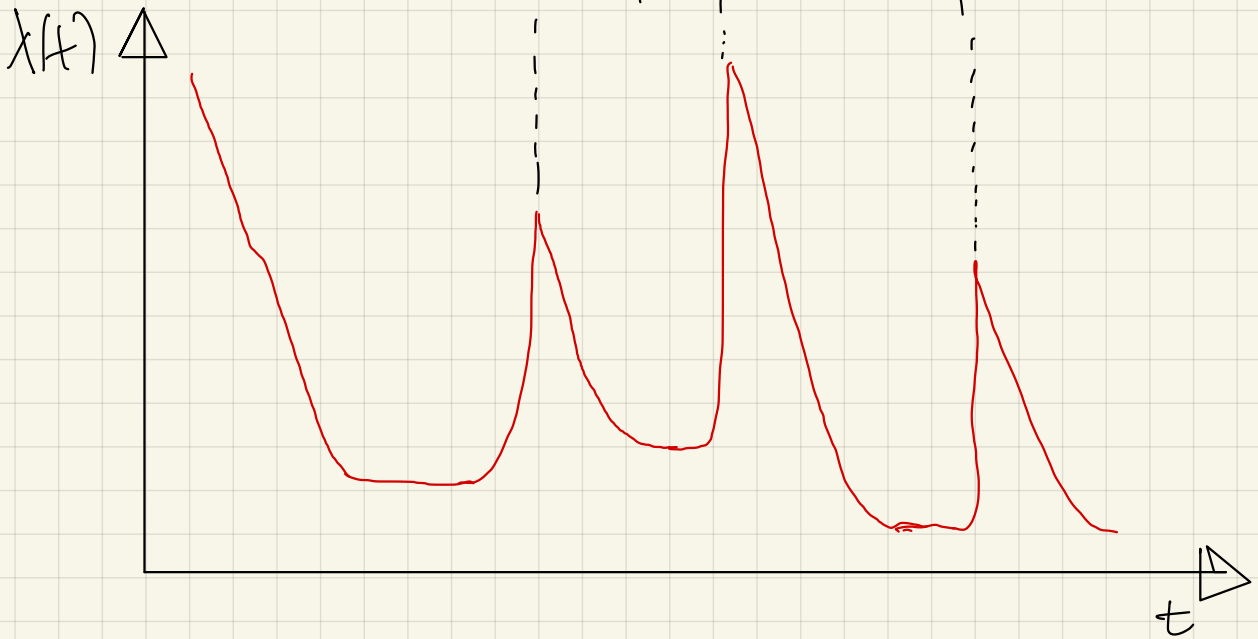
intensitatea defectiunilor $\lambda(t)$

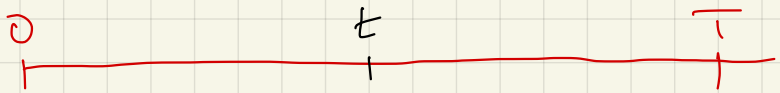


→ curba caracteristica bolii bathtub curve

$$\lambda = \Pi_L \cdot \Pi_Q \left(C_1 \Pi_T \Pi_V + C_2 \Pi_E \right)$$ → intensitatea defectiunilor pt. circuite integrate

intensitatea defectiunilor pt. SW





$f(t)$ – functia densitate de probabilitate $\quad t \geq 0$

$F(t)$ – functia distributie cumulativa de probabilitate

$$f(t) = \frac{dF(t)}{dt} \qquad F(t) = \int_0^t f(z)\,dz$$

$$f(t) \geq 0 \ (\forall)\ t \geq 0 \qquad \int_0^\infty f(t)\,dt = 1$$

$$F(t) = Prob(t \leq T)$$

$$R(t) = 1 - F(t) = Prob(T > t)$$

$$\lambda(t) = \frac{f(t)}{1 - F(t)}$$

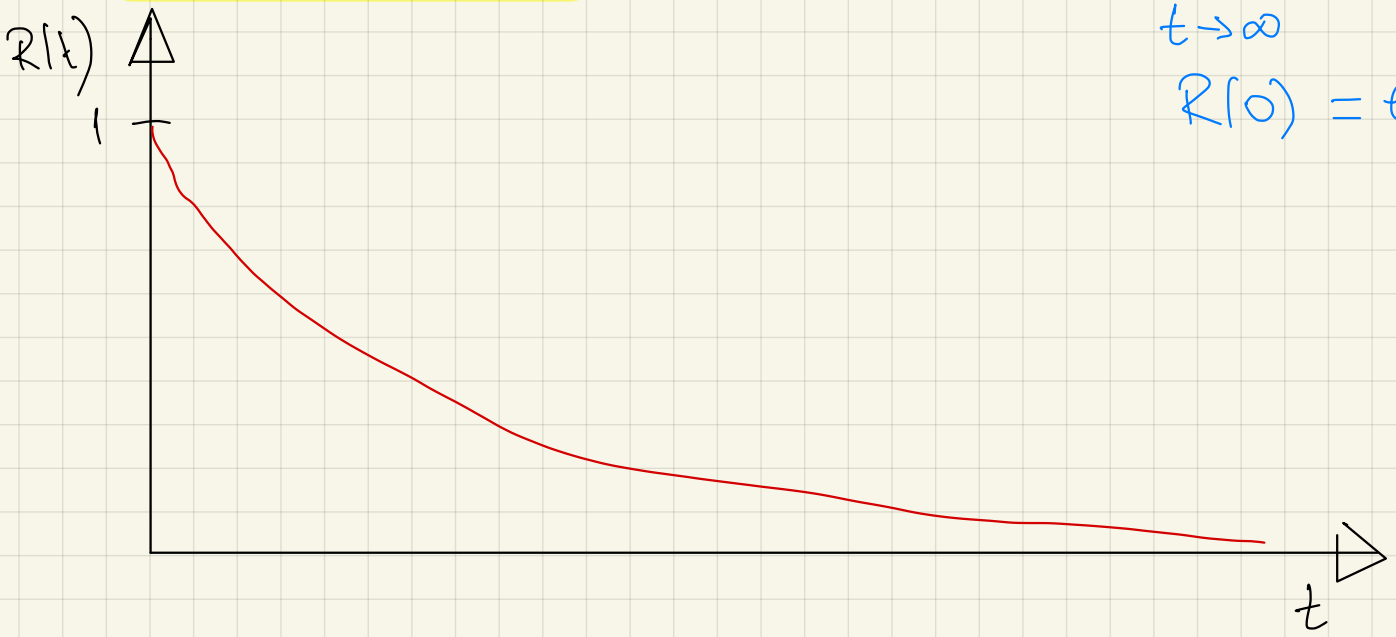
$$f(t) = \frac{dF(t)}{dt} = \frac{d(1-R(t))}{dt} = -\frac{dR(t)}{dt}$$

$$\lambda(t) = -\frac{dR(t)}{dt}\,\frac{1}{R(t)} \Leftrightarrow \frac{1}{R(t)}\,dR(t) = -\lambda(t)\,dt \Rightarrow$$

dacă $\lambda(t) = \lambda = ct.$

$$\Rightarrow \int \frac{1}{R(t)}\,dR(t) = \int -\lambda\,dt \Leftrightarrow \ln(R(t)) = -\lambda t \Rightarrow$$

$$R(t) = e^{-\lambda t}$$



$$\lim_{t\to\infty} R(t) = \lim_{t\to\infty}\left(e^{-\lambda t}\right) = 0$$

$$R(0) = e^{-\lambda \cdot 0} = 1$$

Valoare așteptată

Zar 6 fețe: 1 5 6 2 3 5 1 .... $E[X] = 3{,}5$

$$E[X] = p_1 x_1 + p_2 x_2 + p_3 x_3 + \dots p_n x_n = \frac{p_1 x_1 + p_2 x_2 + \dots p_n x_n}{p_1 + p_2 + \dots p_n}$$

$$E[X] = \frac{1}{6}\cdot 1 + \frac{1}{6} 2 + \frac{1}{6}\cdot 3 \dots \frac{1}{6}\cdot 6 = 3{,}5$$

↳ valoarea așteptată pt. aruncările unui zar cu 6 fețe.

$X \to f(x) \qquad E[X] = \int_{-\infty}^{\infty} x f(x)\,dx$



$$MTBF = E[T] = \int_0^{\infty} t f(t)\,dt$$

$$f(t) = -\frac{dR(t)}{dt} \Rightarrow MTBF = \int_0^{\infty} t\left(-\frac{dR(t)}{dt}\right)dt = -\int_0^{\infty} t\frac{dR(t)}{dt}dt =$$

$$= -tR(t)\Big|_0^{\infty} + \int_0^{\infty} R(t)\,dt = \int_0^{\infty} R(t)\,dt$$

$$\int fg'\,dt = fg + \int f'g\,dt$$

$$\int e^{ax}\,dx = \frac{1}{a}e^{ax}$$

$$MTBF = \int_0^{\infty} R(t)\,dt$$

dacă $R(t) = e^{-\lambda t} \Rightarrow MTBF = \int_0^{\infty} e^{-\lambda t}\,dt = -\frac{1}{\lambda}e^{-\lambda t}\Big|_0^{\infty} = \frac{1}{\lambda}$

$$MTBF = \frac{1}{\lambda}$$

$$f(t) = \lambda\beta t^{\beta-1} e^{-\lambda t^{\beta}} \rightarrow$$ distribuție Weibull

$$R(t) = e^{-\lambda t^{\beta}}$$

$$MTBF = \int_0^{\infty} R(t)\,dt = \frac{\Gamma(\beta^{-1})}{\beta \lambda^{\beta-1}}$$

$$\Gamma(t) = \int_0^{\infty} y^{t-1} e^{-y}\,dy$$ – funcția gamma

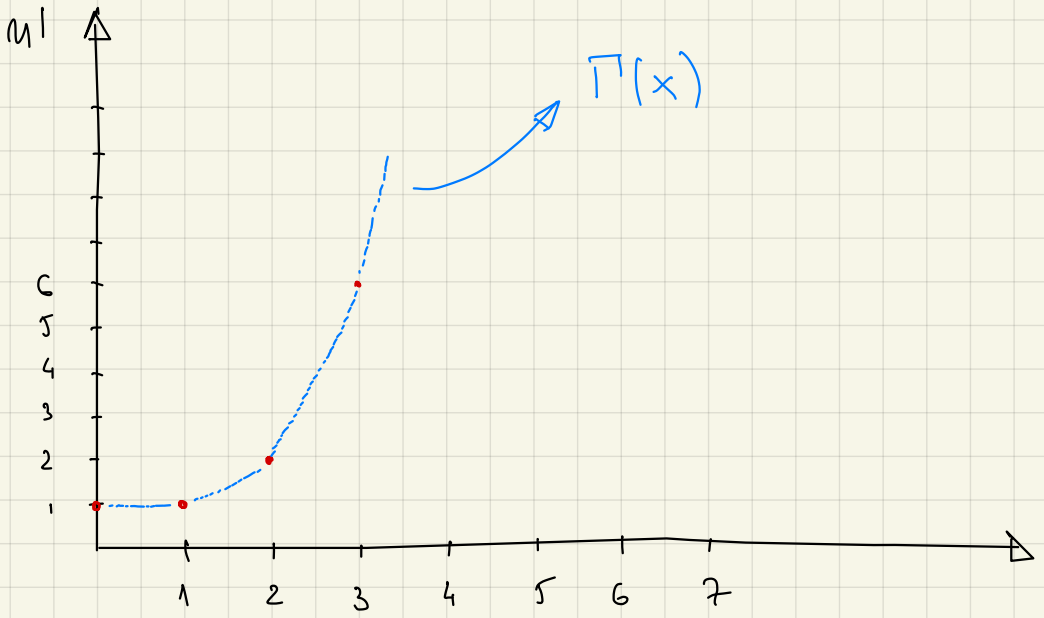
$n! = 1 \cdot 2 \cdot 3 \cdots n$

$\Gamma(n) = n!$ dacă $n \in \mathbb{Z}$

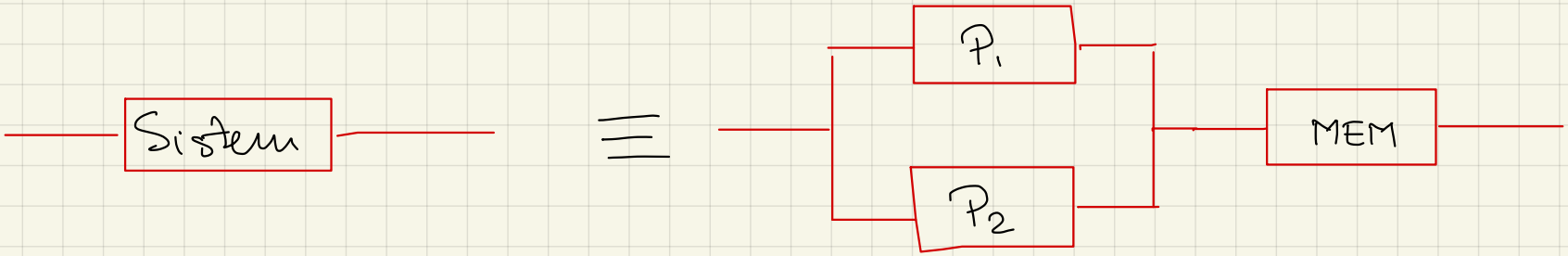
$\Gamma(x) = (x-1)\Gamma(x-1)$ $(\forall)$ $x > 1$

$\Gamma(0) = \Gamma(1) = 1$

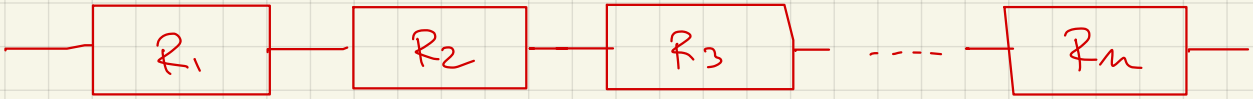
$\Gamma(n) = \Gamma(n-1)!$ $(\forall)$ $n \in \mathbb{N}$



Diagrame bloc pt. fiabilitate



Structuri serie



$R_1(t) = e^{-\lambda_1 t}$ $\quad Q_1(t) = 1 - R_1(t)$

$R_2(t) = e^{-\lambda_2 t}$

$R_n(t) = e^{-\lambda_n t}$ $\quad Q_n(t) = 1 - R_n(t)$

$$R_S(t) = R_1(t) \cdot R_2(t) \cdot R_3(t) \cdots R_n(t)$$

$$R_S(t) = \prod_{i=1}^{n} R_i(t) = \prod_{i=1}^{n} e^{-\lambda_i t} = e^{-\sum_{i=1}^{n} \lambda_i \cdot t} = e^{-\lambda_S \cdot t}$$

$$\lambda_S = \sum_{i=1}^{n} \lambda_i$$

$$Q_S(t) = 1 - R_S(t) = 1 - \prod_{i=1}^{n} \left(1 - Q_i(t)\right)$$

$$\prod_{i=1}^{n} \left(1 - Q_i(t)\right) = 1 - \left(Q_1 + Q_2 + \cdots Q_n\right) + \left(Q_1 Q_2 + Q_1 Q_3 + \cdots Q_{n-1} Q_n\right) - \cdots \simeq$$

$R_i(t) \geq 90\% \Rightarrow Q_i(t) \leq 10\% \Rightarrow Q_i Q_j \simeq 0$

$$\simeq 1 - (Q_1 + Q_2 + \dots Q_M) = 1 - \sum_{i=1}^{M} Q_i$$

$$Q_S = 1 - \prod_{i=1}^{n} (1 - Q_i) \cong \sum_{i=1}^{n} Q_i$$

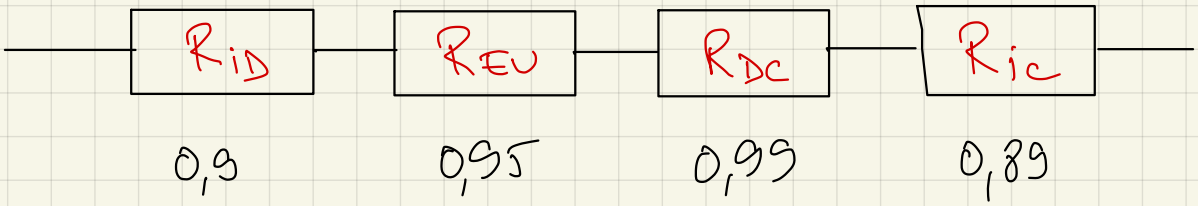
$$MTBF = \int_0^{\infty} R(t)\,dt \qquad R(t)_i = e^{-\lambda_i t} \Rightarrow MTBF_i = \frac{1}{\lambda_i} \Leftrightarrow \lambda_i = \frac{1}{MTBF_i}$$

$$MTBF_S = \frac{1}{\lambda_S} = \frac{1}{\sum_{i=1}^{n} \lambda_i} = \frac{1}{\sum_{i=1}^{n} \frac{1}{MTBF_i}}$$

$$MTBF_S < MTBF_i \ (\forall)\ i = \overline{1,n}$$

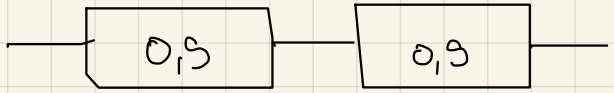
Module identice: $R_1 = R_2 = \dots R_M = R = e^{-\lambda t}$

$$\lambda_S = M\lambda \qquad R_S(t) = e^{-M\lambda t} \qquad MTBF_S = \frac{1}{M\lambda} = \frac{MTBF}{M}$$

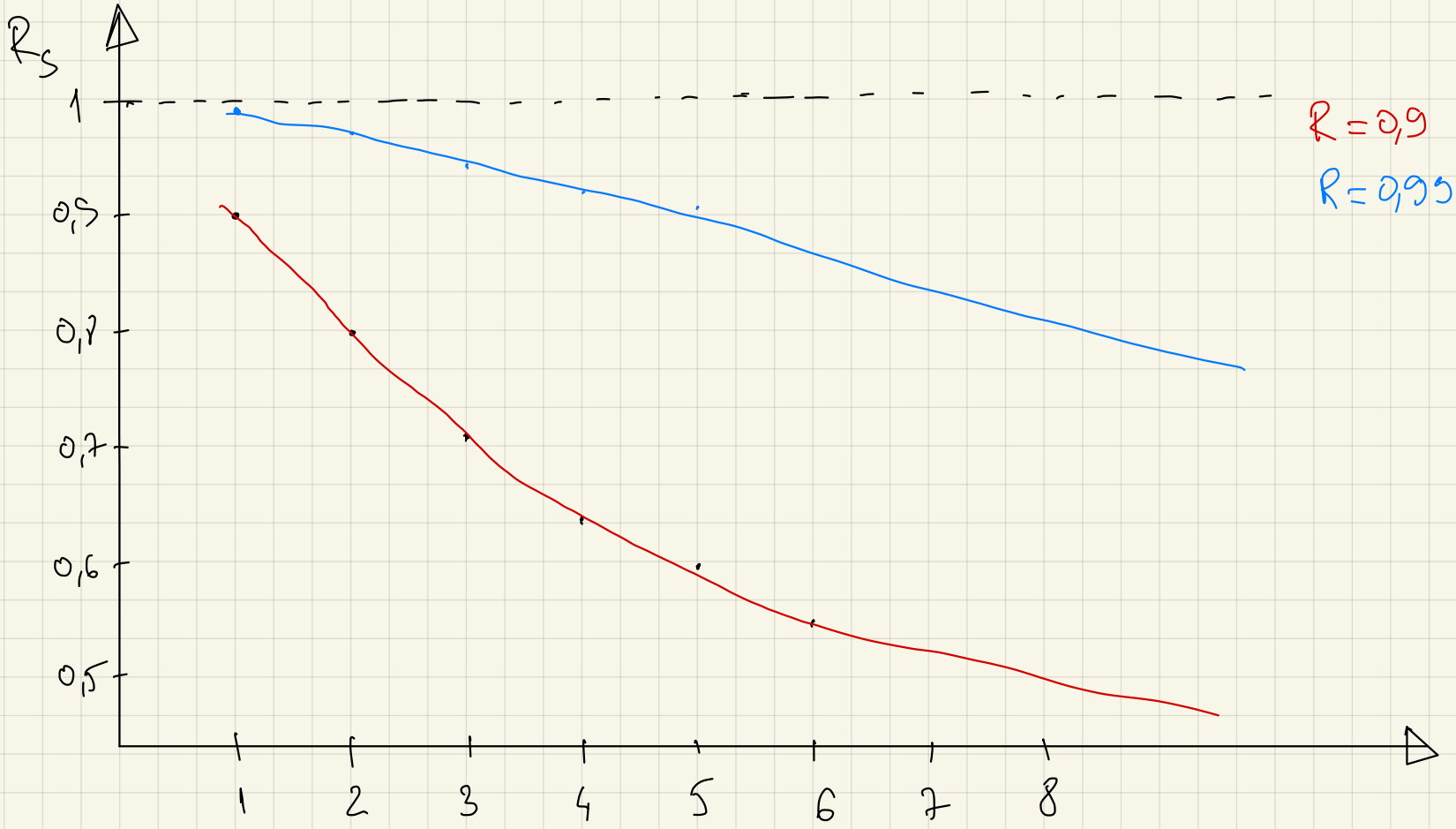


$$R_S = R^n$$
$$0{,}955 = R^{10^3}$$
$$R = ?$$

$$R_S = 0{,}9 \cdot 0{,}95 \cdot 0{,}99 \cdot 0{,}89 = 0{,}75 \ (75\%)$$



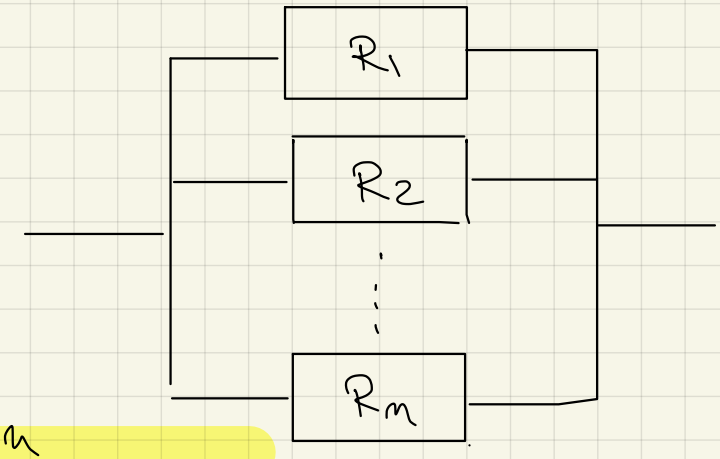
$$R_S = 0{,}9^2 = 0{,}81$$



IC - Cursul 7

Notițe de curs

Fiabilitatea structurii paralele



$R_p = ?$ $\quad R_p = 1 - Q_p$

$Q_p = Q_1 \cdot Q_2 \cdot \dots Q_m$

$1 - R_p = (1-R_1)(1-R_2) \dots (1-R_m)$

$$R_p = 1 - (1-R_1)(1-R_2)\dots(1-R_m) = 1 - \prod_{i=1}^{m}(1-R_i)$$

$R_i(t) = e^{-\lambda_i t}$

$$R_p(t) = 1 - \prod_{i=1}^{m}(1 - e^{-\lambda_i t}) = \sum_{i=1}^{m} e^{-\lambda_i t} - \sum_{\substack{i=1,j=1 \\ i \neq j}}^{m} e^{-(\lambda_i+\lambda_j)t} + \dots$$

$$\dots + (-1)^{m+1} \prod_{k=1}^{m} e^{-\lambda_k t}$$

$$MTBF_p = \int_0^\infty R_p(t)\,dt = \sum_{i=1}^{m} \int_0^\infty e^{-\lambda_i t}\,dt - \sum_{\substack{i=1,j=1 \\ i \neq j}}^{m} \int_0^\infty e^{-(\lambda_i+\lambda_j)t}\,dt + \dots$$

$$+ (-1)^{m+1} \int_0^\infty e^{-\sum_{k=1}^{m} \lambda_k t}\,dt$$

$$\int_0^{\infty} e^{-\lambda t}\,dt = \dots = \frac{1}{\lambda}$$

$$\sum_{i=1}^{n} \frac{1}{\lambda_i} - \sum_{\substack{i=1,\ j=1 \\ i \neq j}}^{n} \frac{1}{\lambda_i + \lambda_j} + \dots + (-1)^{n+1} \frac{1}{\sum_{k=1}^{n} \lambda_k} = MTBF_p$$

$$\lambda_i = \lambda_j = \dots \lambda_n = \lambda$$

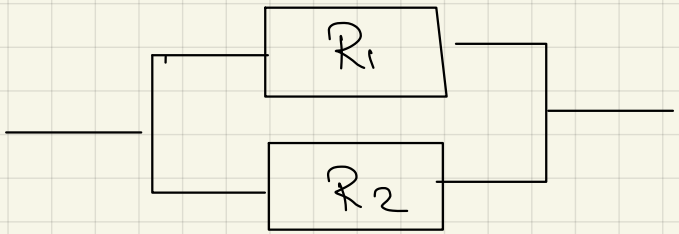
$$MTBF_p = \frac{n}{\lambda} - \frac{n}{2\lambda} + \dots + (-1)^{n+1} \frac{1}{n\lambda} = \frac{1}{\lambda} \sum_{i=1}^{n} (-1)^{n+1} \frac{C_n^i}{i}$$

$$\sum_{i=1}^{n} (-1)^{n+1} \frac{C_n^i}{i} = \sum_{i=1}^{n} \frac{1}{i}$$

$$MTBF_p = \frac{1}{\lambda} \sum_{i=1}^{n} \frac{1}{i} = \frac{1}{\lambda_p} \Rightarrow \lambda_p = \frac{\lambda}{\sum_{i=1}^{n} \frac{1}{i}}$$

Exemplu:

$$R_p = 1 - \prod_{i=1}^{2}(1-R_i) = 1-(1-R_1)(1-R_2) =$$



$$= R_1 + R_2 - R_1R_2$$

$R_1 = 0,9$ (90%)

$R_2 = 0,95$ (95%)

$R_p = 0,9 + 0,95 - 0,9 \cdot 0,95 = 1,85 - 0,855 = 0,995$ (99,5%)

doca: $R_1(t) = e^{-\lambda_1 t}$ $R_2(t) = e^{-\lambda_2 t}$

$$R_p(t) = e^{-\lambda_1 t} + e^{-\lambda_2 t} - e^{-(\lambda_1+\lambda_2)t}$$

$$MTBF_p = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} - \frac{1}{\lambda_1+\lambda_2}$$



$R_S = R_1R_2 = 0,9 \cdot 0,95 = 0,855$ (85,5%)

doua: $R_1(t) = R_2(t) = e^{-\lambda t} = R(t)$

$$R_p(t) = 2R(t) - R^2(t) = 2e^{-\lambda t} - e^{-2\lambda t}$$

$$MTBF_p = \frac{2}{\lambda} - \frac{1}{2\lambda} = \frac{3}{2\lambda} \Rightarrow \lambda_p = \frac{2}{3}\lambda$$

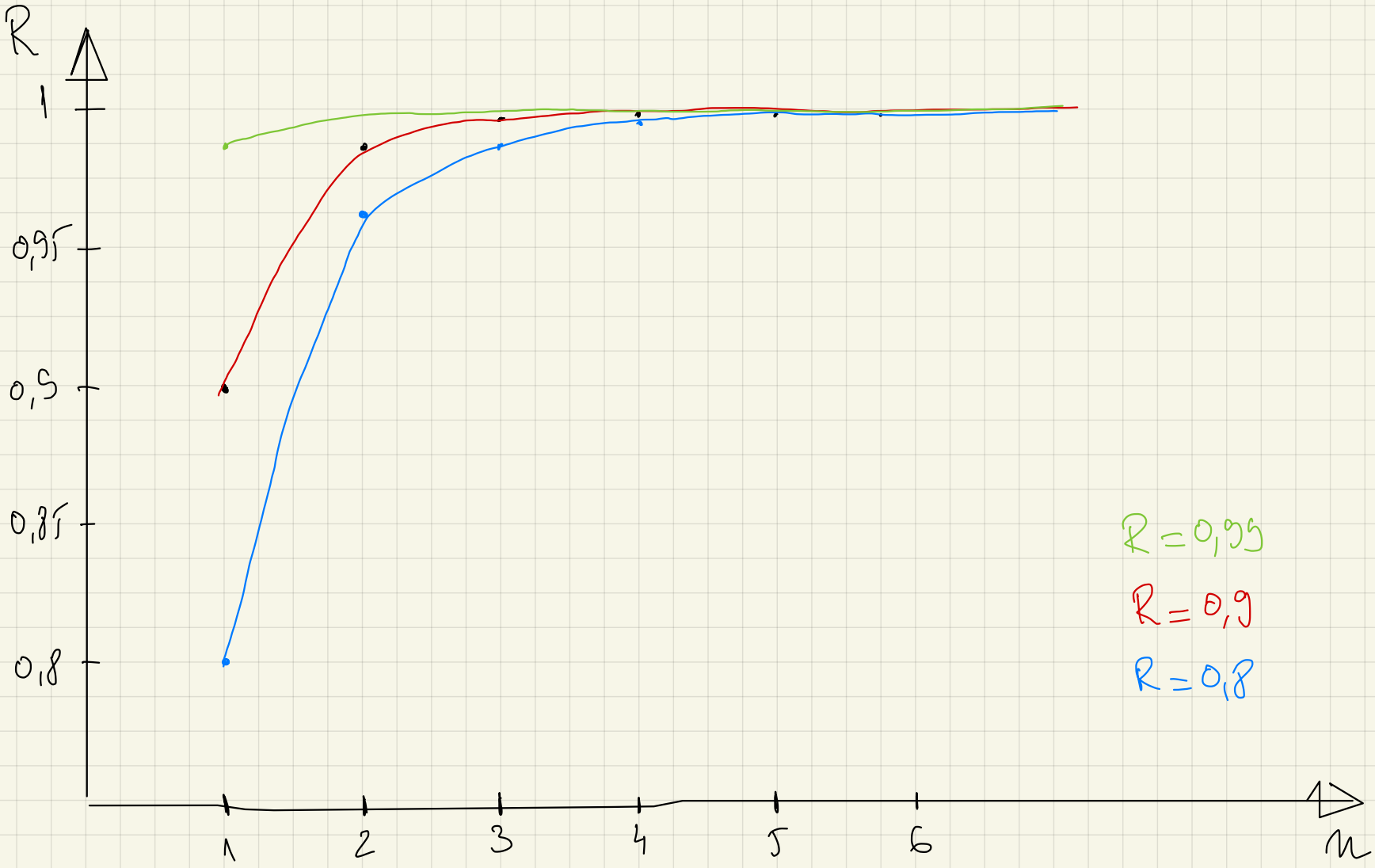
$$R_p = 2R - R^2 > R \quad (\forall) \quad R \in [0,1)$$

Exemplu :

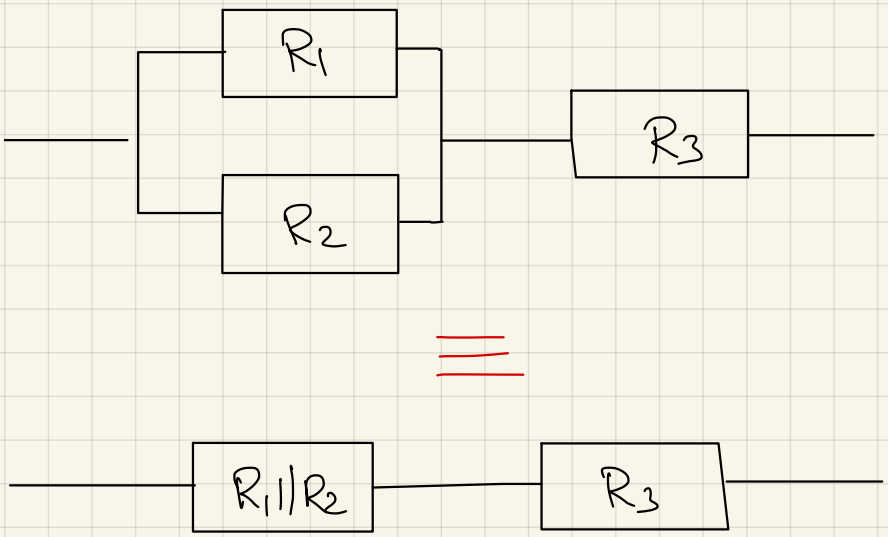
2 module 90% : $R_p = 1 - (1-0,9)^2 = 99\%$

3 module 90% : $R_p = 1 - (1-0,9)^3 = 99,9\%$

4 module 90% : $R_p = 1 - (1-0,9)^4 = 99,99\%$



Exemplu:



$$R_{TOTAL} = (R_1 \| R_2) \cdot R_3 = \left(1 - (1-R_1)(1-R_2)\right) R_3 =$$

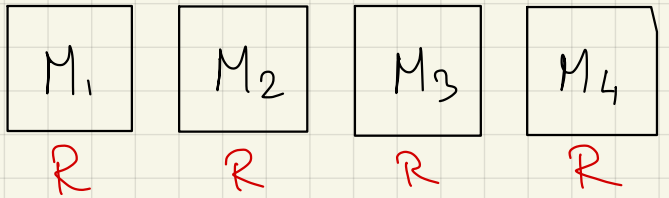
$$= (R_1 + R_2 - R_1 R_2) R_3 = R_1 R_3 + R_2 R_3 - R_1 R_2 R_3$$

$R_1 = 99,5\%$ $R_2 = 98,7\%$ $R_3 = 97,3\%$

$R_T = 0,995 \cdot 0,973 + \ldots \simeq 0,97$ (97%)

Fiabilitatea structurilor n din m

$$R_T = R^4 + 4(1-R)R^3 + 6(1-R)^2R^2 + 4(1-R)^3R$$



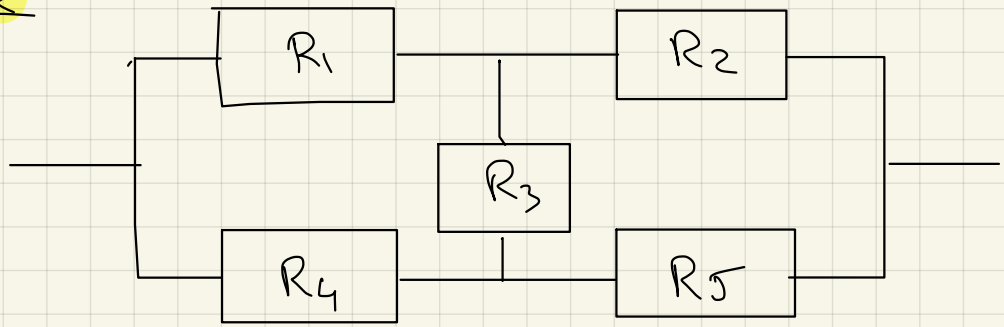
→ 4 motoare pt. un avion 747

în general: $R_{n/m} = C_m^n R^n (1-R)^{m-n}$

dacă: $n = m \rightarrow R_{n/m} = R_S = R^m$ – fiab serie

$n = 1 \rightarrow R_{n/m} = R_P = 1-(1-R)^m$ – fiab paralel

Fiabilitatea structurilor nedecomposabile



1: R_3 – funcționează ($R_3 = 1$)

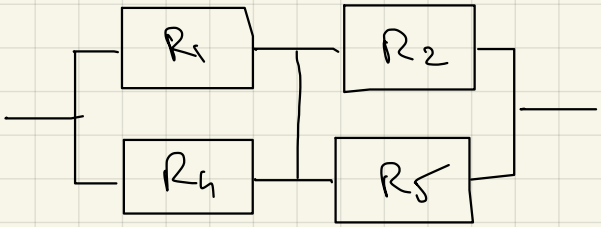
$P(S|R_3) = \ldots$

2: R_3 – nu funcționează ($R_3 = 0$)

$P(S|\overline{R_3}) = \ldots$

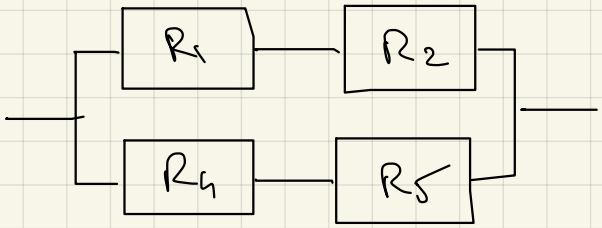
$$R_{TOTAL} = P(S|R_3) \cdot R_3 + P(S|\overline{R_3}) \cdot (1 - R_3)$$

$P(S|R_3)$:



$$(R_1 \| R_4) \cdot (R_2 \| R_5) = (R_1 + R_4 - R_1R_4)(R_2 + R_5 - R_2R_5)$$

$P(S|\overline{R_3})$:



$$(R_1 \cdot R_2) \| (R_4 \cdot R_5) = R_1R_2 + R_4R_5 - R_1R_2R_4R_5$$

$$R_{TOTAL} = (R_1 + R_4 - R_1R_4)(R_2 + R_5 - R_2R_5) \cdot R_3 + (R_1R_2 + R_4R_5 - R_1R_2R_4R_5)(1 - R_3)$$

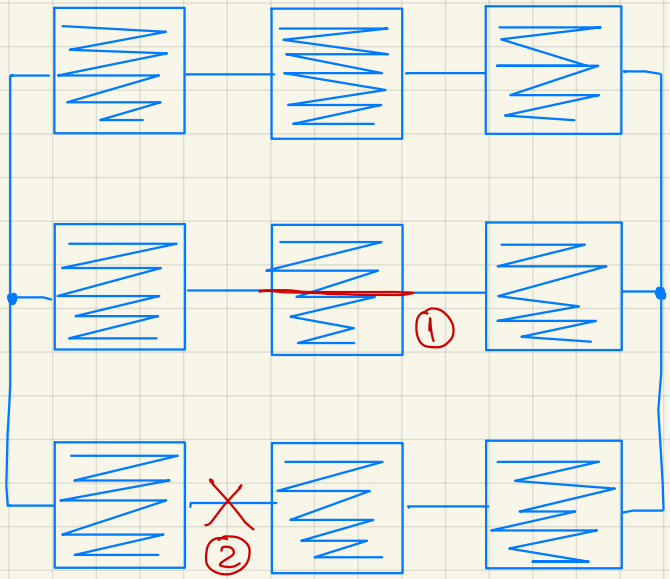
Dacă $R_1 = R_2 = \ldots = R_5 = R \Rightarrow R_{TOTAL} = R^2(2R^3 - 5R^2 + 2R + 2)$

Rezervor bimodală

Failure mode prevalent : — defect prin scurtcircuit
— defect prin întrerupere

Exemplu : panouri fotovoltaice 9 celule PV, cum le legăm a.i. fiabilitatea să fie mai mare? (modul de defectiune prevalent pt PV este întrerupere)

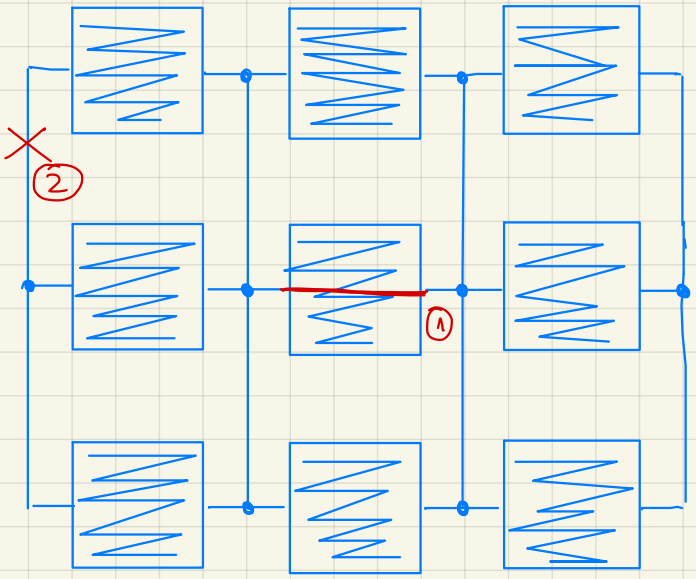
serie - paralel



① scurtcircuit : pierdem 1 celulă din 9

② întrerupere : pierdem 3 celule din 9

paralel - serie

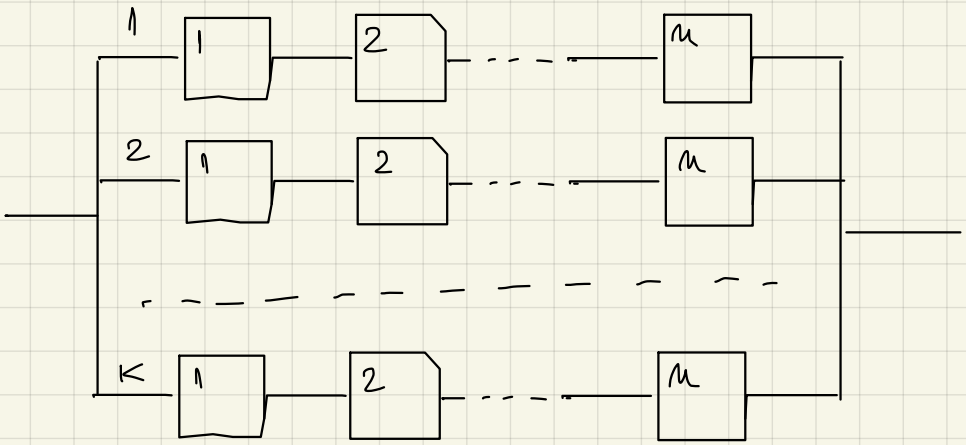


① scurtcircuit: pierdem 3 celule din 9

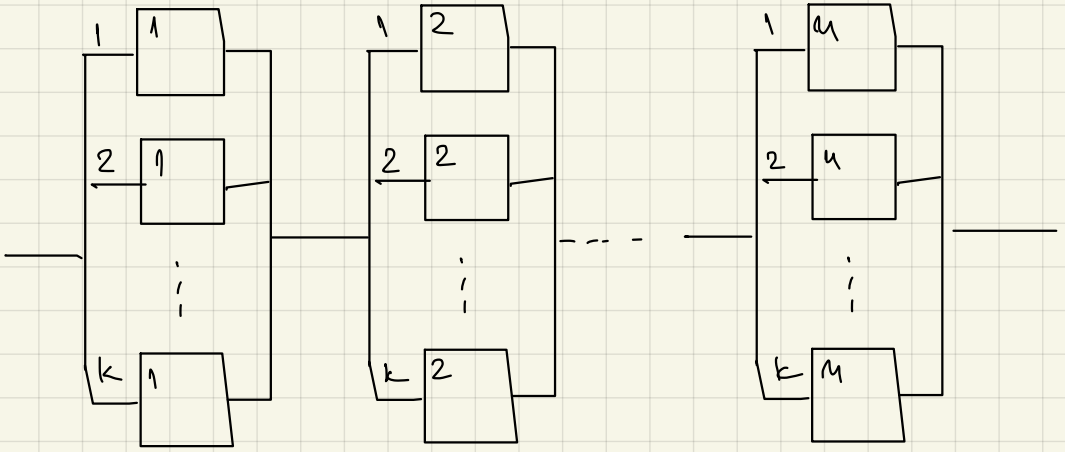
② întrerupere : pierdem 1 celulă din 9

Caz general: $k \times n$ module de fiabilitate identică R

Structura serie/paralel



Structura paralel/serie



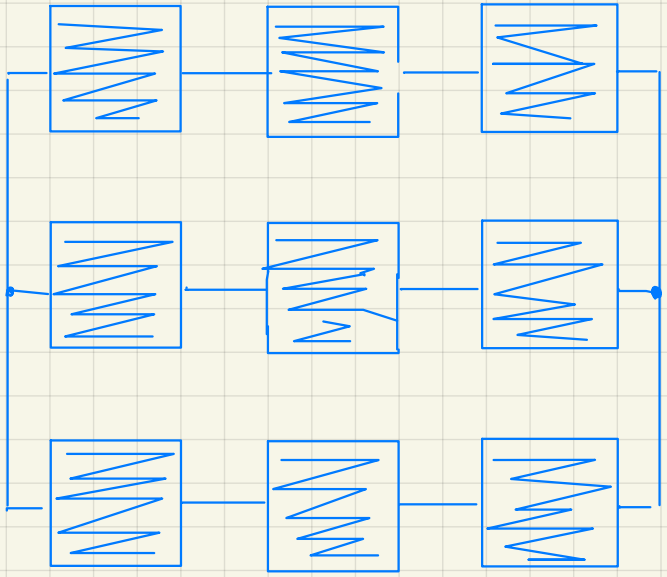
$$R_{SP} = \underbrace{R^n \| R^n \| \dots \| R^n}_{k \text{ ori}} = 1 - (1 - R^n)^k$$

$$R_{PS} = \left[1 - (1 - R)^k\right]^n$$

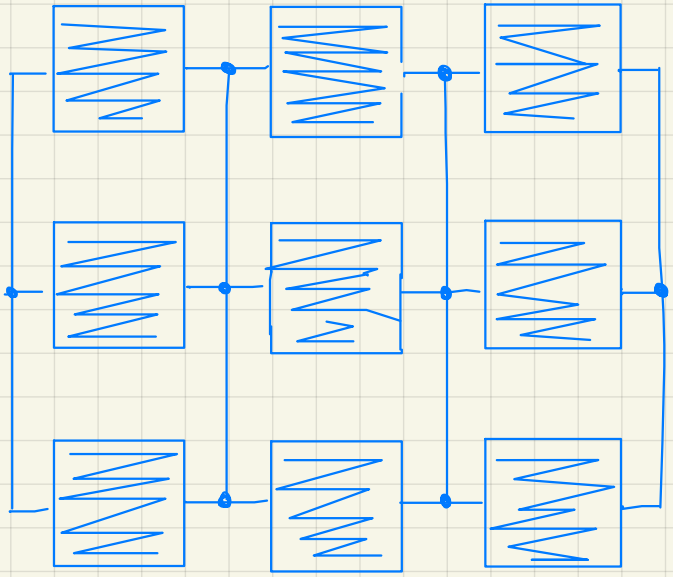
Exemplu:

9 celule PV legate în cele 2 configurații

S-P



P-S



$R = 90\%$, $K = M = 3$

$$R_{PS} = \left(1-(1-R)^3\right)^3 = \left(1-(1-0{,}9)^3\right)^3 = \left(1-0{,}001\right)^3 = 0{,}999^3 \simeq 0{,}997 \ (99{,}7\%)$$

$$R_{SP} = 1-\left(1-R^3\right)^3 = \left(1-\left(1-0{,}9^3\right)^3\right) = 1-0{,}271^3 \simeq 0{,}98 \ (98\%)$$

IC - Cursul 8

Notițe de curs

Structuri cu votare majoritară

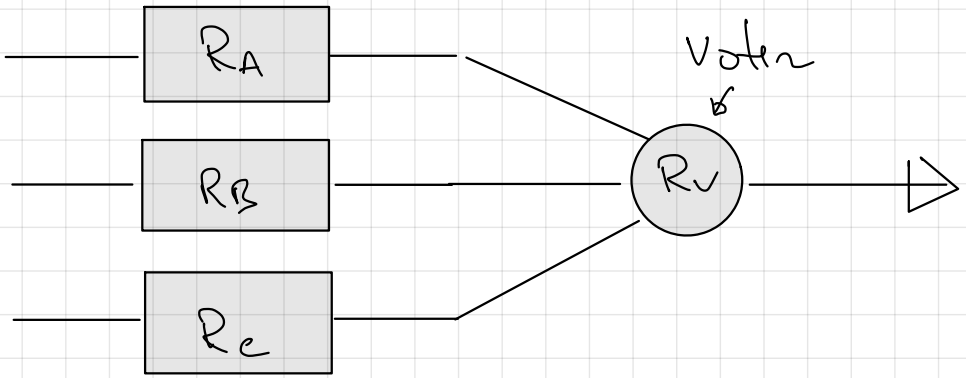
$$R_{2/3} = R_V \cdot \left(R_A R_B R_C + (1-R_A) R_B R_C + \dots \right)$$

1: $R_A \cdot R_B \cdot R_C$

2: $(1-R_A) \cdot R_B \cdot R_C$

3: $R_A (1-R_B) R_C$

4: $R_A \cdot R_B (1-R_C)$



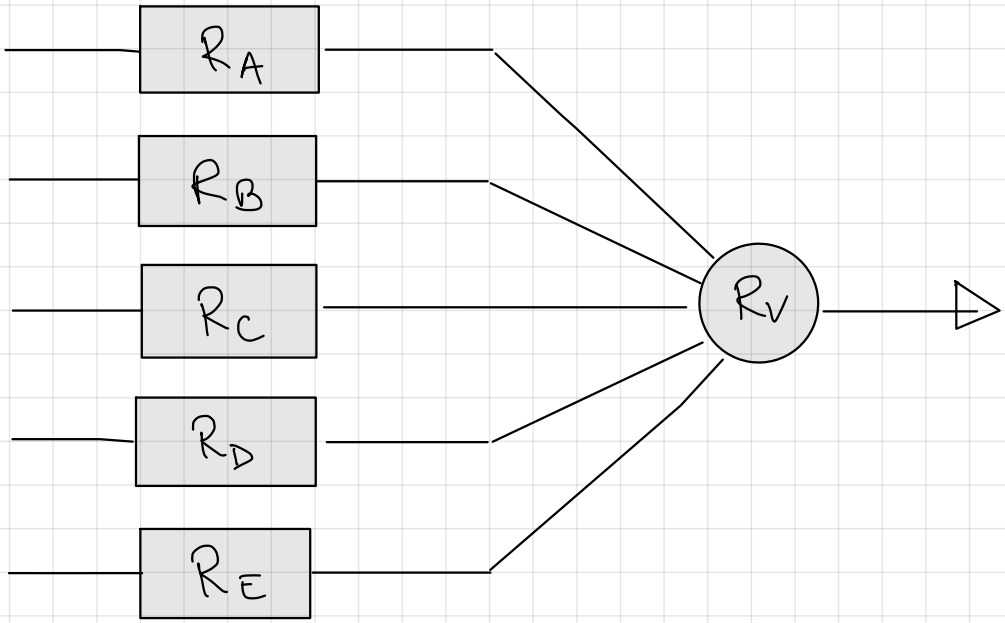
$$= R_V \cdot \left(R_A R_B + R_A R_C + R_B R_C - 2 R_A R_B R_C \right)$$

$$= R_A R_B + R_A R_C + R_B R_C - 2 R_A R_B R_C$$

$R_V >> R_A, R_B, R_C$

$R_V \simeq 1$

$R_{3/5} = ?$



$$R_A = R_B = R_C = R$$

$$R_{2/3} = C_3^2 R^2 (1-R) + C_3^3 R^3 = 3R^2(1-R) + R^3$$

$$R_{3/5} = C_5^3 R^3 (1-R)^2 + C_5^4 R^4 (1-R) + C_5^5 R^5$$

$$R_{RVM} = \left[\sum_{i=0}^{M-1} C_{2M-1}^{i} R^{2M-1-i} (1-R)^i \right] R_V$$

$$C_M^k = \frac{M!}{k!(M-k)!}$$

$$C_M^M = \frac{M!}{M!\,0!} = 1$$

$$C_M^0 = \frac{M!}{0!\,M!} = 1$$

$$C_M^{M-1} = \frac{M!}{(M-1)!\,1!} = M$$

$$R(t) = e^{-\lambda t}$$

$$R_{2/3} = 3e^{-2\lambda t} - 2e^{-3\lambda t}$$

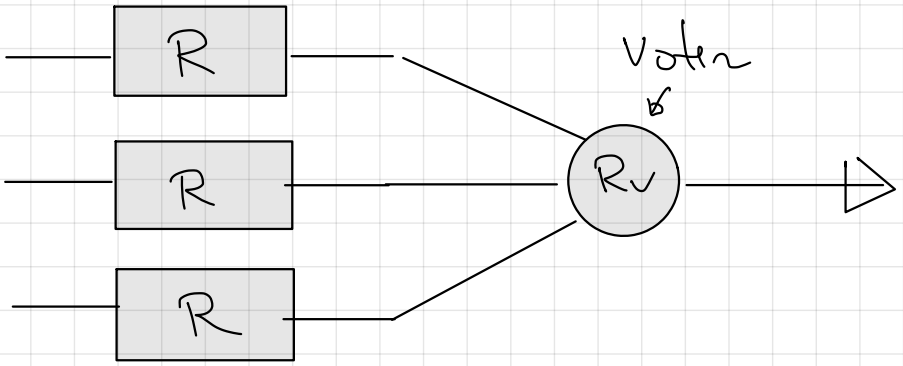
$$R_{3/5} = 10e^{-3\lambda t} - 15e^{-4\lambda t} + 6e^{-5\lambda t}$$

$$R_{2/3}(t) > R(t) ? \iff 3R^2 - 2R^3 > R \xrightarrow{R>0}$$

$$\Rightarrow 3R - 2R^2 > 1 \iff 2R^2 - 3R + 1 < 0 \quad , \quad 1, \frac{1}{2} \rightarrow \text{wobcim} \Rightarrow$$

$$\Rightarrow R \in \left(\frac{1}{2}, 1\right) \Rightarrow R > 50\,\%$$

1. $R = 10\%$ $(0,1)$



$R_{2/3} = 3R^2 - 2R^3 = 3 \cdot 0,01 - 2 \cdot 0,001$

$= 0,03 - 0,002 \simeq 3\%$

2. $R = 90\%$ $(0,9)$

$R_{2/3} = 3 \cdot 0,9^2 - 2 \cdot 0,9^3 = 3 \cdot 0,81 - 2 \cdot 0,729 = 2,43 - 1,458 = 0,98$

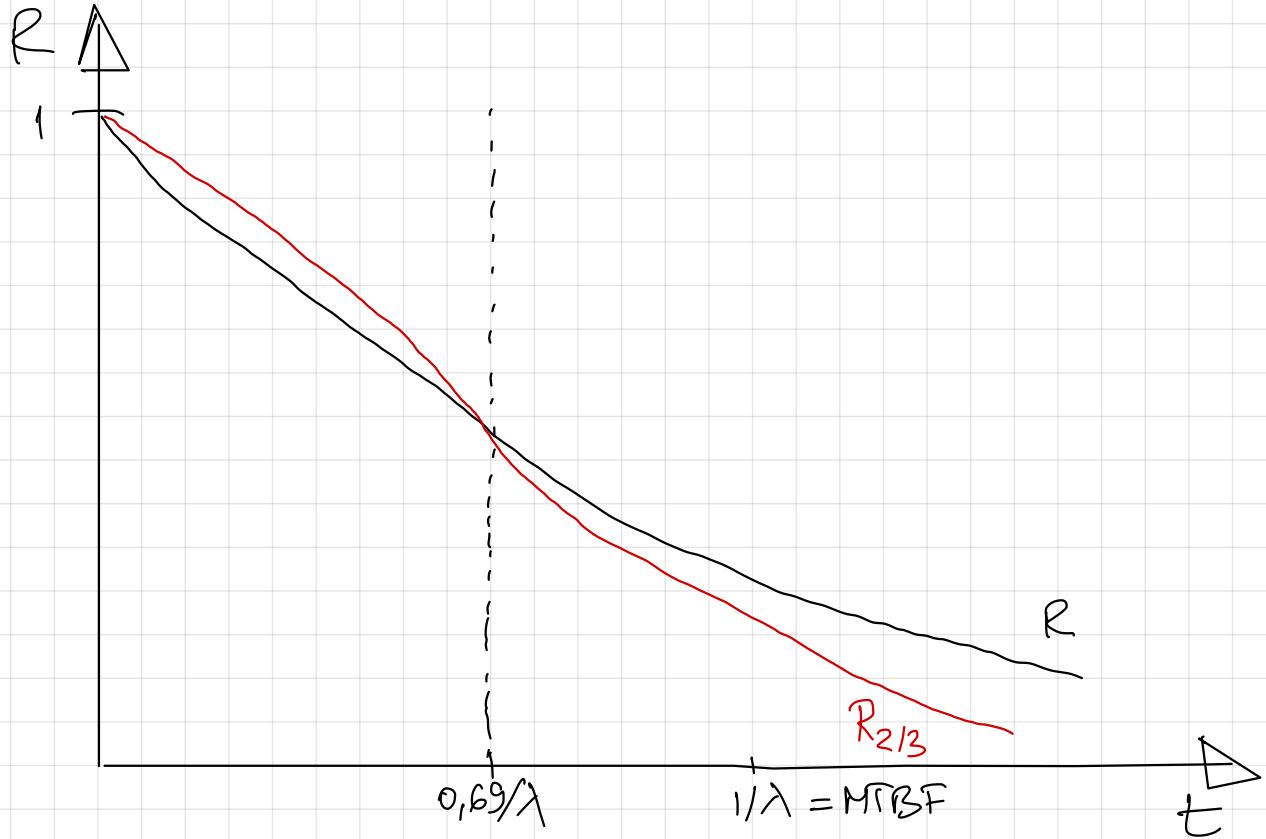
98%

$R_{2/3} > R \iff R > \frac{1}{2} \Rightarrow e^{-\lambda t} > \frac{1}{2} \Rightarrow \lambda t < \ln 2$

$t < \frac{\ln 2}{\lambda} \simeq \frac{0,69}{\lambda}$

$\frac{1}{\lambda} = MTBF$

$\frac{0,69}{\lambda}$



$$MTBF_{2/3} = \int_0^\infty R_{2/3}(t)\,dt = \int_0^\infty \left(3R^2(t) - 2R^3(t)\right)dt =$$

$R(t) = e^{-\lambda t}$

$$= \int_0^\infty \left(3e^{-2\lambda t} - 2e^{-3\lambda t}\right)dt = 3\int_0^\infty e^{-2\lambda t}\,dt - 2\int_0^\infty e^{-3\lambda t}\,dt =$$

$$\int_0^\infty e^{-ax}\,dx = \frac{1}{a}$$

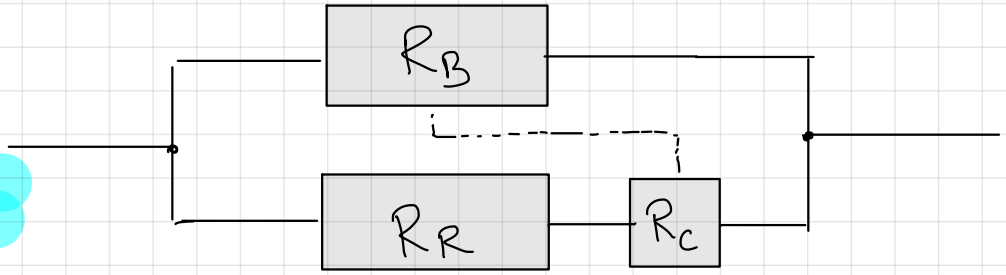
$$= \frac{3}{} 3 \cdot \frac{1}{2\lambda} - \frac{2}{} 2 \cdot \frac{1}{3\lambda} = \frac{5}{6} \cdot \underbrace{\frac{1}{\lambda}}_{MTBF} = \frac{5}{6} \cdot MTBF \Rightarrow MTBF_{2/3} < MTBF$$

$$MTBF_{3/5} = \int_0^\infty R_{3/5}(t)\,dt = \int_0^\infty \left(10e^{-3\lambda t} - 15e^{-4\lambda t} + 6e^{-5\lambda t}\right)dt =$$

$$= 10\,\frac{1}{3\lambda} - 15 \cdot \frac{1}{4\lambda} + 6 \cdot \frac{1}{5\lambda} < \frac{1}{\lambda} \quad (MTBF)$$

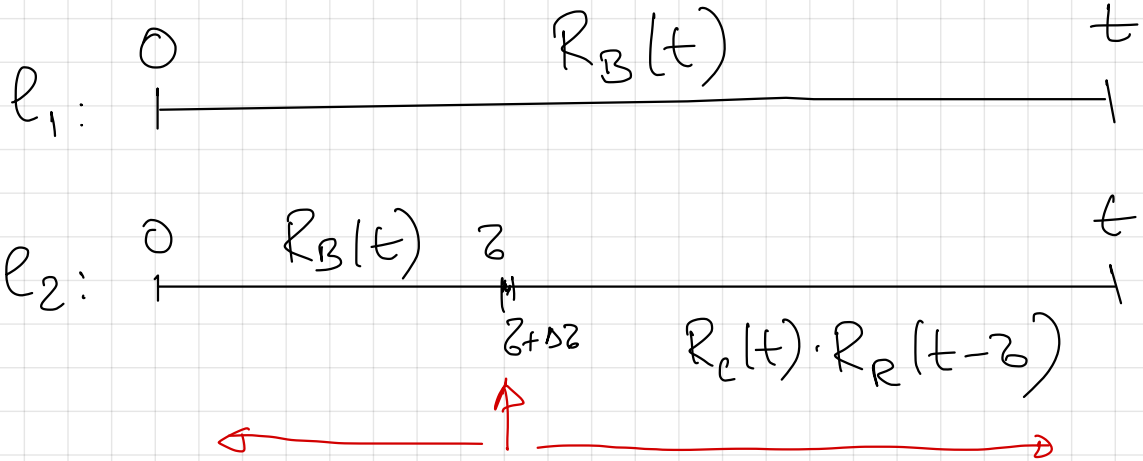
Structuri cu un singur element de rezervă

$R_T = ?$



ℓ_1: Baza funcționează în $(0,t)$

$p(\ell_1) = R_B(t)$



ℓ_2: Baza funcționează în $(0,z)$, se defectează în $[z, z+\Delta z]$, și elem. de rezervă comută în stare de funcționare

$p_b \rightarrow$ prob. să se defecteze elem. bază în $[z, z+\Delta z] \Rightarrow$

$p_b = f_b(z)\,dz$, unde $f_b(z) = \frac{dF_b(z)}{dz} = -\frac{dR_B(z)}{dz}$

$p_R \rightarrow$ prob. ca elem de rezervă să funcționeze în intervalul (z,t)

$$p_1(t) = R_R(t-z), \quad R_C \simeq 1$$

z - se produce oricand în $(0, t)$

$$p(e_2) = \underset{\simeq 1}{R_C} \cdot \int_0^t p_b \cdot p_2 \, dz = \int_0^t f_b(z) R_R(t-z) \, dz$$

$$R_S(t) = p(e_1) + p(e_2) = R_B(t) + \int_0^t f_b(z) R_R(t-z) \, dz$$

$$R_B(t) = e^{-\lambda_B t} \qquad R_R = e^{-\lambda_R t}$$

$$f_b(z) = -\frac{dR_B(z)}{dz} = -(-\lambda_B) e^{-\lambda_B t} = \lambda_B e^{-\lambda_B t}$$

$$R_S(t) = e^{-\lambda_B t} + \int_0^t \lambda_B e^{-\lambda_B z} e^{-\lambda_R (t-z)} \, dz =$$

$$= e^{-\lambda_B t} + \lambda_B e^{-\lambda_R t} \int_0^t e^{-\lambda_B z} \cdot e^{\lambda_R z} \, dz = e^{-\lambda_B t} + \lambda_B e^{-\lambda_R t} \cdot \int_0^t e^{(\lambda_R - \lambda_B) z} \, dz$$

$$= e^{-\lambda_B t} + \lambda_B e^{-\lambda_R t} \cdot \frac{1}{\lambda_R - \lambda_B} e^{(\lambda_R - \lambda_B) z} \Big|_0^t = \ldots = \frac{\lambda_R}{\lambda_R - \lambda_B} e^{-\lambda_B t} - \frac{\lambda_B}{\lambda_R - \lambda_B} e^{-\lambda_R t}$$

$$R_S(t) = \frac{\lambda_R}{\lambda_R - \lambda_B} R_B(t) - \frac{\lambda_B}{\lambda_R - \lambda_B} R_R(t) \supset R_D(t) \; ?$$

$$MTBF_S = \int_0^\infty R_S(t)\,dt = \int_0^\infty \left(\frac{\lambda_R}{\lambda_R - \lambda_D} e^{-\lambda_D t} - \frac{\lambda_D}{\lambda_R - \lambda_D} e^{-\lambda_R t} \right) dt =$$

$$= \frac{\lambda_R}{\lambda_R - \lambda_B} \int_0^\infty e^{-\lambda_B t}\,dt - \frac{\lambda_B}{\lambda_R - \lambda_B} \int_0^\infty e^{-\lambda_R t}\,dt = \frac{\lambda_R}{\lambda_B} \cdot \frac{1}{\lambda_R - \lambda_B} - \frac{\lambda_B}{\lambda_R} \frac{1}{\lambda_R - \lambda_B}$$

$$= \frac{\lambda_R^2 - \lambda_B^2}{\lambda_R \lambda_B (\lambda_R - \lambda_B)} = \frac{(\lambda_R - \lambda_B)(\lambda_R + \lambda_B)}{\lambda_R \lambda_B (\lambda_R - \lambda_B)} = \frac{\lambda_R + \lambda_B}{\lambda_R \cdot \lambda_B} = \frac{1}{\lambda_B} + \frac{1}{\lambda_R} =$$

$$= MTBF_B + MTBF_R$$

$$MTBF_S = MTBF_B + MTBF_R$$

Резервиран сб: $R_B(t) = R_R(t) = R(t) = e^{-\lambda t}$

$$R_S(t) = (1+\lambda t)e^{-\lambda t} = (1+\lambda t)R(t)$$

$$MTBF_S = 2 \cdot MTBF = 2 \cdot \frac{1}{\lambda}$$

Още:

$$R_S(t) > R(t) ? \Rightarrow (1+\lambda t)e^{-\lambda t} > e^{-\lambda t} \Big/ \cdot e^{\lambda t} \Leftrightarrow 1+\lambda t > 0 \Rightarrow$$

$$\Rightarrow \lambda t > -1 \Rightarrow t > -\frac{1}{\lambda}$$

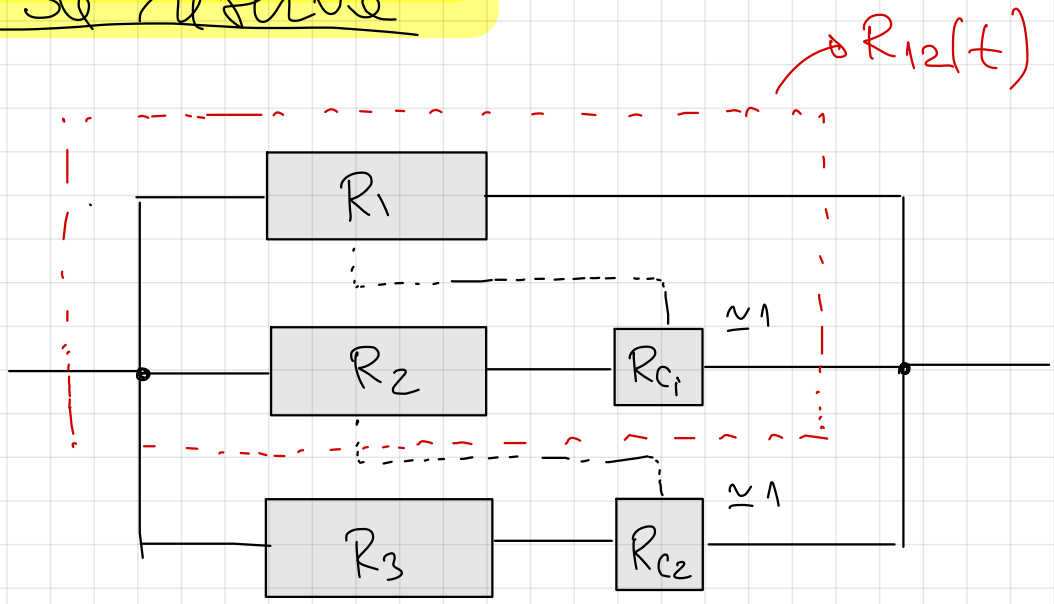
Дан $t > 0 \Rightarrow R_S(t) > R(t)\ (\forall)\ t > 0$

Structură cu două elemente de rezervă

$$R_{12}(t) = R_1(t) + \int_0^t f_1(\tau) R_2(t-\tau)\, d\tau$$

$$R_{123}(t) = R_{12}(t) + \int_0^t f_{12}(\tau) R_3(t-\tau)\, d\tau$$

$$f_{12}(\tau) = -\frac{d R_{12}(\tau)}{d\tau}$$



$$R_1(t) = R_2(t) = R_3(t) = e^{-\lambda t}$$

$$R_{12}(t) = (1+\lambda t) e^{-\lambda t}, \quad f_{12}(t) = \frac{d R_{12}(t)}{dt} = \lambda^2 t e^{-\lambda t}$$

$$R_S(t) = \left(1 + \lambda t + \frac{\lambda^2 t^2}{2}\right) e^{-\lambda t}$$

$$MTBF_S = 3 \cdot \frac{1}{\lambda} = 3 \cdot MTBF$$

Structură cu k-1 elemente de rezervă

$$R_{S,i}(t) = R_{S,i-1}(t) + \int_0^t f_{S,i-1}(t)\, R_i(t-\tau)\, d\tau$$

$$f_{S,i-1}(t) = -\frac{d R_{S,i-1}(t)}{dt}$$

$$R_{S,k} = \frac{\lambda_2 \cdot \lambda_3 \cdots \lambda_k}{(\lambda_2-\lambda_1)\cdots(\lambda_k-\lambda_1)} e^{-\lambda_1 t} + \cdots + \frac{\lambda_1 \lambda_2 \cdots \lambda_{k-1}}{(\lambda_1-\lambda_k)\cdots(\lambda_{k-1}-\lambda_k)} e^{-\lambda_k t} =$$

$$= \sum_{j=1}^{k} \left[\prod_{\substack{i=1 \\ i \neq j}}^{k} \frac{\lambda_i}{\lambda_i - \lambda_j} e^{-\lambda_i t} \right]$$

$$MTBF = \sum_{i=1}^{k} MTBF_i$$

dacă toate modulele sunt identice:

$$\lambda_1 = \lambda_2 = \lambda_3 = \dots = \lambda_{k-1} = \lambda_k = \lambda$$

$$R_{S,k} = \sum_{j=1}^{k} \frac{(\lambda t)^{j-1}}{(j-1)!} e^{-\lambda t}$$

$$MTBF_{S,k} = \frac{k}{\lambda} = k \cdot MTBF$$

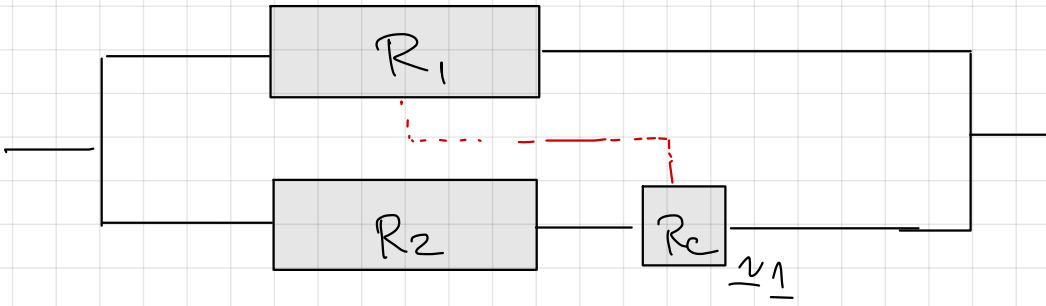
$$\frac{x^0}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \dots = \sum_{i=0}^{\infty} \frac{x^i}{i!} = e^x$$

$k \to \infty$ (foarte multe elem. de back-up)

$$\sum_{j=1}^{k} \frac{(\lambda t)^{j-1}}{(j-1)!} \to e^{\lambda t}$$

$$R_{S,k} = e^{\lambda t} \cdot e^{-\lambda t} = 1$$

Structură cu elem. de rezervă activ



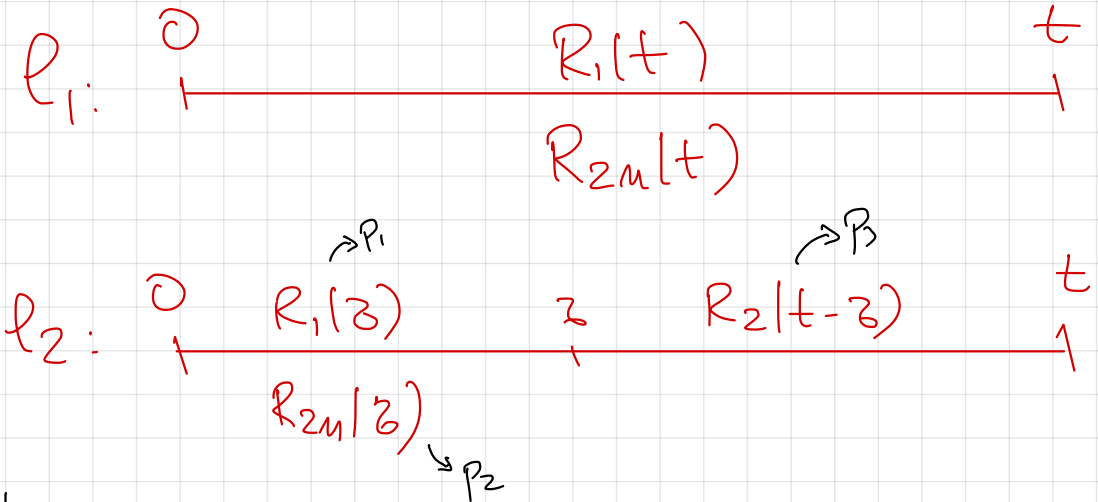
$$R_S(t) = p(e_1) + p(e_2)$$

$$p(e_1) = R_1(t)$$

$$p(e_2) = \int_0^t p_1 p_2 p_3 \, d\tau =$$

$$= \int_0^t f_1(\tau) R_{2m}(\tau) R_2(t-\tau) \, d\tau$$

$$f_1(\tau) = -\frac{dR_1(\tau)}{d\tau}$$



$$R_1(t) = e^{-\lambda_1 t} \qquad R_2(t) = e^{-\lambda_2 t} \qquad R_{2m}(t) = e^{-\lambda_{2m} t}$$

$$R_S(t) = e^{-\lambda_1 t} + \frac{\lambda_1}{\lambda_1 + \lambda_{2m} - \lambda_2}\left[e^{-\lambda_2 t} - e^{-(\lambda_1 + \lambda_{2m})t}\right]$$

dacă avem module identice: $R_1(t) = R_2(t) = e^{-\lambda t}$ $\qquad R_{2m}(t) = e^{-\lambda_m t}$

$$R_S(t) = \left[1 + \frac{\lambda}{\lambda_m}\left(1 - e^{-\lambda_m t}\right)\right] e^{-\lambda t}$$

$$MTBF = \frac{1}{\lambda} + \frac{1}{\lambda + \lambda_m}$$

IC - Cursul 9

Notițe de curs

Modele Markov

- starea sistemului $(x_1, x_2 \dots x_n)$ – discrete
- timpul de observare $(t_1, t_2 \dots t_n)$

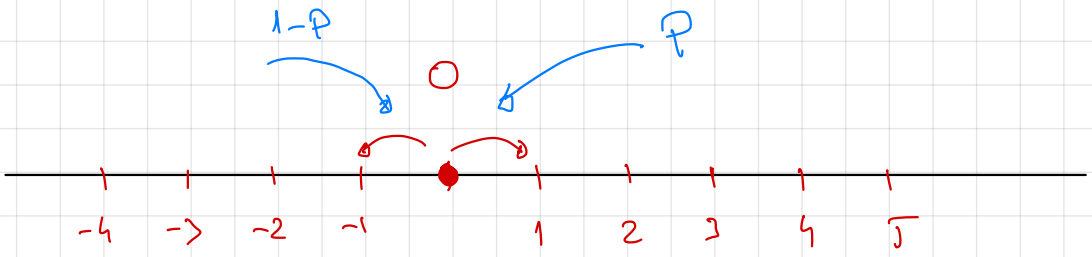
Lanturi Markov

$X(1), X(2) \dots X(n) \dots$

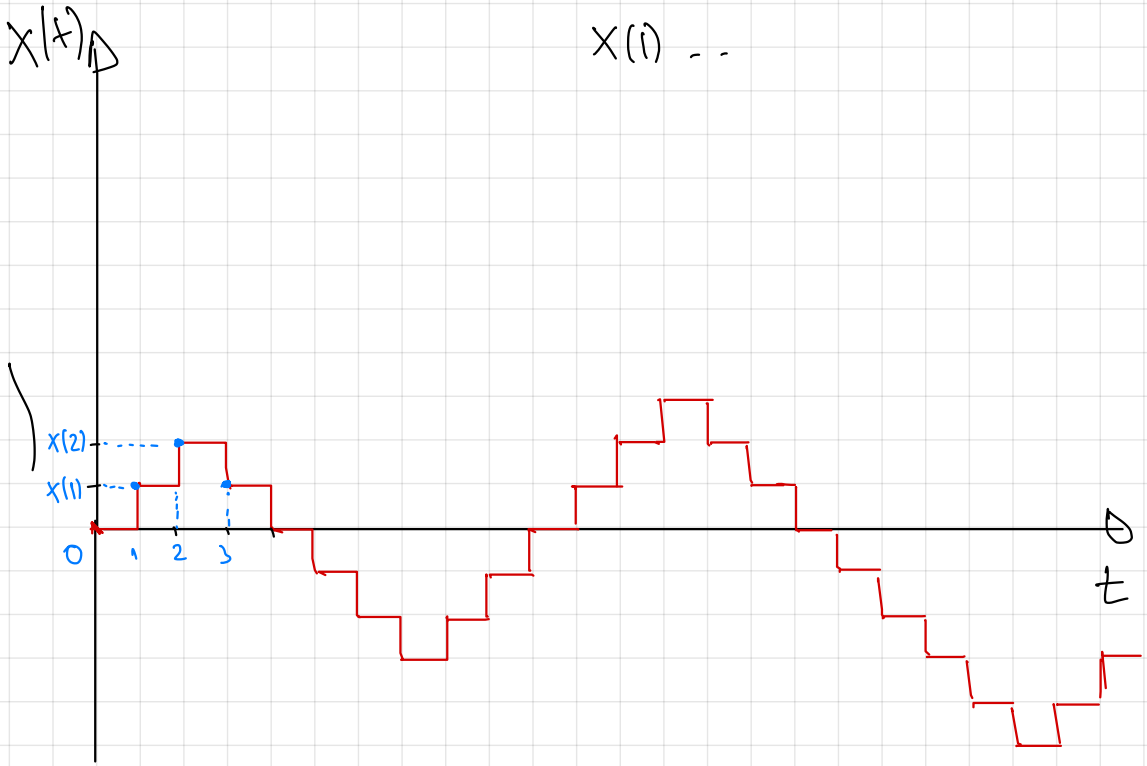
$$p_i(k) = P(X(k) = S_i)$$

$$\sum_{i=1}^{n} p_i(k) = 1$$

$$p_{ij} = P\{X(k) = S_i \mid X(k+1) = S_j\}$$



$X(1) \dots$

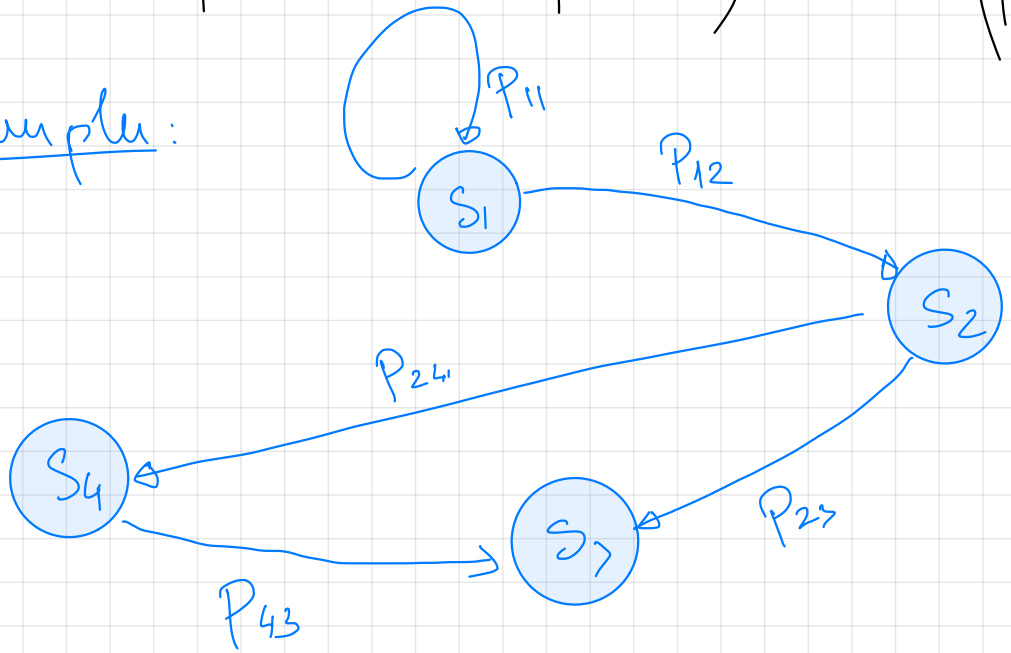


$$\begin{pmatrix} p_{11} & p_{12} & \dots & p_{1M} \\ p_{21} & p_{22} & \dots & p_{2M} \\ & - & - & - \\ p_{M1} & p_{M2} & \dots & p_{MM} \end{pmatrix} = A$$

$$p_i(k) = \sum_{j=1}^{M} p_j(k-1) p_{ji}$$

$$\left(p_1(t) \quad p_2(t) \quad \dots \quad p_M(t)\right) \cdot A = \left(p_1(t+\Delta t) \quad p_2(t+\Delta t) \quad \dots \quad p_M(t+\Delta t)\right)$$

Example:



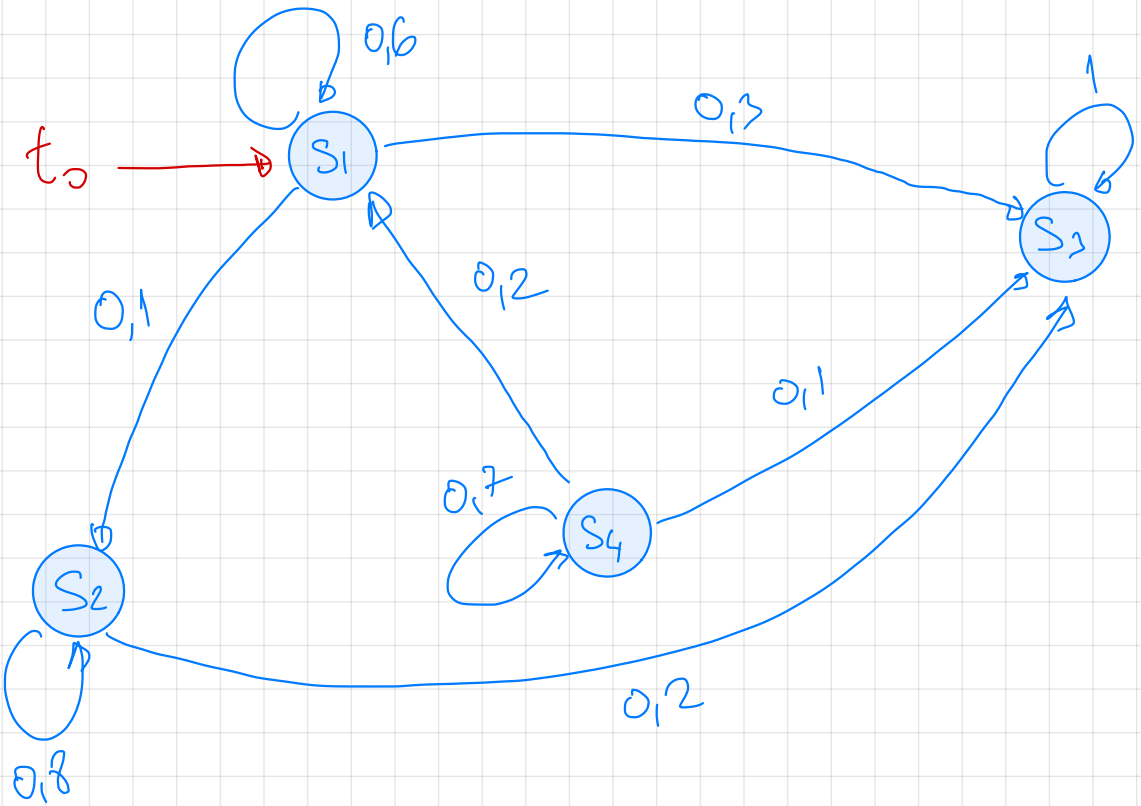
Ex: Calculator (C), imprimantă (I), Plotter (P)

$t_0 \rightarrow$ S_1 – C+I+P – funcționale

S_2 – C+P – funcționale, I – defectă

S_3 – C+I – funcționale, P – defect

S_4 – C+I+P – defecte



$$A = \begin{pmatrix} 0,6 & 0,1 & 0,3 & 0 \\ 0 & 0,8 & 0,2 & 0 \\ 0 & 0 & 1 & 0 \\ 0,2 & 0 & 0,1 & 0,7 \end{pmatrix}$$

$$\left(P_1(t)\ P_2(t)\ P_3(t)\ P_4(t)\right) = P(t)$$

$$\left(P_1(t+\Delta t)\ P_2(t+\Delta t)\ P_3(t+\Delta t)\ P_4(t+\Delta t)\right) = P(t+\Delta t)$$

$$P(t) \cdot A = P(t+\Delta t)$$

$$P(t) = \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix}$$

$$P(t+\Delta t) = \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0,6 & 0,1 & 0,3 & 0 \\ 0 & 0,8 & 0,2 & 0 \\ 0 & 0 & 1 & 0 \\ 0,2 & 0 & 0,1 & 0,7 \end{pmatrix} = \begin{pmatrix} 0,6 & 0,1 & 0,3 & 0 \end{pmatrix}$$

După două intervale de timp:

$$P(t+2\Delta t) = P(t+\Delta t) \cdot A = \begin{pmatrix} 0,36 & 0,14 & 0,5 & 0 \end{pmatrix}$$

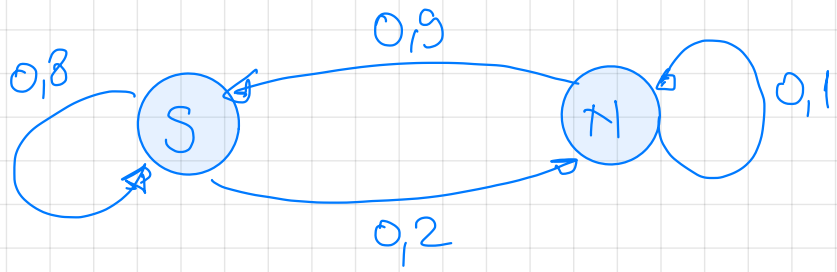
$$P(t+2\Delta t) = P(t+\Delta t) \cdot A = P(t) \cdot A^2$$

După n intervale de timp:

$$P(t+n\Delta t) = P(t) \cdot A^n$$

Ex: Starea vremii

Stări: Soare (S), Nor (N)



	S	N
S	0,8	0,2
N	0,9	0,1

$$\Rightarrow A = \begin{pmatrix} 0,8 & 0,2 \\ 0,9 & 0,1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 0,8 & 0,2 \\ 0,9 & 0,1 \end{pmatrix} = \begin{matrix} S & N \\ \begin{pmatrix} 0,8 & 0,2 \end{pmatrix} \end{matrix} \begin{pmatrix} 0,8 & 0,2 \\ 0,9 & 0,1 \end{pmatrix} = \begin{pmatrix} 0,64+0,18 & 0,16+0,02 \end{pmatrix} =$$

$$= \begin{matrix} S & N \\ \begin{pmatrix} 0,82 & 0,18 \end{pmatrix} \end{matrix} \quad \ldots$$

Steady state: $(q_1 \; q_2) \cdot A = (q_1 \; q_2) \Rightarrow (q_1 \; q_2) A - (q_1 \; q_2) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} I_2 = (0 \; 0) \Rightarrow$

$$\Rightarrow (q_1 \; q_2)(A - I_2) = (0 \; 0) \Rightarrow (q_1 \; q_2) \begin{pmatrix} -0,2 & 0,2 \\ 0,9 & -0,9 \end{pmatrix} = (0 \; 0) \Rightarrow$$

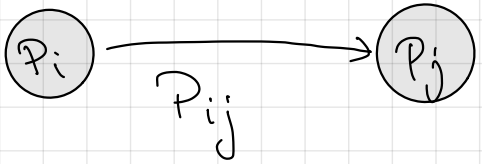
$$\Rightarrow \begin{cases} 0,2 q_1 - 0,9 q_2 = 0 \\ q_1 + q_2 = 1 \end{cases} \Rightarrow \begin{cases} 2q_1 - 9q_2 = 0 \\ q_2 = 1 - q_1 \end{cases} \Rightarrow 2q_1 - 9 + 9q_1 = 0$$

$$11 q_1 = 9 \Rightarrow q_1 = \frac{9}{11}$$

$$q_2 = \frac{2}{11}$$

$\left(\frac{9}{11} \quad \frac{2}{11} \right)$ → vector steady-state

Procese Markov



λ_{ij} – densitate de probabilitate de tranziție

$$\lambda_{ij} = \lim_{\Delta t \to 0} \frac{P\{X(t+\Delta t) = S_j \mid X(t) = S_i\}}{\Delta t}$$

dacă $\Delta t \to 0 \Rightarrow P_{ij} \simeq \lambda_{ij} \Delta t$

$$P(t) = (P_1(t) \; P_2(t) \; \dots \; P_M(t))$$

$$A = \begin{pmatrix} P_{11} & P_{12} & \dots & P_{1M} \\ P_{21} & P_{22} & \dots & P_{2M} \\ \dots & \dots & \dots & \dots \\ P_{M1} & P_{M2} & \dots & P_{MM} \end{pmatrix} = \begin{pmatrix} \lambda_{11}\Delta t & \lambda_{12}\Delta t & \dots & \lambda_{1M}\Delta t \\ \lambda_{21}\Delta t & \lambda_{22}\Delta t & \dots & \lambda_{2M}\Delta t \\ \dots & \dots & \dots & \dots \\ \lambda_{M1}\Delta t & \lambda_{M2}\Delta t & \dots & \lambda_{MM}\Delta t \end{pmatrix} =$$

$$= \begin{pmatrix} 1-\sum_{i=2}^{n}\lambda_{1i}\Delta t & \lambda_{12}\Delta t & \dots & \lambda_{1n}\Delta t \\ \lambda_{21}\Delta t & 1-\sum_{\substack{i=1\\ i\neq 2}}^{n}\lambda_{2i}\Delta t & \dots & \lambda_{2n}\Delta t \\ \dots & \dots & \dots & \dots \\ \lambda_{n1}\Delta t & \lambda_{n2}\Delta t & \dots & 1-\sum_{i=1}^{n-1}\lambda_{ni}\Delta t \end{pmatrix} = A$$

explicităm relația $P(t)\cdot A = P(t+\Delta t) \Rightarrow$

$$\Rightarrow (p_1(t)\ p_2(t)\ \dots\ p_n(t))\cdot A = (p_1(t+\Delta t)\ p_2(t+\Delta t)\ \dots\ p_n(t+\Delta t))$$

$$p_1(t+\Delta t) = p_1(t)\left(1-\sum_{i=2}^{n}\lambda_{1i}\Delta t\right) + p_2(t)\lambda_{21}\Delta t + \dots p_n(t)\lambda_{n1}\Delta t \Leftrightarrow$$

$$p_1(t+\Delta t) - p_1(t) = \sum_{i=2}^{n} p_i(t)\lambda_{i1}\Delta t - p_1(t)\sum_{i=2}^{n}\lambda_{1i}\Delta t \ \Big|:\Delta t \Rightarrow$$

$$\Rightarrow \frac{p_1(t+\Delta t)-p_1(t)}{\Delta t} = \sum_{i=2}^{n} p_i(t)\lambda_{i1} - p_1(t)\sum_{i=2}^{n}\lambda_{1i}$$

$$\lim_{\Delta t \to 0} \frac{p_1(t+\Delta t) - p_1(t)}{\Delta t} = \sum_{i=2}^{n} p_i(t)\lambda_{i1} - p_1(t)\sum_{i=2}^{n}\lambda_{1i}$$

$$\frac{dp_1(t)}{dt} = \sum_{i=2}^{n} p_i(t)\lambda_{i1} - p_1(t)\sum_{i=2}^{n}\lambda_{1i}$$

$$\frac{dp_j(t)}{dt} = \sum_{\substack{i=1 \\ i\neq j}}^{n} p_i(t)\lambda_{ij} - p_j(t)\sum_{\substack{i=1 \\ i\neq j}}^{n}\lambda_{ji}, \quad (\forall) j = 1 \dots n$$

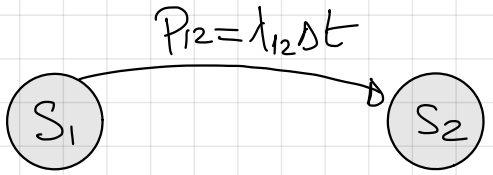
↓

Sistem ecuații Chapman – Kolmogorov

Forma matriceală

$$P(t) \cdot A^* = P'(t), \text{ unde } A^* = \begin{pmatrix} -\sum_{i=2}^{n}\lambda_{1i} & \lambda_{12} & \lambda_{13} & \dots & \lambda_{1n} \\ & - & - & - & - \\ \lambda_{n1} & \lambda_{n2} & \dots & & -\sum_{i=1}^{n-1}\lambda_{ni} \end{pmatrix}$$

Sistem cu două stări



S_1 - funcționare
S_2 - defect

$$A^* = \begin{pmatrix} -\lambda_{12} & \lambda_{12} \\ 0 & 0 \end{pmatrix} \qquad A = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} = \begin{pmatrix} 1-\lambda_{12}\Delta t & \lambda_{12}\Delta t \\ 0 & 1 \end{pmatrix}$$

$$P(t)\cdot A = P(t+\Delta t) \quad \dots$$

$$P(t)\cdot A^* = P'(t) \Rightarrow \begin{pmatrix} P_1(t) & P_2(t) \end{pmatrix} \cdot \begin{pmatrix} -\lambda_{12} & \lambda_{12} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} P_1'(t) & P_2'(t) \end{pmatrix}$$

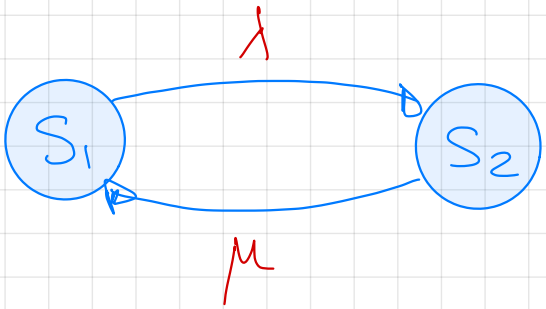
$$\begin{cases} P_1'(t) = -\lambda_{12} P_1(t) \\ P_2'(t) = \lambda_{12} P_1(t) \end{cases}$$

$$\frac{dP_1(t)}{dt} = -\lambda_{12} P_1(t) \Leftrightarrow \frac{dP_1(t)}{P_1(t)} = -\lambda_{12} dt \Leftrightarrow \int \frac{1}{P_1(t)} dP_1(t) = -\int \lambda_{12}\, dt \Rightarrow$$

$$\Rightarrow \ln(P_1(t)) = -\lambda_{12} t \Rightarrow P_1(t) = e^{-\lambda_{12} t}, \lambda_{12} = \lambda = ct \Rightarrow P_1(t) = e^{-\lambda t}$$

$$P_2(t) = 1 - P_1(t) = 1 - e^{-\lambda t}$$

Sistem cu două stări + reparare



S_1 – funcționare
S_2 – defect

$$A^* = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix}$$

$$P(t) \cdot A^* = P'(t) \Rightarrow \begin{cases} P_1'(t) = \mu P_2(t) - \lambda P_1(t) \\ P_2'(t) = \lambda P_1(t) - \mu P_2(t) \end{cases}$$

$$1 = P_1(t) + P_2(t)$$

Inițial, sistemul funcționează

$P_1(0) = 1 \quad P_2(0) = 0$

$$P_1'(t) = -\lambda P_1(t) + \mu(1 - P_1(t)) = \mu - (\lambda+\mu) P_1(t)$$

$$P_1'(t) + (\lambda+\mu) P_1(t) = \mu \Rightarrow P_1(t) = \frac{\mu}{\mu+\lambda} + \frac{\lambda}{\mu+\lambda} e^{-(\lambda+\mu)t}$$

$$\lim_{t \to \infty} P_1(t) = \frac{\mu}{\mu+\lambda} \Big| \mu\lambda = \frac{\frac{1}{\lambda}}{\frac{1}{\lambda} + \frac{1}{\mu}} = \frac{MTBF}{MTBF + MTR} = A$$

↳ disponibilitatea sistemului

IC - Cursul 10

Notițe de curs

Procese Markov

$\frac{dy}{dt} + p(t)\,y = g(t)$ – ec. diferențială liniară de gr. 1

$\mu(t) \to \mu(t)p(t) = \mu'(t) \iff \frac{\mu'(t)}{\mu(t)} = p(t) \,/\int \Rightarrow \int \frac{1}{\mu(t)}\mu'(t) = \int p(t)\,dt \Rightarrow$

$\Rightarrow \ln(\mu(t)) + k = \int p(t)\,dt \Rightarrow \ln \mu(t) = \int p(t)\,dt + k \Rightarrow$

$\Rightarrow \mu(t) = e^{\int p(t)\,dt + k} = e^k \cdot e^{\int p(t)\,dt} = k \cdot e^{\int p(t)\,dt}$

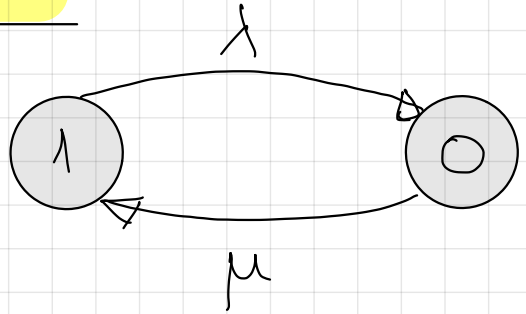
$\mu(t)\cdot y'(t) + \mu(t)p(t)y(t) = g(t)\mu(t) \iff \mu(t)y'(t) + \mu'(t)y(t) = g(t)\mu(t)$

$\Rightarrow (\mu(t)y(t))' = \mu(t)g(t) \,/\int \Rightarrow \int (y(t)\mu(t))'\,dt = \int \mu(t)g(t)\,dt \iff$

$\iff y(t)\mu(t) + C = \int \mu(t)g(t)\,dt \Rightarrow y(t) = \frac{\int \mu(t)g(t)\,dt + C}{\mu(t)} \Rightarrow$

$$\Rightarrow y(t) = \frac{\int k e^{\int p(t)\,dt} \cdot g(t)\,dt + C}{k \cdot e^{\int p(t)\,dt}} = \frac{\int e^{\int p(t)\,dt} g(t)\,dt + K}{e^{\int p(t)\,dt}}$$

Procas Markov pt. un sistem cu reparare



$$\left(P_1(t)\ \ P_0(t)\right)\cdot A^* = \left(P_1'(t)\ \ P_0'(t)\right)$$

$$A^* = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix}$$

Condiții inițiale:

$P_1(0) = 1 \quad P_0(0) = 0$

$$\begin{cases} P_1'(t) = -\lambda P_1(t) + \mu P_0(t) \\ P_0'(t) = \lambda P_1(t) - \mu P_0(t) \end{cases}$$

$$\int e^{ax}dx = \frac{1}{a}e^{ax} + c$$

$P_1(t) + P_0(t) = 1 \Rightarrow P_0(t) = 1 - P_1(t)$

$$P_1'(t) = -\lambda P_1(t) + \mu\left(1 - P_1(t)\right) \Rightarrow P_1'(t) + (\lambda+\mu)P_1(t) = \mu$$

$k \cdot e^{\int(\lambda+\mu)dt} = k \cdot e^{(\lambda+\mu)t}$

$g(t) = \mu \qquad p(t) = (\lambda+\mu)$

$$P_1(t) = \frac{\int e^{(\lambda+\mu)t}\,\mu\,dt + k}{e^{(\lambda+\mu)t}} = \frac{\mu\frac{1}{\lambda+\mu}e^{(\lambda+\mu)t} + k}{e^{(\lambda+\mu)t}} = \frac{\frac{\mu}{\lambda+\mu}e^{(\lambda+\mu)t} + \frac{\lambda}{\lambda+\mu}}{e^{(\lambda+\mu)t}}$$

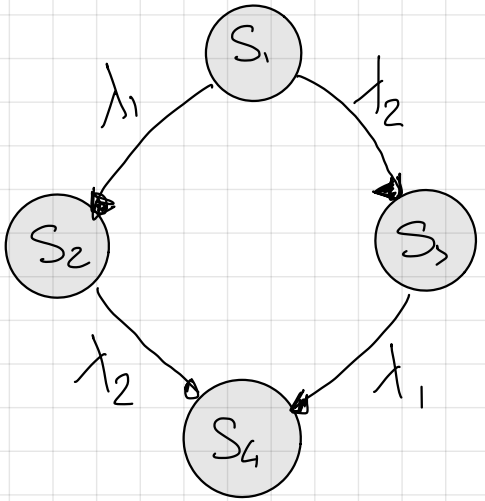
$P_1(0) = 1 \Rightarrow \frac{\mu}{\lambda+\mu} + k = 1 \Rightarrow k = 1 - \frac{\mu}{\lambda+\mu} = \frac{\lambda}{\lambda+\mu}$

$$= \frac{\mu\, e^{(\lambda+\mu)t} + \lambda}{e^{(\lambda+\mu)t}} \cdot \frac{1}{\lambda+\mu} = \left(\mu + \frac{\lambda}{e^{(\lambda+\mu)t}}\right) \cdot \frac{1}{\lambda+\mu}$$

$$\lim_{t\to\infty} P_1(t) = \frac{\mu}{\lambda+\mu} \quad \mu\lambda \quad = \frac{\frac{1}{\lambda}}{\frac{1}{\mu}+\frac{1}{\lambda}} = \frac{MTBF}{MTR+MTBF} = A$$

Sistem cu două componente

Stare	Comp. 1	Comp. 2
S_1	Op.	Op.
S_2	F	Op.
S_3	Op.	F
S_4	F	F



$$\left(P_1(t)\ \ P_2(t)\ \ P_3(t)\ \ P_4(t)\right)\cdot A^* = \left(P_1'(t)\ \ P_2'(t)\ \ P_3'(t)\ \ P_4'(t)\right)$$

$$A^* = \begin{pmatrix} -\lambda_1-\lambda_2 & \lambda_1 & \lambda_2 & 0 \\ 0 & -\lambda_2 & 0 & \lambda_2 \\ 0 & 0 & -\lambda_1 & \lambda_1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{cases} P_1'(t) = -(\lambda_1+\lambda_2)P_1(t) \\ P_2'(t) = \lambda_1 P_1(t) - \lambda_2 P_2(t) \\ P_3'(t) = \lambda_2 P_1(t) - \lambda_1 P_3(t) \\ P_4'(t) = \lambda_2 P_2(t) + \lambda_1 P_1(t) \end{cases} \Rightarrow$$

$$P_1(t)+P_2(t)+P_3(t)+P_4(t) = 1$$

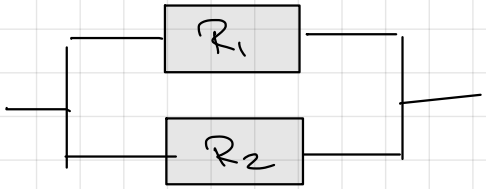
$$\frac{P_1'(t)}{P_1(t)} = -(\lambda_1+\lambda_2) \Big/ \int \Rightarrow \int \frac{P_1'(t)}{P_1(t)}dt = -\int(\lambda_1+\lambda_2)dt \Rightarrow$$

$$P_1(t) = e^{-(\lambda_1+\lambda_2)t}$$

$$P_2(t) = e^{-\lambda_2 t} - e^{-(\lambda_1+\lambda_2)t}$$

$$P_3(t) = e^{-\lambda_1 t} - e^{-(\lambda_1+\lambda_2)t}$$

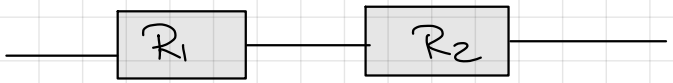
Paralel



$$R_P = 1 - (1-R_1)(1-R_2)$$
$$= 1 - (1 - R_2 - R_1 + R_1R_2)$$
$$= R_1 + R_2 - R_1R_2$$
$$= e^{-\lambda_1 t} + e^{-\lambda_2 t} - e^{-(\lambda_1+\lambda_2)t}$$

$$P_1(t) + P_2(t) + P_3(t) = e^{-\lambda_1 t} + e^{-\lambda_2 t} - e^{-(\lambda_1+\lambda_2)t}$$

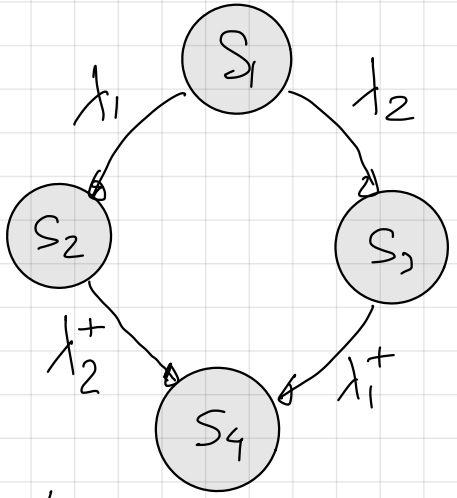
Serie



$$R_S = R_1R_2 = e^{-\lambda_1 t} \cdot e^{-\lambda_2 t} = e^{-(\lambda_1+\lambda_2)t}$$
$$\| $$
$$P_1(t)$$

Load sharing system

$$A^* = \begin{pmatrix} -\lambda_1-\lambda_2 & \lambda_1 & \lambda_2 & 0 \\ 0 & -\lambda_2^+ & 0 & \lambda_2^+ \\ 0 & 0 & -\lambda_1^+ & \lambda_1^+ \\ 0 & 0 & 0 & 0 \end{pmatrix}$$



$$\frac{dP_1(t)}{dt} = -(\lambda_1+\lambda_2)P_1(t) \Rightarrow P_1(t) = e^{-(\lambda_1+\lambda_2)t}$$

$$\frac{dP_2(t)}{dt} = \lambda_1 P_1(t) - \lambda_2^+ P_2(t) \Rightarrow P_2(t) = \frac{\lambda_1}{\lambda_1+\lambda_2-\lambda_2^+}\left(e^{-\lambda_2^+ t} - e^{-(\lambda_1+\lambda_2)t}\right)$$

$$\frac{dP_3(t)}{dt} = \lambda_2 P_1(t) - \lambda_1^+ P_3(t) \Rightarrow P_3(t) = \frac{\lambda_1}{\lambda_1+\lambda_2-\lambda_1^+}\left(e^{-\lambda_1^+ t} - e^{-(\lambda_1+\lambda_2)t}\right)$$

$$P_4(t) = 1 - P_1(t) - P_2(t) - P_3(t)$$

$$R(t) = ? = P_1(t) + P_2(t) + P_3(t) \qquad MTBF = ? = \int R(t)\,dt \ldots.$$

$\lambda_1 = 10/\text{am}$ $\quad \lambda_1^+ = 50/\text{am}$

$\lambda_2 = 20/\text{am}$ $\quad \lambda_2^+ = 100/\text{am}$

Sistem cu 3 componente cu standby

Stare	Comp. 1	Comp. 2	Comp. 3
S_1	Op.	Stby.	Stby.
S_2	F	Op.	Stby.
S_3	F	F	Op.
S_4	F	F	F



p.p. Comp. identice $\Rightarrow$ cu același λ

$$A^* = \begin{pmatrix} -\lambda & \lambda & 0 & 0 \\ 0 & -\lambda & \lambda & 0 \\ 0 & 0 & -\lambda & \lambda \\ 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow P(t) \cdot A^* = P'(t)$$

$$\begin{cases} P_1'(t) = -\lambda P_1(t) \Rightarrow P_1(t) = e^{-\lambda t} \\ P_2'(t) = \lambda P_1(t) - \lambda P_2(t) \Rightarrow P_2'(t) = \lambda e^{-\lambda t} - \lambda P_2(t) \Rightarrow P_2'(t) + \lambda P_2(t) = \lambda e^{-\lambda t} \\ P_3'(t) = \lambda P_2(t) - \lambda P_3(t) \\ P_4'(t) = \lambda P_3(t) \dots \end{cases}$$

$$P_2(t) = \lambda t e^{-\lambda t}$$

$$P_3(t) = \frac{(\lambda t)^2 e^{-\lambda t}}{2}$$

$$R(t) = P_1(t) + P_2(t) + P_3(t) = e^{-\lambda t} + \lambda t e^{-\lambda t} + \frac{(\lambda t)^2}{2} e^{-\lambda t} =$$

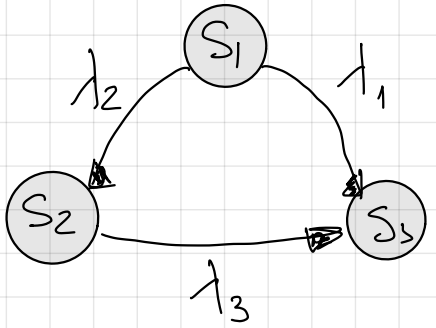
$$= e^{-\lambda t}\left(1 + \lambda t + \frac{(\lambda t)^2}{2}\right)$$

$$MTBF = \int R(t)\,dt = \int \left(e^{-\lambda t} + \lambda t e^{-\lambda t} + \frac{(\lambda t)^2}{2} e^{-\lambda t}\right) dt =$$

$$= \int e^{-\lambda t} dt + \lambda \int t e^{-\lambda t} dt + \frac{\lambda^2}{2} \int t^2 e^{-\lambda t} dt = \frac{3}{\lambda} = 3 \cdot MTBF_i$$

Sistem cu funcționare degradată

S_1: Complet operațional
S_2: stare degradată
S_3: defect



$$A^* = \begin{pmatrix} -\lambda_1-\lambda_2 & \lambda_2 & \lambda_1 \\ 0 & -\lambda_3 & \lambda_3 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{cases} P_1'(t) = -(\lambda_1+\lambda_2)P_1(t) \\ P_2'(t) = \lambda_2 P_1(t) - \lambda_3 P_2(t) \\ P_3'(t) = \lambda_1 P_1(t) + \lambda_3 P_2(t) \end{cases} \Rightarrow P_2'(t) + \lambda_3 P_2(t) = \lambda_2 e^{-(\lambda_1+\lambda_2)t}$$

$$P_1(t) = e^{-(\lambda_1+\lambda_2)t}$$

$$P_2(t) = \frac{\lambda_2}{\lambda_1+\lambda_2-\lambda_3}\left(e^{-\lambda_3 t} - e^{-(\lambda_1+\lambda_2)t}\right)$$

$\lambda_1 = 2/\text{an}$
$\lambda_2 = 5/\text{an}$
$\lambda_3 = 10/\text{an}$

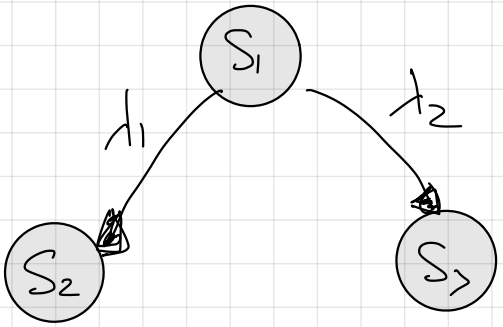
$$R(t) = P_1(t) + P_2(t) = \ldots$$

$$MTBF = \int R(t)\,dt = \int e^{-(\lambda_1+\lambda_2)t}\,dt + \int \frac{\lambda_2}{\lambda_1+\lambda_2-\lambda_3}\left(e^{-\lambda_3 t} - e^{-(\lambda_1+\lambda_2)t}\right)dt$$

$$= \frac{1}{\lambda_1+\lambda_2} + \frac{\lambda_2}{\lambda_1+\lambda_2-\lambda_3}\left(\frac{1}{\lambda_3} - \frac{1}{\lambda_1+\lambda_2}\right) = \frac{1}{2+5} + \frac{5}{2+5-10}\left(\frac{1}{10} - \frac{1}{2+5}\right) = \ldots$$

Sistema three - state

S_1 : operational
S_2 : Fail - open
S_3 : Fail - short



$$A^* = \begin{pmatrix} -\lambda_1-\lambda_2 & \lambda_1 & \lambda_2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\begin{cases} P_1'(t) = -(\lambda_1+\lambda_2) P_1(t) \\ P_2'(t) = \lambda_1 P_1(t) \\ P_3'(t) = \lambda_2 P_1(t) \end{cases} \Rightarrow \begin{cases} P_1(t) = e^{-(\lambda_1+\lambda_2)t} \\ P_2(t) = \lambda_1 e^{-(\lambda_1+\lambda_2)t} \\ P_3(t) = \lambda_2 e^{-(\lambda_1+\lambda_2)t} \end{cases}$$

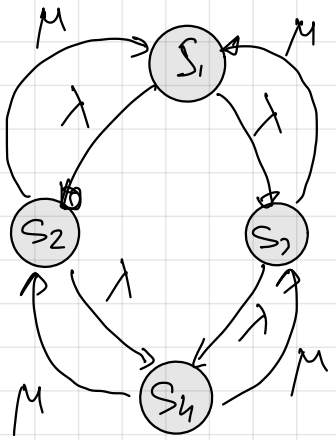
$$R(t) = P_1(t) = e^{-(\lambda_1+\lambda_2)t}$$

$$MTBF = \frac{1}{\lambda_1+\lambda_2}$$

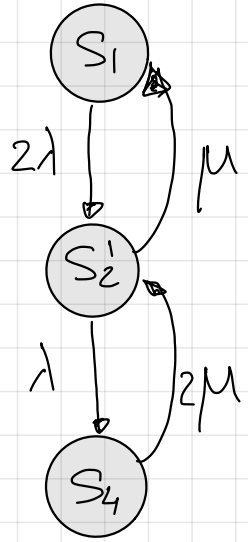
Sistem cu 2 componente si cu 2 depanatori

Stare	Comp. 1	Comp. 2
S_1	Op.	Op.
S_2	Repan	Op.
S_3	Op.	Repan
S_4	Rep.	Rep.

S_2' = S_2 ∪ S_3



$$A^* = \begin{pmatrix} -2\lambda & 2\lambda & 0 \\ \mu & -\mu-\lambda & \lambda \\ 0 & 2\mu & -2\mu \end{pmatrix}$$



$$\begin{cases} P_1'(t) = -2\lambda P_1(t) + \mu P_2(t) \\ P_2'(t) = 2\lambda P_1(t) - (\lambda+\mu) P_2(t) + 2\mu P_4(t) \\ P_4'(t) = \lambda P_2(t) - 2\mu P_4(t) \end{cases}$$

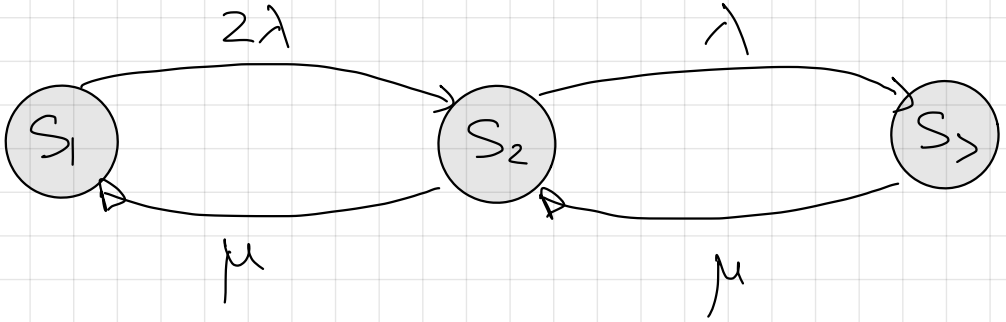
$$P_1(t) + P_2(t) + P_4(t) = 1$$

$\lambda = 3/\text{an} \qquad \mu = 5/\text{an}$

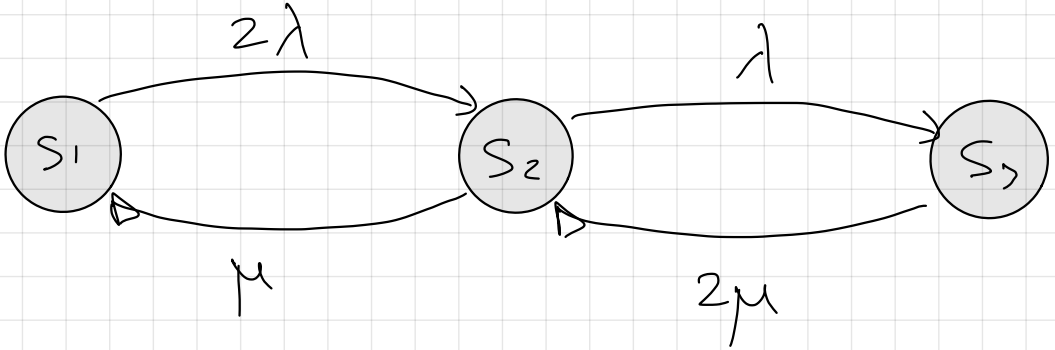
$$R(t) = P_1(t) + P_2(t) = \dots$$

$$\int R(t)\,dt = MTBF = \dots$$

S_1 – ambele operaționale S_2 – 1 sistem defect S_3 – ambele defecte



1 depanator



2 depanatori